Atom optics and quantum technologies

Fisica Quantistica

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ATOM OPTICS AND QUANTUM TECHNOLOGIES

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Quantum theory of atom-light interactions

General problem of perturbation of a two-state system

Let us consider a system described by an unperturbed Hamiltonian operator $\hat{H}_0$ with two unperturbed eigenstates $|\varphi_1\rangle$ and $|\varphi_2\rangle$ such that

$$\hat{H}_0|\varphi_1\rangle = E_1|\varphi_1\rangle$$
$$\hat{H}_0|\varphi_2\rangle = E_2|\varphi_2\rangle$$

The two eigenstates provide a complete basis, thus their linear combination can be used to represent any state. This basis is orthonormal, meaning that

$$\langle\varphi_i|\varphi_j\rangle = \delta_{ij}$$

The **spectral decomposition of a state** consists in writing the general state as

$$|\psi\rangle = \sum_{i=1}^{2}|\varphi_i\rangle\langle\varphi_i|\psi\rangle = \hat{\mathbb{I}}|\psi\rangle$$

where $\langle\varphi_i|\psi\rangle$ is the component of the general state $|\psi\rangle$ along $|\varphi_i\rangle$.

We have defined the identity operator

$$\hat{\mathbb{I}} = \sum_{i=1}^{2}|\varphi_i\rangle\langle\varphi_i|$$

and the $|\psi\rangle = \hat{\mathbb{I}}|\psi\rangle$ relation is called resolution of identity.

This operator can also be applied to another operator, which gives us its **matrix representation**. For example, in the case of the uncoupled Hamiltonian:

$$\hat{H}_0 = \hat{\mathbb{I}}\hat{H}_0\hat{\mathbb{I}} = \sum_{i=1}^{2}|\varphi_i\rangle\langle\varphi_i|\,\hat{H}_0\sum_{j=1}^{2}|\varphi_j\rangle\langle\varphi_j| = \sum_{i,j=1}^{2}|\varphi_i\rangle\langle\varphi_i|\hat{H}_0|\varphi_j\rangle\langle\varphi_j| = \sum_{i,j=1}^{2}H_{0,ij}|\varphi_i\rangle\langle\varphi_j|$$

where we have defined the matrix element of the operator $\hat{H}_0$

$$H_{0,ij} = \langle\varphi_i|\hat{H}_0|\varphi_j\rangle$$

such that

$$\hat{H}_0 \quad \leftrightarrow \quad \begin{bmatrix} H_{0,11} & H_{0,12} \\ H_{0,21} & H_{0,22} \end{bmatrix} = \begin{bmatrix} \langle\varphi_1|\hat{H}_0|\varphi_1\rangle & \langle\varphi_1|\hat{H}_0|\varphi_2\rangle \\ \langle\varphi_2|\hat{H}_0|\varphi_1\rangle & \langle\varphi_2|\hat{H}_0|\varphi_2\rangle \end{bmatrix}$$

In particular, on our basis

$$\hat{H}_0 \quad \leftrightarrow \quad \begin{bmatrix} E_1 & 0 \\ 0 & E_2 \end{bmatrix}$$

which is a diagonal matrix: this is a general result any time we use as basis the eigenstates of the operator whose matrix we are computing.

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Case study: Hydrogen

By studying the perturbation of the hydrogen atom, we want to show that in nature it is always a molecule, H_2^+.

Let us consider two protons very far apart so that they don't interact, and an electron close to one of the protons.

The Hamiltonian for this system is

$$\hat{H}_0 = \hat{E}_k + \hat{E}_{p,1} + \hat{E}_{p,2} \simeq \hat{E}_k + \hat{E}_{p,1}$$

where E_k is the kinetic energy of the electron and $E_{p,i}$ the potential energy due to proton i and where we have neglected the potential energy of the second proton since it is much further away.

This corresponds to the Hamiltonian of an hydrogen atom for which the solution is the atomic orbital ground state $|\varphi_1\rangle$ $(1s)$ with energy E_1.

We also need to consider the opposite situation, where the electron is close to the other photon. The solution of this problem, $|\varphi_2\rangle$, is the same function as $|\varphi_1\rangle$ except it depends on the coordinate of proton 2, $\vec{r}_2$. The two states are degenerate, $E_1 = E_2$.

Dynamics of the unperturbed system and stationary states

Let us consider the dynamic of the system described by the unperturbed Hamiltonian.

$$\hat{H}_0|\varphi_1\rangle = E_1|\varphi_1\rangle$$
$$\hat{H}_0|\varphi_2\rangle = E_2|\varphi_2\rangle$$

The general state can be represented as

$$|\psi(t)\rangle = a_1(t)|\varphi_1\rangle + a_2(t)|\varphi_2\rangle$$

Let us consider the time-dependent Schrödinger equation

$$\boldsymbol{i\hbar\frac{\partial}{\partial t}|\psi\rangle = \hat{H}_0|\psi\rangle}$$

If we multiply by $\langle\varphi_1|$ or $\langle\varphi_2|$ we get for the left side

$$\begin{cases} i\hbar\frac{\partial}{\partial t}\langle\varphi_1|\psi\rangle = i\hbar\frac{\partial}{\partial t}\langle\varphi_1|a_1(t)\varphi_1 + a_2(t)\varphi_2\rangle = i\hbar\frac{\partial}{\partial t}a_1(t) \\ i\hbar\frac{\partial}{\partial t}\langle\varphi_2|\psi\rangle = i\hbar\frac{\partial}{\partial t}\langle\varphi_2|a_1(t)\varphi_1 + a_2(t)\varphi_2\rangle = i\hbar\frac{\partial}{\partial t}a_2(t) \end{cases}$$

and for the right side

$$\begin{cases} \langle\varphi_1|\hat{H}_0|\psi\rangle = a_1(t)\langle\varphi_1|\hat{H}_0|\varphi_1\rangle + a_2(t)\langle\varphi_1|\hat{H}_0|\varphi_2\rangle = a_1(t)E_1 \\ \langle\varphi_2|\hat{H}_0|\psi\rangle = a_1(t)\langle\varphi_2|\hat{H}_0|\varphi_1\rangle + a_2(t)\langle\varphi_2|\hat{H}_0|\varphi_2\rangle = a_2(t)E_2 \end{cases}$$

which gives two coupled equations

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$$\begin{cases} i\hbar\frac{\partial}{\partial t}a_1(t) = a_1(t)E_1 \\ i\hbar\frac{\partial}{\partial t}a_2(t) = a_2(t)E_2 \end{cases} \Rightarrow \begin{cases} \dot{a}_1(t) = -\frac{i}{\hbar}E_1 a_1(t) \\ \dot{a}_2(t) = -\frac{i}{\hbar}E_2 a_2(t) \end{cases}$$

$$\Rightarrow \begin{cases} a_1(t) = a_1(0)e^{-i\frac{E_1}{\hbar}t} \\ a_2(t) = a_2(0)e^{-i\frac{E_2}{\hbar}t} \end{cases}$$

The general state at time t of the unperturbed system is therefore

$$\boldsymbol{|\psi(t)\rangle = a_1(0)e^{-i\frac{E_1}{\hbar}t}|\varphi_1\rangle + a_2(0)e^{-i\frac{E_2}{\hbar}t}|\varphi_2\rangle}$$

When the system is prepared in an eigenstate, for example $|\psi(0)\rangle = |\varphi_1\rangle$, it will remain in such state:

$$a_1(0) = 1 \quad a_2(0) = 0$$

$$|\psi(0)\rangle = e^{-i\frac{E_1}{\hbar}t}|\varphi_1\rangle$$

The eigenstates are stationary states: if the system is put in one of these states, it will stay there forever.

Perturbed system

We will first consider the **time-independent perturbation theory**.

The perturbed Hamiltonian is given by

$$\hat{H} = \hat{H}_0 + \hat{W}$$

where $\hat{W}$ is the Hamiltonian of the perturbation.

The eigenvalue problem takes the form

$$\hat{H}|\psi_+\rangle = E_+|\psi_+\rangle$$
$$\hat{H}|\psi_-\rangle = E_-|\psi_-\rangle$$

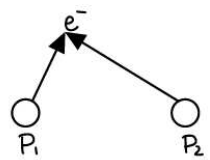
where $|\psi_+\rangle$ and $|\psi_-\rangle$ are the perturbed eigenstates, which are stationary states for the perturbed system described by $\hat{H}$, while E_+ and E_- are their eigenvalues.

The consequences of the perturbation are:

1) E_1 and E_2 are no longer possible energies;
2) $|\varphi_1\rangle$ and $|\varphi_2\rangle$ are no longer stationary states.

Case study: Hydrogen

Let us consider the H_2^+ molecule: by decreasing the distance between the two protons, we increase the perturbation given by the interaction with the second proton.



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The unperturbed state is the atomic orbital $1s$, $|\psi_1\rangle$. The new eigenstates will be $|\psi_+\rangle$ and $|\psi_-\rangle$ and we are going to focus on the one with the lower energy, $|\psi_-\rangle$.

The H_2^+ molecule is favoured if $E_- < E_1 = E_2$: in this case, the formation of the molecule happens spontaneously.

We are now going to diagonalise the Hamiltonian to find the eigenstates and energies of the perturbed system.

We <u>work on the basis of the eigenstates of the uncoupled Hamiltonian</u>, $|\varphi_1\rangle$ and $|\varphi_2\rangle$.

$$H_{ij} = \langle\varphi_i|\hat{H}|\varphi_j\rangle = \langle\varphi_i|\hat{H}_0|\varphi_j\rangle + \langle\varphi_i|\hat{W}|\varphi_j\rangle$$

$$\begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} = \begin{bmatrix} E_1 + W_{11} & W_{12} \\ W_{21} & E_2 + W_{22} \end{bmatrix}$$

The <u>diagonalisation</u> of $\hat{H}$ is equivalent to solving the eigenvalue equation

$$\hat{H}|\psi_+\rangle = E_+|\psi_+\rangle$$
$$\hat{H}|\psi_-\rangle = E_-|\psi_-\rangle$$

since on this basis the matrix is diagonal.

Let us start from the expression of the matrix associated to the perturbed Hamiltonian

$$\begin{aligned}
\begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} &= \begin{bmatrix} \frac{1}{2}(H_{11} + H_{11}) + \frac{1}{2}(H_{22} - H_{22}) & H_{12} \\ H_{21} & \frac{1}{2}(H_{11} - H_{11}) + \frac{1}{2}(H_{22} + H_{22}) \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{2}(H_{11} + H_{22}) + \frac{1}{2}(H_{11} - H_{22}) & H_{12} \\ H_{21} & \frac{1}{2}(H_{11} + H_{22}) - \frac{1}{2}(H_{11} - H_{22}) \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{2}(H_{11} + H_{22}) & 0 \\ 0 & \frac{1}{2}(H_{11} + H_{22}) \end{bmatrix} + \begin{bmatrix} \frac{1}{2}(H_{11} - H_{22}) & H_{12} \\ H_{21} & -\frac{1}{2}(H_{11} - H_{22}) \end{bmatrix} \\
&= \frac{1}{2}(H_{11} + H_{22})[\mathbb{I}] + \frac{1}{2}(H_{11} - H_{22}) \begin{bmatrix} 1 & 2\frac{H_{12}}{H_{11} - H_{22}} \\ 2\frac{H_{21}}{H_{11} - H_{22}} & -1 \end{bmatrix}
\end{aligned}$$

$$\Rightarrow \quad [H] = h_1[\mathbb{I}] + h_2[K]$$
$$\hat{H} = h_1 + h_2\hat{K}$$

where $h_1 = \frac{1}{2}(H_{11} + H_{22})$ and $h_2 = \frac{1}{2}(H_{11} - H_{22})$.

Therefore, $\hat{H}$ and $\hat{K}$ share the same eigenfunctions $|\psi_+\rangle$ and $|\psi_-\rangle$ but with different eigenvalues E_+, E_- and k_+, k_-:

$$\begin{cases} \hat{H}|\psi_{+,-}\rangle = E_{+,-}|\psi_{+,-}\rangle \\ \hat{H}|\psi_{+,-}\rangle = (h_1 + h_2\hat{K})|\psi_{+,-}\rangle = (h_1 + h_2 k_{+,-})|\psi_{+,-}\rangle \end{cases}$$

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The relation between the eigenvalues is thus

$$E_{+,-} = h_1 + h_2 k_{+,-}$$

which means that we can start by <u>computing the eigenvalues of $\widehat{K}$</u>.

Introducing the variables θ and φ such that

$$\begin{cases} \boldsymbol{\tan\theta = \dfrac{2|H_{12}|}{H_{11} - H_{22}}}, & \boldsymbol{0 \le \theta \le \pi} \\ \boldsymbol{H_{21} = |H_{12}| e^{i\varphi}}, & \boldsymbol{0 \le \varphi \le 2\pi} \end{cases}$$

and keeping in mind the relation $H_{12} = H_{21}^* = |H_{12}| e^{-i\varphi}$ (since $\widehat{H}$ is Hermitian) we can write

$$[\boldsymbol{K}] = \begin{bmatrix} 1 & \dfrac{2H_{12}}{H_{11} - H_{22}} \\ \dfrac{2H_{21}}{H_{11} - H_{22}} & -1 \end{bmatrix} = \begin{bmatrix} 1 & \dfrac{2|H_{12}| e^{-i\varphi}}{H_{11} - H_{22}} \\ \dfrac{2|H_{12}| e^{i\varphi}}{H_{11} - H_{22}} & -1 \end{bmatrix} = \begin{bmatrix} \boldsymbol{1} & \boldsymbol{\tan\theta\, e^{-i\varphi}} \\ \boldsymbol{\tan\theta\, e^{i\varphi}} & \boldsymbol{-1} \end{bmatrix}$$

To find the eigenvalues we must solve

$$0 = \det([K] - k[\mathbb{I}]) = \det \begin{bmatrix} 1-k & \tan\theta\, e^{-i\varphi} \\ \tan\theta\, e^{i\varphi} & -1-k \end{bmatrix} = -(1-k)(1+k) - \tan^2\theta$$
$$= k^2 - 1 - \tan^2\theta$$

$$k^2 = 1 + \tan^2\theta = 1 + \frac{\sin^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}$$

$$k_{+,-} = \pm\frac{1}{\cos\theta}$$

$$\tan^2\theta = \frac{4|H_{12}|^2}{(H_{11} - H_{22})^2}$$

$$\frac{1}{\cos^2\theta} = 1 + \tan^2\theta = 1 + \frac{4|H_{12}|^2}{(H_{11} - H_{22})^2} = \frac{(H_{11} - H_{22})^2 + 4|H_{12}|^2}{(H_{11} - H_{22})^2}$$

$$\frac{1}{\cos\theta} = \pm\frac{\sqrt{(H_{11} - H_{22})^2 + 4|H_{12}|^2}}{H_{11} - H_{22}}$$

We can show that only the positive sign can be accepted:

$$\tan\theta = \frac{2|H_{12}|}{H_{11} - H_{22}} \qquad \begin{cases} \text{if } H_{11} - H_{22} > 0, & \tan\theta > 0 \quad 0 \le \theta \le \pi/2 \quad \cos\theta > 0 \\ \text{if } H_{11} - H_{22} < 0, & \tan\theta < 0 \quad \pi/2 \le \theta \le \pi \quad \cos\theta < 0 \end{cases}$$

Thus, $1/\cos\theta$ should have the same sign as the denominator: there is a plus sign.

The eigenvalues are then

$$k_{+,-} = \pm\frac{1}{\cos\theta} = \pm\frac{\sqrt{(H_{11} - H_{22})^2 + 4|H_{12}|^2}}{H_{11} - H_{22}}$$

We have used the extra steps to show the role of $1/\cos\theta$, while in a more direct way we could have computed:

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$$k_{+,-} = \pm\sqrt{1+\tan^2\theta} = \pm\sqrt{1+\frac{4|H_{12}|^2}{(H_{11}-H_{22})^2}} = \pm\frac{\sqrt{(H_{11}-H_{22})^2+4|H_{12}|^2}}{H_{11}-H_{22}}$$

To find the eigenstates of $\widehat{K}$ we must solve the equation for $|\psi_+\rangle = a|\varphi_1\rangle + b|\varphi_2\rangle$

$$[K]|\psi_+\rangle = k_+|\psi_+\rangle$$

$$\begin{bmatrix} 1 & \tan\theta\, e^{-i\varphi} \\ \tan\theta\, e^{i\varphi} & -1 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = k_+\begin{bmatrix} a \\ b \end{bmatrix}$$

and the analogous equation for $|\psi_-\rangle = c|\varphi_1\rangle + d|\varphi_2\rangle$

$$[K]|\psi_-\rangle = k_-|\psi_-\rangle$$

$$\begin{bmatrix} 1 & \tan\theta\, e^{-i\varphi} \\ \tan\theta\, e^{i\varphi} & -1 \end{bmatrix}\begin{bmatrix} c \\ d \end{bmatrix} = k_-\begin{bmatrix} c \\ d \end{bmatrix}$$

For $|\psi_+\rangle$ we get

$$\begin{cases} a + b\tan\theta\, e^{-i\varphi} = k_+ a = \dfrac{1}{\cos\theta}a \\ a\tan\theta\, e^{i\varphi} + b = k_+ b = \dfrac{1}{\cos\theta}b \end{cases}$$

which are two linearly dependent equations, since the determinant is equal to zero. From the first one we get

$$a\cos\theta + b\sin\theta\, e^{-i\varphi} = a$$

$$a(1-\cos\theta) = b\sin\theta\, e^{-i\varphi}$$

$$2a\sin^2\left(\frac{\theta}{2}\right) = 2b\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)e^{-i\varphi}$$

$$a\left[\sin\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}\right] = b\left[\cos\left(\frac{\theta}{2}\right)e^{-i\frac{\varphi}{2}}\right]$$

A solution is

$$\begin{cases} a = \cos\left(\dfrac{\theta}{2}\right)e^{-i\frac{\varphi}{2}} \\ b = \sin\left(\dfrac{\theta}{2}\right)e^{i\frac{\varphi}{2}} \end{cases}$$

$$\boldsymbol{|\psi_+\rangle = \cos\left(\frac{\theta}{2}\right)e^{-i\frac{\varphi}{2}}|\varphi_1\rangle + \sin\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}|\varphi_2\rangle}$$

With the same calculations we can find

$$\boldsymbol{|\psi_-\rangle = -\sin\left(\frac{\theta}{2}\right)e^{-i\frac{\varphi}{2}}|\varphi_1\rangle + \cos\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}|\varphi_2\rangle}$$

We have already found the eigenvalues of $\widehat{K}$, from which we can find the ones of the Hamiltonian:

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$$E_{+,-} = h_1 + h_2 k_{+,-} = \frac{1}{2}(H_{11} + H_{22}) + \frac{1}{2}(H_{11} - H_{22})\left(\pm\frac{\sqrt{(H_{11} - H_{22})^2 + 4|H_{12}|^2}}{H_{11} - H_{22}}\right)$$
$$= \frac{1}{2}(H_{11} + H_{22}) \pm \frac{1}{2}\sqrt{(H_{11} - H_{22})^2 + 4|H_{12}|^2}$$

We now need to <u>find the matrix elements of $\hat{H}$</u>, which appear in the expressions we have just retrieved.

It can be assumed that the $\hat{W}$ matrix is antidiagonal

$$[W] = \begin{bmatrix} 0 & W_{12} \\ W_{21} & 0 \end{bmatrix}$$

Let us consider the case of electric dipole interaction, keeping in mind that we are working with <u>time-independent perturbation</u> theory:

$$\hat{W} = -\hat{\vec{\mu}}_e \cdot \vec{E} = -\hat{\vec{\mu}}_e \cdot \vec{\varepsilon} E_0$$

where we have introduced the polarisation versor $\vec{\varepsilon}$.

$$W_{ij} = \langle \varphi_i | -\hat{\vec{\mu}}_e \cdot \vec{\varepsilon} E_0 | \varphi_j \rangle = \langle \varphi_i | e\hat{\vec{r}} \cdot \vec{\varepsilon} E_0 | \varphi_j \rangle = e\langle \varphi_i | \hat{\vec{r}} | \varphi_j \rangle \cdot \vec{\varepsilon} E_0 = e\, \vec{r}_{ij} \cdot \vec{\varepsilon} E_0$$

We can notice that

$$\vec{r}_{ij} = \langle \varphi_i | \hat{\vec{r}} | \varphi_j \rangle = \int \varphi_i^*(\vec{r}) \vec{r} \varphi_j(\vec{r}) dr^3$$

where $\vec{r}$ has odd parity and the eigenstates have defined parity: thus, the matrix of $\hat{r}$ is antidiagonal, which implies that the matrix associated to $\hat{W}$ is antidiagonal, too.

Moreover, is we arbitrarily choose the phases of $\varphi_{i,j}$ so that $\vec{r}_{ij}$ is real, it holds $\vec{r}_{12} = \vec{r}_{21}^* = \vec{r}_{21}$.

The matrix of $\hat{H}$ is then

$$\begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} = \begin{bmatrix} E_1 & W_{12} \\ W_{21} & E_2 \end{bmatrix}$$

With these matrix elements we get

$$\begin{cases} \tan\theta = \dfrac{2|H_{12}|}{H_{11} - H_{22}} = \dfrac{2|W_{21}|}{E_1 - E_2} \\ H_{21} = |H_{12}| e^{i\varphi} \quad \rightarrow \quad W_{21} = |W_{21}| e^{i\varphi} \end{cases}$$

$$E_{+,-} = \frac{1}{2}(H_{11} + H_{22}) \pm \frac{1}{2}\sqrt{(H_{11} - H_{22})^2 + 4|H_{12}|^2}$$
$$= \frac{1}{2}(E_1 + E_2) \pm \frac{1}{2}\sqrt{(E_1 - E_2)^2 + 4|W_{21}|^2}$$

What is the physical meaning of these values of the energies?

Defining

$$\begin{cases} E_m = \dfrac{1}{2}(E_1 + E_2) \\ \Delta = \dfrac{1}{2}(E_1 - E_2) \end{cases}$$

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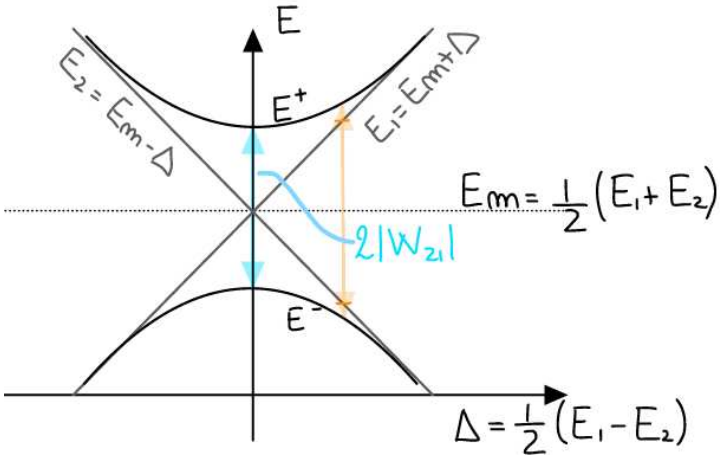
we can write

$$\boldsymbol{E_+ = E_m + \sqrt{\Delta^2 + |W_{21}|^2}}$$
$$E_- = E_m - \sqrt{\Delta^2 + |W_{21}|^2}$$

and express E_1 and E_2 with respect to E_m and Δ:

$$E_2 = 2E_m - E_1$$
$$2\Delta = E_1 - E_2 = E_1 - (2E_m - E_1)$$
$$\begin{cases} E_1 = E_m + \Delta \\ E_2 = E_m - \Delta \end{cases}$$

In figure, the energies are represented as a function of Δ:



When Δ is very large, the perturbed and unperturbed energies become almost equal, while when Δ is zero their difference is maximised. For any Δ, however, it holds

$$E_+ - E_- > E_1 - E_2$$

Therefore, as a consequence of the coupling ($W_{21} \neq 0$), the two energy levels repel each other at all values of Δ. This effect is even more pronounced for $\Delta = 0$ (which represents the degeneracy condition $E_1 = E_2$), where

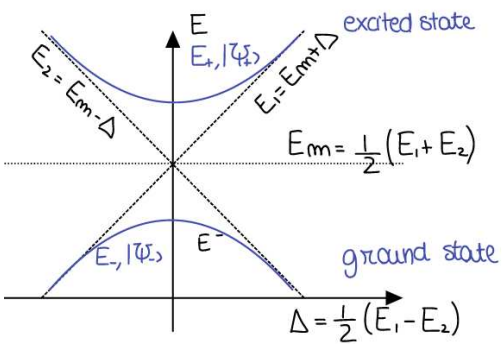
$$E_+ - E_- = 2|W_{21}|$$

Case study: Hydrogen

Let us consider once again the hydrogen molecule. Since the two unperturbed states are equivalent, they are degenerate and $\Delta = 0$. When the perturbation is introduced, meaning that the protons are moved quite close to each other, the energy of the perturbed ground state is then

$$E_- = E_m - \sqrt{\Delta^2 + |W_{21}|^2} = E_m - |W_{21}|$$
$$= E_1 - |W_{21}| < E_1 = E_2$$

where we have considered that in the degenerate case $\Delta = 0$ and $E_m = \frac{1}{2}(E_1 + E_2) = E_1 = E_2$.



We have thus demonstrated that the formation of the molecule is favoured.

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Weak and strong coupling

Energies

We have found that the energies are given by:

$$E_+ = E_m + \sqrt{\Delta^2 + |W_{21}|^2}$$
$$E_- = E_m - \sqrt{\Delta^2 + |W_{21}|^2}$$

In the **weak coupling** case

$$|W_{21}| \ll \Delta$$

$$E_{+,-} = E_m \pm \Delta \sqrt{1 + \frac{|W_{21}|^2}{\Delta^2}} \simeq E_m \pm \Delta \left[1 + \frac{|W_{21}|^2}{2\Delta^2}\right]$$

$$E_+ \simeq E_m + \Delta + \frac{|W_{21}|^2}{2\Delta}$$
$$E_- \simeq E_m - \Delta - \frac{|W_{21}|^2}{2\Delta}$$

Which considering the relations

$$E_m + \Delta = \frac{1}{2}(E_1 + E_2) + \frac{1}{2}(E_1 - E_2) = E_1$$
$$E_m - \Delta = \frac{1}{2}(E_1 + E_2) - \frac{1}{2}(E_1 - E_2) = E_2$$

can be rewritten as

$$E_+ \simeq E_1 + \frac{|W_{21}|^2}{2\Delta}$$
$$E_- \simeq E_2 - \frac{|W_{21}|^2}{2\Delta}$$

We can see that the perturbed energies are equal to the unperturbed energies plus or minus a <u>small second order perturbation term</u>.

In the **strong coupling** case

$$|W_{21}| \gg \Delta$$
$$E_{+,-} \simeq E_m \pm |W_{21}|$$

We can see that the <u>perturbation term</u> this time is <u>of the first order</u>, thus much stronger.

Eigenfunctions

We have previously found that the eigenstates are:

$$|\psi_+\rangle = \cos\left(\frac{\theta}{2}\right) e^{-i\frac{\varphi}{2}} |\varphi_1\rangle + \sin\left(\frac{\theta}{2}\right) e^{i\frac{\varphi}{2}} |\varphi_2\rangle$$
$$|\psi_-\rangle = -\sin\left(\frac{\theta}{2}\right) e^{-i\frac{\varphi}{2}} |\varphi_1\rangle + \cos\left(\frac{\theta}{2}\right) e^{i\frac{\varphi}{2}} |\varphi_2\rangle$$

where

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$$\boldsymbol{\tan\theta} = \frac{2|H_{12}|}{H_{11}-H_{22}} = \frac{2|W_{21}|}{E_1-E_2} = \frac{|\boldsymbol{W_{21}}|}{\boldsymbol{\Delta}}$$

In the **weak coupling** case

$$\tan\theta = \frac{|W_{21}|}{\Delta} \ll 1$$

therefore we can approximate

$$\theta \ll 1 \quad \Rightarrow \quad \begin{aligned} \sin\left(\frac{\theta}{2}\right) &\simeq \frac{\theta}{2} \simeq \frac{\tan\theta}{2} = \frac{|W_{21}|}{2\Delta} \\ \cos\left(\frac{\theta}{2}\right) &\simeq 1 \end{aligned}$$

and we get

$$|\psi_+\rangle = e^{-i\frac{\varphi}{2}}|\varphi_1\rangle + \frac{|W_{21}|}{2\Delta}e^{i\frac{\varphi}{2}}|\varphi_2\rangle = e^{-i\frac{\varphi}{2}}\left[|\varphi_1\rangle + \frac{|W_{21}|}{2\Delta}e^{i\varphi}|\varphi_2\rangle\right]$$

$$|\psi_-\rangle = -\frac{|W_{21}|}{2\Delta}e^{-i\frac{\varphi}{2}}|\varphi_1\rangle + e^{i\frac{\varphi}{2}}|\varphi_2\rangle = e^{i\frac{\varphi}{2}}\left[-\frac{|W_{21}|}{2\Delta}e^{-i\varphi}|\varphi_1\rangle + |\varphi_2\rangle\right]$$

As with the energies, in the weak coupling case the eigenstates are only slightly changed with respect to their unperturbed state: there is a small cross-contamination.

In the **strong coupling** case

$$\tan\theta = \frac{|W_{21}|}{\Delta} \gg 1$$

therefore we can approximate

$$\theta \simeq \frac{\pi}{2} \quad \Rightarrow \quad \sin\left(\frac{\theta}{2}\right) = \cos\left(\frac{\theta}{2}\right) = \frac{1}{\sqrt{2}}$$

and we get

$$|\psi_+\rangle = \frac{1}{\sqrt{2}}\left[e^{-i\frac{\varphi}{2}}|\varphi_1\rangle + e^{i\frac{\varphi}{2}}|\varphi_2\rangle\right]$$

$$|\psi_-\rangle = \frac{1}{\sqrt{2}}\left[-e^{-i\frac{\varphi}{2}}|\varphi_1\rangle + e^{i\frac{\varphi}{2}}|\varphi_2\rangle\right]$$

In this case the interaction is very strong and the eigenfunctions don't correspond to an unperturbed state each, as they did before.

Dynamics of the two-state system with a time-independent perturbation

$$\hat{H} = \hat{H}_0 + \hat{W} \quad \leftrightarrow \quad [H] = [H_0] + [W] = \begin{bmatrix} E_1 & W_{12} \\ W_{21} & E_2 \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix}$$

$$\hat{H}|\psi_+\rangle = E_+|\psi_+\rangle$$
$$\hat{H}|\psi_-\rangle = E_-|\psi_-\rangle$$

Given a general state on the <u>bare states basis</u>

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$$|\psi(t)\rangle = a_1(t)|\varphi_1\rangle + a_2(t)|\varphi_2\rangle$$

given the equation that describes the time evolution of $|\psi\rangle$

$$i\hbar\frac{\partial}{\partial t}|\psi\rangle = \widehat{H}|\psi\rangle$$

and multiplying by $\langle\varphi_1|$ and $\langle\varphi_2|$ we get

$$\begin{matrix} i\hbar\frac{\partial}{\partial t}\langle\varphi_1|\psi\rangle = \langle\varphi_1|\widehat{H}|\psi\rangle \\ i\hbar\frac{\partial}{\partial t}\langle\varphi_2|\psi\rangle = \langle\varphi_2|\widehat{H}|\psi\rangle \end{matrix} \quad \rightarrow \quad \begin{matrix} i\hbar\dot{a}_1(t) = a_1H_{11} + a_2H_{12} \\ i\hbar\dot{a}_2(t) = a_1H_{21} + a_2H_{22} \end{matrix}$$

$$\begin{cases} \dot{a}_1(t) = -\frac{i}{\hbar}(a_1H_{11} + a_2H_{12}) \\ \dot{a}_2(t) = -\frac{i}{\hbar}(a_1H_{21} + a_2H_{22}) \end{cases}$$

This system of first order differential equations can be written in matrix form as

$$\begin{bmatrix} \dot{\boldsymbol{a}}_{\boldsymbol{1}}(\boldsymbol{t}) \\ \dot{\boldsymbol{a}}_{\boldsymbol{2}}(\boldsymbol{t}) \end{bmatrix} = -\frac{\boldsymbol{i}}{\hbar}[\boldsymbol{H}]\begin{bmatrix} \boldsymbol{a}_{\boldsymbol{1}}(\boldsymbol{t}) \\ \boldsymbol{a}_{\boldsymbol{2}}(\boldsymbol{t}) \end{bmatrix}$$

From this system we can find the solution for the general state at any time:

$$|\boldsymbol{\psi}(\boldsymbol{t})\rangle = \boldsymbol{A}|\boldsymbol{\psi}_+\rangle\boldsymbol{e}^{\boldsymbol{\lambda}_+\boldsymbol{t}} + \boldsymbol{B}|\boldsymbol{\psi}_-\rangle\boldsymbol{e}^{\boldsymbol{\lambda}_-\boldsymbol{t}}$$

where $\lambda_\pm$ are the eigenvalues of the matrix $-\frac{i}{\hbar}[H]$ and A and B are the coefficients at $t = 0$.

Recalling that

$$|\psi_+\rangle = a|\varphi_1\rangle + b|\varphi_2\rangle = \cos\left(\frac{\theta}{2}\right)e^{-i\frac{\varphi}{2}}|\varphi_1\rangle + \sin\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}|\varphi_2\rangle$$

$$|\psi_-\rangle = c|\varphi_1\rangle + d|\varphi_2\rangle = -\sin\left(\frac{\theta}{2}\right)e^{-i\frac{\varphi}{2}}|\varphi_1\rangle + \cos\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}|\varphi_2\rangle$$

we can write

$$\begin{bmatrix} a_1(t) \\ a_2(t) \end{bmatrix} = A\begin{bmatrix} a \\ b \end{bmatrix}e^{\lambda_+t} + B\begin{bmatrix} c \\ d \end{bmatrix}e^{\lambda_-t}$$

Writing $\lambda_\pm$ as a function of $E_\pm$

$$\lambda_\pm = -\frac{i}{\hbar}E_\pm = -\frac{i}{\hbar}\left(E_m \pm \sqrt{\Delta^2 + |W_{12}|^2}\right)$$

the general state is

$$|\psi(t)\rangle = A|\psi_+\rangle e^{-i\frac{E_+}{\hbar}t} + B|\psi_-\rangle e^{-i\frac{E_-}{\hbar}t} = A\begin{bmatrix} a \\ b \end{bmatrix}e^{-i\frac{E_+}{\hbar}t} + B\begin{bmatrix} c \\ d \end{bmatrix}e^{-i\frac{E_-}{\hbar}t}$$

which can be written more explicitly as

$$\begin{cases} a_1(t) = Aae^{-i\frac{E_+}{\hbar}t} + Bce^{-i\frac{E_-}{\hbar}t} = A\cos\left(\frac{\theta}{2}\right)e^{-i\frac{\varphi}{2}}e^{-i\frac{E_+}{\hbar}t} - B\sin\left(\frac{\theta}{2}\right)e^{-i\frac{\varphi}{2}}e^{-i\frac{E_-}{\hbar}t} \\ a_2(t) = Abe^{-i\frac{E_+}{\hbar}t} + Bde^{-i\frac{E_-}{\hbar}t} = A\sin\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}e^{-i\frac{E_+}{\hbar}t} + B\cos\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}e^{-i\frac{E_-}{\hbar}t} \end{cases}$$

To find specific solutions for the problem, we need to impose specific conditions.

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Let us consider the <u>initial condition</u> that at $t = 0$ the system is in the state $|\varphi_1\rangle$.

$$|\varphi_1\rangle = |\psi(0)\rangle = A|\psi_+\rangle + B|\psi_-\rangle$$

Inverting the relations for $|\psi_\pm\rangle$ with respect to $|\varphi_{1,2}\rangle$ we get

$$|\varphi_1\rangle = d|\psi_+\rangle - b|\psi_-\rangle = \cos\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}|\psi_+\rangle - \sin\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}|\psi_-\rangle$$

The solution is then

$$\begin{cases} A = \cos\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}} \\ B = -\sin\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}} \end{cases}$$

The state for the given initial condition is then

$$|\psi(t)\rangle = \cos\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}|\psi_+\rangle e^{-i\frac{E_+}{\hbar}t} - \sin\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}|\psi_-\rangle e^{-i\frac{E_-}{\hbar}t}$$

which gives a transition probability

$$\begin{aligned} P_{12}(t) = |\langle\varphi_2|\psi\rangle|^2 = |a_2(t)|^2 &= \left|A\sin\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}e^{-i\frac{E_+}{\hbar}t} + B\cos\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}e^{-i\frac{E_-}{\hbar}t}\right|^2 \\ &= \left|\cos\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}\sin\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}e^{-i\frac{E_+}{\hbar}t} - \sin\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}\cos\left(\frac{\theta}{2}\right)e^{i\frac{\varphi}{2}}e^{-i\frac{E_-}{\hbar}t}\right|^2 \\ &= \left[2\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\theta}{2}\right)\right]^2\left|\frac{e^{-i\frac{E_+}{\hbar}t} - e^{-i\frac{E_-}{\hbar}t}}{2}\right|^2 = \sin^2\theta\left|\frac{1 - e^{i\frac{E_+ - E_-}{\hbar}t}}{2}\right|^2 \\ &= \sin^2\theta\sin^2\left(\frac{E_+ - E_-}{2\hbar}t\right) \end{aligned}$$

We have therefore an oscillation with a frequency

$$\omega = \frac{E_+ - E_-}{\hbar}$$

Noticing that

$$\tan\theta = \frac{2|W_{21}|}{E_1 - E_2} = \frac{|W_{21}|}{\Delta}$$

$$\frac{\sin^2\theta}{\cos^2\theta} = \tan^2\theta = \frac{|W_{21}|^2}{\Delta^2}$$

$$\Delta^2\sin^2\theta = |W_{21}|^2\cos^2\theta = |W_{21}|^2(1 - \sin^2\theta)$$

$$\sin^2\theta\,(\Delta^2 + |W_{21}|^2) = |W_{21}|^2$$

$$\sin^2\theta = \frac{|W_{21}|^2}{\Delta^2 + |W_{21}|^2}$$

and

$$E_+ - E_- = \left(E_m + \sqrt{\Delta^2 + |W_{21}|^2}\right) - \left(E_m - \sqrt{\Delta^2 + |W_{21}|^2}\right) = 2\sqrt{\Delta^2 + |W_{21}|^2}$$

we can rewrite the transition probability as

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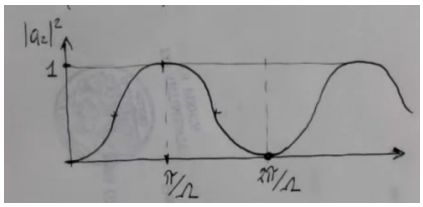
$$\boldsymbol{P_{12}(t)} = \sin^2\theta \sin^2\left(\frac{E_+ - E_-}{2\hbar}t\right) = \frac{|\boldsymbol{W_{21}}|^2}{\boldsymbol{\Delta^2} + |\boldsymbol{W_{21}}|^2}\mathbf{sin^2}\left(\frac{\mathbf{2}\sqrt{\boldsymbol{\Delta^2} + |\boldsymbol{W_{21}}|^2}}{\mathbf{2}\hbar}\boldsymbol{t}\right)$$
$$= \frac{|\boldsymbol{W_{21}}|^2}{\boldsymbol{\Delta^2} + |\boldsymbol{W_{21}}|^2}\mathbf{sin^2}\left(\frac{\boldsymbol{\Omega}}{\mathbf{2}}\boldsymbol{t}\right)$$

which describes the so called **Rabi oscillations** and where we have defined the Rabi frequency

$$\boldsymbol{\Omega} = \frac{\boldsymbol{E_+} - \boldsymbol{E_-}}{\hbar} = \mathbf{2}\frac{\sqrt{\boldsymbol{\Delta^2} + |\boldsymbol{W_{21}}|^2}}{\hbar}$$

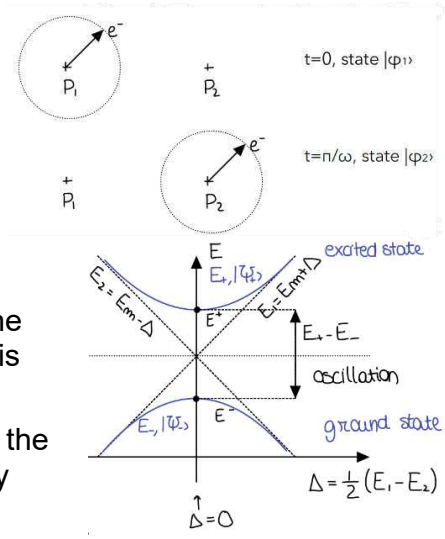
which is the resonance frequency of the system.

Case study: Hydrogen



In the case of the H_2^+ molecule $\Delta = 0$, thus

$$P_{12} = \sin^2\left(\frac{\Omega}{2}t\right) \quad \Omega = \frac{2|W_{21}|}{\hbar}$$



Consider the dynamic of the electron: initially it is in the ground state of proton 1, while after $t = \frac{\pi}{\Omega}$ it is in the ground state of proton 2. Therefore, the centre of negative charge oscillates between the positions of the two protons with Rabi frequency

$$\Omega = \frac{E_+ - E_-}{\hbar}$$

which is equal to the characteristic frequency of the system of the H_2^+ molecule. Thus, we must find this frequency in the electromagnetic spectrum.

It is worth noticing that when the system is in $|\varphi_2\rangle$ the energy is not E_2 since the only measurable energy levels for the perturbed system are $E_\pm$.

Time-dependent perturbation theory: generalised Rabi oscillations and light shift

Consider a system perturbed by the electric dipole interaction with an <u>electromagnetic wave</u>:

$$\widehat{W} = -\hat{\vec{\mu}}_e \cdot \vec{E}$$
$$\hat{\vec{\mu}}_e = -e\hat{\vec{r}}$$
$$\vec{E} = \vec{E}(\vec{r}, t) = \vec{\varepsilon}E_0\cos\left(\omega t - \vec{k}\cdot\vec{r}\right)$$

In electric dipole approximation (or <u>long wavelength approximation</u>) the spatial variation of the position of the electron is much smaller than the wavelength of the

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radiation, therefore we can consider $\vec{r}$ constant and make it coincide with the origin of our coordinates.

$$\vec{E}(\vec{r},t) \simeq \vec{E}(0,t) = \vec{\varepsilon}E_0 \cos(\omega t)$$

$$\widehat{W} = -\hat{\vec{\mu}}_e \cdot \vec{E} \simeq e\hat{\vec{r}} \cdot \vec{\varepsilon}E_0 \cos(\omega t)$$

$$W_{ij} = \langle \varphi_i | \widehat{W} | \varphi_j \rangle = e \langle \varphi_i | \hat{\vec{r}} | \varphi_j \rangle \cdot \vec{\varepsilon}E_0 \cos(\omega t) = e\vec{r}_{ij} \cdot \vec{\varepsilon}E_0 \cos(\omega t)$$

where we have introduced the elements $\vec{r}_{ij}$ of the matrix

$$[\vec{r}] = \begin{bmatrix} 0 & \vec{r}_{12} \\ \vec{r}_{21} & 0 \end{bmatrix} \quad \Rightarrow \quad [W] = \begin{bmatrix} 0 & W_{12} \\ W_{21} & 0 \end{bmatrix} = \begin{bmatrix} 0 & e\vec{r}_{12} \cdot \vec{\varepsilon}E_0 \cos(\omega t) \\ e\vec{r}_{21} \cdot \vec{\varepsilon}E_0 \cos(\omega t) & 0 \end{bmatrix}$$

To find the dynamic of the system we must solve

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \widehat{H} |\psi\rangle$$

and we use the ansatz

$$|\boldsymbol{\psi(t)}\rangle = \boldsymbol{a_1(t)} e^{-i\frac{E_1}{\hbar}t} |\boldsymbol{\varphi_1}\rangle + \boldsymbol{a_2(t)} e^{-i\frac{E_2}{\hbar}t} |\boldsymbol{\varphi_2}\rangle$$

Multiplying by $\langle \varphi_1 |$ and $\langle \varphi_2 |$ we get

$$i\hbar \frac{\partial}{\partial t} \langle \varphi_1 | \psi \rangle = \langle \varphi_1 | \widehat{H} | \psi \rangle = \langle \varphi_1 | \widehat{H}_0 | \psi \rangle + \langle \varphi_1 | \widehat{W} | \psi \rangle$$

$$i\hbar \frac{\partial}{\partial t} \langle \varphi_2 | \psi \rangle = \langle \varphi_2 | \widehat{H} | \psi \rangle = \langle \varphi_2 | \widehat{H}_0 | \psi \rangle + \langle \varphi_2 | \widehat{W} | \psi \rangle$$

The first equation can be written as

$$i\hbar \frac{\partial}{\partial t} \left(a_1 e^{-i\frac{E_1}{\hbar}t} \right) = a_1 e^{-i\frac{E_1}{\hbar}t} E_1 + a_2 e^{-i\frac{E_2}{\hbar}t} W_{12} = a_1 E_1 e^{-i\frac{E_1}{\hbar}t} + a_2 e\vec{r}_{12} \cdot \vec{\varepsilon}E_0 \cos(\omega t)\, e^{-i\frac{E_2}{\hbar}t}$$

Let us define the <u>matrix element of the Rabi frequency</u>

$$\boldsymbol{\Omega_{12}} = -\frac{\boldsymbol{\vec{\mu}_{e,12} \cdot \vec{E}_0}}{\hbar} = \frac{\boldsymbol{e\vec{r}_{12} \cdot \vec{\varepsilon}E_0}}{\hbar}$$

Since $\vec{r}_{12} = \vec{r}_{21}$, it holds $\Omega_{12} = \Omega_{21} = \Omega$. We get:

$$i\hbar \frac{\partial}{\partial t} \left(a_1 e^{-i\frac{E_1}{\hbar}t} \right) = a_1 E_1 e^{-i\frac{E_1}{\hbar}t} + a_2 \hbar \Omega_{12} \cos(\omega t)\, e^{-i\frac{E_2}{\hbar}t}$$

$$i\hbar \dot{a}_1 e^{-i\frac{E_1}{\hbar}t} + i\hbar a_1 \left(-i\frac{E_1}{\hbar} \right) e^{-i\frac{E_1}{\hbar}t} = a_1 E_1 e^{-i\frac{E_1}{\hbar}t} + a_2 \hbar \Omega_{12} \cos(\omega t)\, e^{-i\frac{E_2}{\hbar}t}$$

$$i\hbar \dot{a}_1 e^{-i\frac{E_1}{\hbar}t} = a_2 \hbar \Omega_{12} \cos(\omega t)\, e^{-i\frac{E_2}{\hbar}t}$$

Considering also the analogous result for the second equation we get

$$\begin{cases} i\hbar \dot{a}_1 = a_2 \hbar \Omega_{12} \cos(\omega t)\, e^{-i\frac{E_2 - E_1}{\hbar}t} \\ i\hbar \dot{a}_2 = a_1 \hbar \Omega_{21} \cos(\omega t)\, e^{i\frac{E_2 - E_1}{\hbar}t} \end{cases}$$

Defining the resonance frequency of the system

$$\omega_0 = \frac{E_2 - E_1}{\hbar}$$

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we have

$$\dot{a}_1 = -\frac{i}{\hbar} a_2 \hbar \Omega_{12} \cos(\omega t)\, e^{-i\omega_0 t} = -\frac{i}{\hbar} a_2 \hbar \Omega_{12} \left(\frac{e^{i\omega t} + e^{-i\omega t}}{2} \right) e^{-i\omega_0 t}$$

$$\dot{a}_2 = -\frac{i}{\hbar} a_1 \hbar \Omega_{21} \cos(\omega t)\, e^{i\omega_0 t} = -\frac{i}{\hbar} a_1 \hbar \Omega_{21} \left(\frac{e^{i\omega t} + e^{-i\omega t}}{2} \right) e^{i\omega_0 t}$$

We can now implement the **rotating wave approximation** (RWA), which consists in neglecting the terms rotating at high frequency, which average out.

Defining $\Delta\omega = \omega - \omega_0$ we get

$$\begin{cases} \dot{a}_1 = -\frac{i}{\hbar} a_2 \frac{\hbar \Omega_{12}}{2} e^{i\Delta\omega t} \\ \dot{a}_2 = -\frac{i}{\hbar} a_1 \frac{\hbar \Omega_{21}}{2} e^{-i\Delta\omega t} \end{cases}$$

Implementing now the **rotating-frame transformation** (RFT) we change the time reference introducing new coefficients $b_1(t)$ and $b_2(t)$ defined as

$$\begin{cases} \boldsymbol{a_1(t) = b_1(t) e^{i\frac{\Delta\omega}{2} t}} \\ \boldsymbol{a_2(t) = b_2(t) e^{-i\frac{\Delta\omega}{2} t}} \end{cases}$$

By substituting in the first equation we get

$$\dot{b}_1 e^{i\frac{\Delta\omega}{2} t} + b_1 \left(i \frac{\Delta\omega}{2} \right) e^{i\frac{\Delta\omega}{2} t} = -\frac{i}{\hbar} b_2 e^{-i\frac{\Delta\omega}{2} t} \frac{\hbar \Omega_{12}}{2} e^{i\Delta\omega t}$$

$$\Rightarrow \quad \dot{b}_1 + \frac{i}{\hbar} \left(\frac{\hbar \Delta\omega}{2} \right) b_1 = -\frac{i}{\hbar} \left(\frac{\hbar \Omega_{12}}{2} \right) b_2$$

Doing the same for the second equation we get the system

$$\begin{cases} \dot{b}_1 = -\frac{i}{\hbar} \left(\frac{\hbar \Delta\omega}{2} \right) b_1 - \frac{i}{\hbar} \left(\frac{\hbar \Omega}{2} \right) b_2 \\ \dot{b}_2 = -\frac{i}{\hbar} \left(\frac{\hbar \Omega}{2} \right) b_1 + \frac{i}{\hbar} \left(\frac{\hbar \Delta\omega}{2} \right) b_2 \end{cases}$$

This system can be expressed in matrix form as

$$\begin{bmatrix} \dot{b}_1 \\ \dot{b}_2 \end{bmatrix} = -\frac{i}{\hbar} [\widetilde{H}] \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

where

$$\boldsymbol{[\widetilde{H}] = \begin{bmatrix} \frac{\hbar \Delta\omega}{2} & \frac{\hbar \Omega}{2} \\ \frac{\hbar \Omega}{2} & -\frac{\hbar \Delta\omega}{2} \end{bmatrix}}$$

is the **effective Hamiltonian matrix**.

This system is formally similar to the one obtained for the time-independent perturbation problem: the rotating frame transformation allows us to change the time frame so that the problem appears similar to the time-independent problem if we substitute the Hamiltonian matrix with an effective one.

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Time independent	Time dependent
$$\begin{bmatrix}\dot{a}_1\\ \dot{a}_2\end{bmatrix} = -\frac{i}{\hbar}[H]\begin{bmatrix}a_1\\ a_2\end{bmatrix}$$ $$[H] = \begin{bmatrix}E_1 & W_{12}\\ W_{21} & E_2\end{bmatrix} = \begin{bmatrix}E_1 & 0\\ 0 & E_2\end{bmatrix} + \begin{bmatrix}0 & W_{12}\\ W_{21} & 0\end{bmatrix} = [H_0] + [W]$$	$$\begin{bmatrix}\dot{b}_1\\ \dot{b}_2\end{bmatrix} = -\frac{i}{\hbar}[\widetilde{H}]\begin{bmatrix}b_1\\ b_2\end{bmatrix}$$ $$[\widetilde{H}] = \begin{bmatrix}\frac{\hbar\Delta\omega}{2} & \frac{\hbar\Omega}{2}\\ \frac{\hbar\Omega}{2} & -\frac{\hbar\Delta\omega}{2}\end{bmatrix} = \begin{bmatrix}\frac{\hbar\Delta\omega}{2} & 0\\ 0 & -\frac{\hbar\Delta\omega}{2}\end{bmatrix} + \begin{bmatrix}0 & \frac{\hbar\Omega}{2}\\ \frac{\hbar\Omega}{2} & 0\end{bmatrix} = [\widetilde{H}_0] + [\widetilde{W}]$$
$$\lvert\psi(t)\rangle = a_1\lvert\varphi_1\rangle + a_2\lvert\varphi_2\rangle$$ where $\lvert\varphi_1\rangle$ and $\lvert\varphi_2\rangle$ are the eigenstates of $\widehat{H}_0$.	$$\lvert\tilde{\psi}(t)\rangle = b_1\lvert\tilde{\varphi}_1\rangle + b_2\lvert\tilde{\varphi}_2\rangle$$ where $\lvert\tilde{\varphi}_1\rangle$ and $\lvert\tilde{\varphi}_2\rangle$ are the eigenstates of $\widehat{\widetilde{H}}_0$.
$$E_m = \frac{1}{2}(E_1 + E_2)$$ $$\Delta = \frac{1}{2}(E_1 - E_2)$$ $$E_+ = E_m + \sqrt{\Delta^2 + \lvert W_{12}\rvert^2}$$ $$E_- = E_m - \sqrt{\Delta^2 + \lvert W_{12}\rvert^2}$$	$$\tilde{E}_m = \frac{1}{2}\left(\frac{\hbar\Delta\omega}{2} - \frac{\hbar\Delta\omega}{2}\right) = 0$$ $$\widetilde{\Delta} = \frac{1}{2}\left(\frac{\hbar\Delta\omega}{2} + \frac{\hbar\Delta\omega}{2}\right) = \frac{\hbar\Delta\omega}{2}$$ $$\tilde{E}_+ = \tilde{E}_m + \sqrt{\widetilde{\Delta}^2 + \lvert\widetilde{W}_{12}\rvert^2} = \sqrt{\left(\frac{\hbar\Delta\omega}{2}\right)^2 + \left(\frac{\hbar\Omega}{2}\right)^2}$$ $$\tilde{E}_- = \tilde{E}_m - \sqrt{\widetilde{\Delta}^2 + \lvert\widetilde{W}_{12}\rvert^2} = -\sqrt{\left(\frac{\hbar\Delta\omega}{2}\right)^2 + \left(\frac{\hbar\Omega}{2}\right)^2}$$
$$P_{12}(t) = \lvert\langle\varphi_2\lvert\psi\rangle\rvert^2 = \lvert a_2(t)\rvert^2 = \frac{\lvert W_{21}\rvert^2}{\Delta^2 + \lvert W_{21}\rvert^2}\sin^2\left(\frac{\sqrt{\Delta^2 + \lvert W_{21}\rvert^2}}{\hbar}t\right) = \frac{\lvert W_{21}\rvert^2}{\Delta^2 + \lvert W_{21}\rvert^2}\sin^2\left(\frac{\Omega}{2}t\right)$$	$$\tilde{P}_{12}(t) = \left\lvert\langle\tilde{\varphi}_2\lvert\tilde{\psi}\rangle\right\rvert^2 = \lvert b_2(t)\rvert^2 = \frac{\lvert\widetilde{W}_{21}\rvert^2}{\widetilde{\Delta}^2 + \lvert\widetilde{W}_{21}\rvert^2}\sin^2\left(\frac{1}{\hbar}\sqrt{\widetilde{\Delta}^2 + \lvert\widetilde{W}_{21}\rvert^2}t\right) = \frac{\Omega^2}{\Delta\omega^2 + \Omega^2}\sin^2\left(\frac{1}{\hbar}\sqrt{\left(\frac{\hbar\Delta\omega}{2}\right)^2 + \left(\frac{\hbar\Omega}{2}\right)^2}t\right) = \frac{\Omega^2}{\Delta\omega^2 + \Omega^2}\sin^2\left(\frac{\sqrt{\Omega^2 + \Delta\omega^2}}{2}t\right)$$ $$\boldsymbol{\Omega' := \sqrt{\Omega^2 + \Delta\omega^2}}$$ $$\boldsymbol{\widetilde{P}_{12}(t) = \frac{\Omega^2}{\Omega'^2}\sin^2\left(\frac{\Omega'}{2}t\right)}$$

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We have obtained **generalised Rabi oscillation**: the frequency of oscillation has changed because of the detuning which takes into account the effect of the electromagnetic driving field.

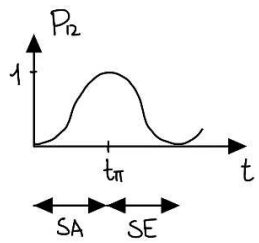
Since $\tilde{P}_{12}(t) = |b_2(t)|^2 = |a_2(t)|^2 = P_{12}(t)$, the probability is the same in the rotated time frame and the original one, thus

$$P_{12}(t) = \tilde{P}_{12}(t) = \frac{\Omega^2}{\Omega'^2}\sin^2\left(\frac{\Omega'}{2}t\right)$$

where we have defined the **generalised Rabi frequency** $\boldsymbol{\Omega'} = \sqrt{\Omega^2 + \Delta\omega^2}$.

At resonance $\Delta\omega = \omega - \omega_0 = 0$, thus

$$P_{12}(t) = \sin^2\left(\frac{\Omega}{2}t\right)$$



At $t_\pi = \frac{\pi}{\Omega}$ we have $P_{12}(t_\pi) = 1$. Thus, in the time interval $[0, t_\pi]$ we have stimulated absorption and in the time interval $[t_\pi, 2t_\pi]$ we have stimulated emission.

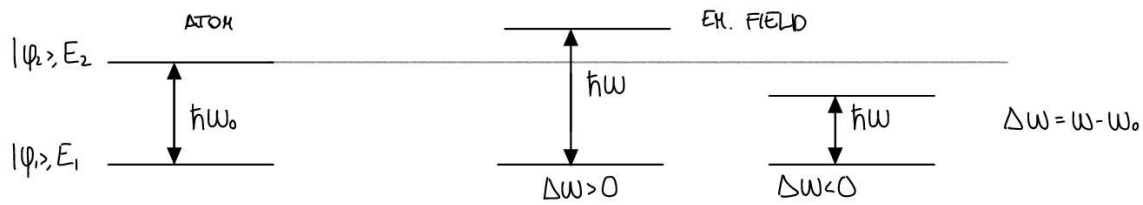
We have worked in the rotating frame reference system, while to find the real energy levels we need to go back to the original time frame.

Notice that we have a shift of the energy levels $\tilde{E}_\pm = \pm\sqrt{\left(\frac{\hbar\Delta\omega}{2}\right)^2 + \left(\frac{\hbar\Omega}{2}\right)^2}$. This shift is proportional to the amplitude of the electromagnetic field through the term $\Omega \propto E_0$. This is a purely quantum mechanical effect called **light shift**.

Since usually $\Omega \ll \Delta\omega$

$$\tilde{E}_\pm = \pm\frac{\hbar\Delta\omega}{2}\sqrt{1 + \left(\frac{\Omega}{\Delta\omega}\right)^2} \simeq \pm\frac{\hbar\Delta\omega}{2}\left[1 + \frac{\Omega^2}{2\Delta\omega^2}\right] = \pm\left(\frac{\hbar\Delta\omega}{2} + \frac{\hbar\Omega^2}{4\Delta\omega}\right)$$

In the physical picture we have the situation in figure



In order to understand the light shift, we start from the description in terms of bare states, which are the eigenstates <u>when the interaction is off</u> ($\Omega = 0$).

$$0 = \Omega = \Omega_{12} = -\frac{\vec{\mu}_{12}\cdot\vec{E}_0}{\hbar} = \frac{e\,\vec{r}_{12}\cdot\vec{\varepsilon}E_0}{\hbar}$$

This is the case, for example, when there is a field but there is no electric dipole, $\vec{\mu}_{12} = 0$. Under this conditions it holds

$$\tilde{E}_\pm = \pm\frac{\hbar\Delta\omega}{2}$$

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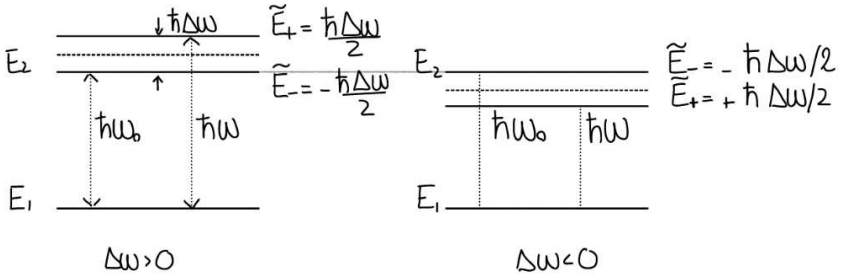
Let us <u>assume that E_1 is the ground state of the overall system</u>, not only of the uncoupled atom. Based on $\Delta\omega$ we have the two configurations in figure



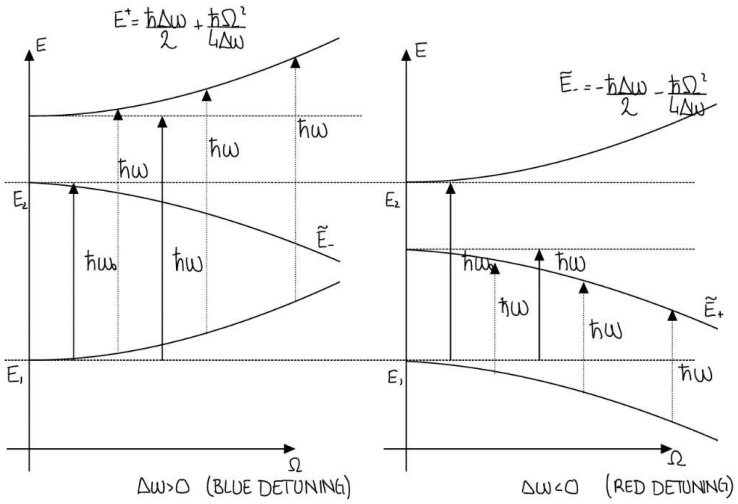
By looking at the picture, which represents the uncoupled system, it is easy to understand how the energies $\tilde{E}_\pm$ in the modified time frame relate to the energies in the original time frame. This way, it is possible to avoid the calculations by extending this relation to the coupled system.

The eigenstates of the <u>perturbed system</u> ($\Delta \neq 0$) are called dressed states.

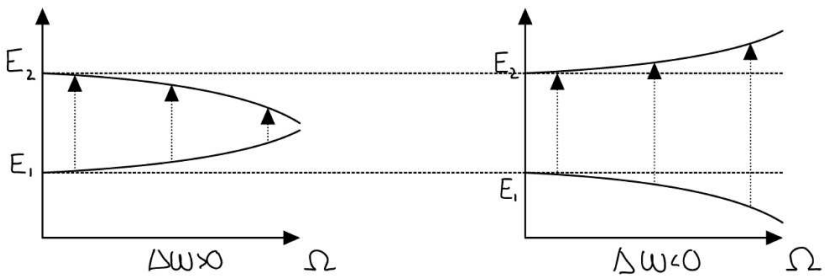
$$\vec{\mu}_{12} \neq 0$$

$$\tilde{E}_\pm = \pm\left(\frac{\hbar\Delta\omega}{2} + \frac{\hbar\Omega^2}{4\Delta\omega}\right)$$

The ground state of the system will also bend in the same way as $\tilde{E}_+$. This is because, while we are increasing the Rabi frequency, the quantum of energy of the electromagnetic field <u>$\hbar\omega$ is still constant</u>, thus the energy difference has to stay fixed.



Let us look closer at the situation for the atom. Based on $\Delta\omega$, the energy levels of the atom get closer or further away for increasing Ω.



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Density operator

The representation based on the density operator is easier to use than the one based on the state vector.

Pure states and mixed states

So far we have considered a system in a pure state, described by the wavefunction $|\psi\rangle$. We have seen that the state $|\psi\rangle$ can be written as a superposition of stationary states $|\varphi_k\rangle$ and quantum mechanics tells us that the squared modulus of the projection of $|\psi\rangle$ on a given stationary state $|\varphi_k\rangle$ gives the probability to find, by a measurement, the system in that particular stationary state. Generally we have to deal with statistical (incoherent) mixtures and we cannot know the exact initial quantum state of a system. For example we may just know the temperature T of the system, so that the state of the system cannot be described by a wavevector $|\psi\rangle$. What it is possible to know, using for example the laws of statistical mechanics, is the probability, p_n, that the system is in the pure state $|\psi_k\rangle$. Note that the probability p_n is completely different from the probability $|c_n|^2$ discussed before in the case of a pure state $|\psi\rangle$. The probability $|c_n|^2$ does not arise from any lack of knowledge, but it is related to the process of quantum mechanical measurement: unless the system is in a particular eigenstate, the measurement places the system in one of its eigenstates with a probability $|c_n|^2$. On the contrary, the probability p_n results from our lack of information about the exact preparation of the system. In this case the quantum state must be described by a weighted mixture (statistical mixture) of states and it is called a **mixed state**.

Let us consider a total number N_t of quantum entities, N_k of which are in state $|\psi_k\rangle$. We can define $p_k = N_k/N_t$ and the **density operator**

$$\boldsymbol{\hat{\rho} = \sum_k p_k |\psi_k\rangle\langle\psi_k|}$$

Consider for example

$$|\psi_1\rangle = |\varphi_1\rangle \quad |\psi_2\rangle = a|\varphi_1\rangle + b|\varphi_2\rangle$$

$$N_1 = 0.3N_t \quad N_2 = 0.7\,N_t$$

$$p_1 = 0.3 \quad p_2 = 0.7$$

$$\hat{\rho} = \sum_{k=1}^{2} p_k|\psi_k\rangle\langle\psi_k| = p_1|\psi_1\rangle\langle\psi_1| + p_2|\psi_2\rangle\langle\psi_2| = 0.3|\psi_1\rangle\langle\psi_1| + 0.7|\psi_2\rangle\langle\psi_2|$$

The $|\psi_k\rangle$ states are pure states or states of maximum knowledge, meaning states where all parameters of the system are well defined.

The general state of the overall system is a mixed state given by the tensor product, the combination, of states of the different subsystems

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \ldots$$

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Spontaneous emission is hard to deal with in the state vector representation. The state function corresponding to the spontaneous emission is

$$|\psi(t)\rangle = \alpha(t)|e,0\rangle + \sum_s \beta_s(t)|g,1_s\rangle$$

where $|e\rangle$ and $|g\rangle$ represents the excited and ground states of the atom, while $|0\rangle$ and $|1_s\rangle$ represent the ground and excited state of the electromagnetic field. Here s represents a specific state of the emitted photon, $s = (\vec{k}, \vec{\varepsilon})$, where $\vec{k}$ is the wave vector of the photon and $\vec{\varepsilon}$ its unit polarisation vector (one of two options for each $\vec{k}$). The sum is over all possible combinations of $\vec{k}$ and $\vec{\varepsilon}$, thus over an infinite number of terms. In this state function representation it is difficult to model spontaneous emission. This is much easier using the representation of the density operator.

Review of properties

Let us consider a system made of many subsystems, each described by a state function $|\psi_k\rangle$. Given a basis of eigenfunctions $\{|\varphi_n\rangle\}$, the spectral decomposition of a state function is

$$|\psi_k\rangle = \hat{\mathbb{I}}|\psi_k\rangle = \sum_n |\varphi_n\rangle\langle\varphi_n|\psi_k\rangle = \sum_n |\varphi_n\rangle c_{n,k}$$

where we have defined the component $c_{n,k}$ of $|\psi_k\rangle$ along $|\varphi_n\rangle$

$$c_{n,k} = \langle\varphi_n|\psi_k\rangle$$

The matrix representation of the density operator, called density matrix, is then

$$\hat{\rho} = \sum_k p_k|\psi_k\rangle\langle\psi_k| \quad \rightarrow \quad [\rho] = \begin{bmatrix} \rho_{11} & \cdots & \rho_{1N} \\ \vdots & \ddots & \vdots \\ \rho_{N1} & \cdots & \rho_{NN} \end{bmatrix}$$

The matrix element is

$$\boldsymbol{\rho_{ij}} = \langle\varphi_i|\hat{\rho}|\varphi_j\rangle = \langle\varphi_i|\textstyle\sum_k p_k|\psi_k\rangle\langle\psi_k|\,|\varphi_j\rangle = \sum_k p_k\langle\varphi_i|\psi_k\rangle\langle\psi_k|\varphi_j\rangle$$

$$= \sum_k p_k\langle\varphi_i|\psi_k\rangle\langle\varphi_j|\psi_k\rangle^* = \sum_{\boldsymbol{k}} \boldsymbol{p_k c_{i,k} c^*_{j,k}}$$

$$\boldsymbol{\rho_{ii}} = \langle\varphi_i|\hat{\rho}|\varphi_i\rangle = \sum_{\boldsymbol{k}} \boldsymbol{p_k}|\boldsymbol{c_{i,k}}|^2$$

Moreover, $\hat{\rho}$ is an Hermitian operator, since

$$\rho_{ji} = \sum_k p_k c_{j,k} c^*_{i,k} = \sum_k p_k\left(c_{i,k}c^*_{j,k}\right)^* = \rho^*_{ij}$$

An advantage of this representation is that it shows similar results for both pure and mixed states, making it easier to switch between the two.

For both pure and mixed states it holds

$$\mathbf{Tr}[\boldsymbol{\rho}] = \mathbf{1}$$

Indeed, since

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$$\rho_{ii} = \langle\varphi_i|\hat{\rho}|\varphi_i\rangle = \langle\varphi_i|(\textstyle\sum_k p_k|\psi_k\rangle\langle\psi_k|)|\varphi_i\rangle = \displaystyle\sum_k p_k\langle\varphi_i|\psi_k\rangle\langle\psi_k|\varphi_i\rangle$$

it holds

$$\begin{aligned}\mathrm{Tr}[\rho] = \sum_i \rho_{ii} &= \sum_i\sum_k p_k\langle\varphi_i|\psi_k\rangle\langle\psi_k|\varphi_i\rangle = \sum_k\sum_i p_k\langle\psi_k|\varphi_i\rangle\langle\varphi_i|\psi_k\rangle \\ &= \sum_k p_k\langle\psi_k|\left(\sum_i|\varphi_i\rangle\langle\varphi_i|\right)|\psi_k\rangle = \sum_k p_k\langle\psi_k|\psi_k\rangle = \sum_k p_k = 1\end{aligned}$$

Given an observable A, it holds

$$\boldsymbol{\langle\hat{A}\rangle = \mathrm{Tr}[\rho A]}$$

To demonstrate this property, first of all we notice that given a probability p_k of finding the system in a state $|\psi_k\rangle$ and given the expectation value of $\hat{A}$ in this state

$$\langle\hat{A}\rangle_k = \langle\psi_k|\hat{A}|\psi_k\rangle$$

the expectation value for a statistical mixture of these states is

$$\langle\hat{A}\rangle = \sum_k p_k\langle\hat{A}\rangle_k = \sum_k p_k\langle\psi_k|\hat{A}|\psi_k\rangle$$

Then we compute

$$\begin{aligned}\mathrm{Tr}[\rho A] = \sum_n\langle\varphi_n|\hat{\rho}\hat{A}|\varphi_n\rangle &= \sum_n\sum_k p_k\langle\varphi_n|\psi_k\rangle\langle\psi_k|\hat{A}|\varphi_n\rangle = \sum_n\sum_k p_k\langle\psi_k|\hat{A}|\varphi_n\rangle\langle\varphi_n|\psi_k\rangle \\ &= \sum_k p_k\langle\psi_k|\hat{A}\left(\sum_n|\varphi_n\rangle\langle\varphi_n|\right)|\psi_k\rangle = \sum_k p_k\langle\psi_k|\hat{A}|\psi_k\rangle = \langle A\rangle\end{aligned}$$

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Optical Bloch equations

We want to model the dynamic of our system based on the <u>time evolution of the density matrix</u>.

Let us consider a <u>two-level system</u> in a <u>pure state</u>:

$$|\psi\rangle = a_1|\varphi_1\rangle + a_2|\varphi_2\rangle$$

$$\hat{\rho} = \sum_k p_k|\psi_k\rangle\langle\psi_k| = |\psi\rangle\langle\psi|$$

$$[\rho] = \begin{bmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{bmatrix} = \begin{bmatrix} \langle\varphi_1|\hat{\rho}|\varphi_1\rangle & \langle\varphi_1|\hat{\rho}|\varphi_2\rangle \\ \langle\varphi_2|\hat{\rho}|\varphi_1\rangle & \langle\varphi_2|\hat{\rho}|\varphi_2\rangle \end{bmatrix}$$

$$\rho_{12} = \langle\varphi_1|\hat{\rho}|\varphi_2\rangle = \langle\varphi_1|\psi\rangle\langle\psi|\varphi_2\rangle = a_1 a_2^*$$

$$\rho_{11} = \langle\varphi_1|\hat{\rho}|\varphi_1\rangle = \langle\varphi_1|\psi\rangle\langle\psi|\varphi_1\rangle = a_1 a_1^* = |a_1|^2$$

$$[\rho] = \begin{bmatrix} |a_1(t)|^2 & a_1 a_2^* \\ a_2 a_1^* & |a_2(t)|^2 \end{bmatrix}$$

The diagonal elements are called <u>populations</u>, while the other ones are called <u>coherences</u>.

Let us consider now a <u>mixed state</u>, given by pure states $|\psi_k\rangle$ with probability p_k.

$$\hat{\rho} = \sum_k p_k|\psi_k\rangle\langle\psi_k|$$

$$[\rho] = \begin{bmatrix} \rho_{11} & \cdots & \rho_{1N} \\ \vdots & \ddots & \vdots \\ \rho_{N1} & \cdots & \rho_{NN} \end{bmatrix}$$

$$\rho_{11} = \langle\varphi_1|\hat{\rho}|\varphi_1\rangle = \langle\varphi_1|(\textstyle\sum_k p_k|\psi_k\rangle\langle\psi_k|)|\varphi_1\rangle = \sum_k p_k\langle\varphi_1|\psi_k\rangle\langle\psi_k|\varphi_1\rangle = \sum_k p_k\left|c_{1,k}(t)\right|^2$$

$$\rho_{12} = \langle\varphi_1|\hat{\rho}|\varphi_2\rangle = \langle\varphi_1|(\textstyle\sum_k p_k|\psi_k\rangle\langle\psi_k|)|\varphi_2\rangle = \sum_k p_k\langle\varphi_1|\psi_k\rangle\langle\psi_k|\varphi_2\rangle$$

$$= \sum_k p_k\, c_{1,k}(t)c_{2,k}^*(t)$$

This is an extension of the result for the pure states: the matrix element is a weighted sum of the matrix elements for the different pure states.

We want to find the matrix

$$\begin{bmatrix} \dot{\rho}_{11}(t) & \dot{\rho}_{12}(t) \\ \dot{\rho}_{21}(t) & \dot{\rho}_{22}(t) \end{bmatrix}$$

which describes the time evolution of the system.

We are going to derive the optical Bloch equations under the following assumptions: <u>two-states system and pure state</u>. <u>We will then generalise the results</u>.

The time evolution of an operator is given by

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$$\boldsymbol{\frac{d}{dt}\hat{\rho} = \frac{i}{\hbar}[\hat{\rho}, \hat{H}]}$$

therefore

$$\dot{\rho}_{mn} = \left\langle \varphi_m \middle| \frac{d}{dt}\hat{\rho} \middle| \varphi_n \right\rangle = \left\langle \varphi_m \middle| \frac{i}{\hbar}[\hat{\rho}, \hat{H}] \middle| \varphi_n \right\rangle = \frac{i}{\hbar}\langle \varphi_m | [\hat{\rho}, \hat{H}_0 + \hat{W}] | \varphi_n \rangle$$
$$= \frac{i}{\hbar}\langle \varphi_m | [\hat{\rho}, \hat{H}_0] | \varphi_n \rangle + \frac{i}{\hbar}\langle \varphi_m | [\hat{\rho}, \hat{W}] | \varphi_n \rangle$$

$$\langle \varphi_m | [\hat{\rho}, \hat{H}_0] | \varphi_n \rangle = \langle \varphi_m | \hat{\rho}\hat{H}_0 | \varphi_n \rangle - \langle \varphi_m | \hat{H}_0\hat{\rho} | \varphi_n \rangle = E_n \langle \varphi_m | \hat{\rho} | \varphi_n \rangle - E_m \langle \varphi_m | \hat{\rho} | \varphi_n \rangle$$
$$= (E_n - E_m)\langle \varphi_m | \hat{\rho} | \varphi_n \rangle = (E_n - E_m)\rho_{mn}$$

$$\langle \varphi_m | [\hat{\rho}, \hat{W}] | \varphi_n \rangle = \langle \varphi_m | \hat{\rho}\hat{\mathbb{I}}\hat{W} | \varphi_n \rangle - \langle \varphi_m | \hat{W}\hat{\mathbb{I}}\hat{\rho} | \varphi_n \rangle$$
$$= \langle \varphi_m | \hat{\rho}(\textstyle\sum_k |\varphi_k\rangle\langle\varphi_k|)\hat{W} | \varphi_n \rangle - \langle \varphi_m | \hat{W}(\textstyle\sum_k |\varphi_k\rangle\langle\varphi_k|)\hat{\rho} | \varphi_n \rangle$$
$$= \sum_k \langle \varphi_m | \hat{\rho} | \varphi_k \rangle \langle \varphi_k | \hat{W} | \varphi_n \rangle - \sum_k \langle \varphi_m | \hat{W} | \varphi_k \rangle \langle \varphi_k | \hat{\rho} | \varphi_n \rangle$$
$$= \sum_k (\rho_{mk} W_{kn} - W_{mk}\rho_{kn})$$

$$\dot{\rho}_{mn} = \frac{i}{\hbar}(E_n - E_m)\rho_{mn} + \frac{i}{\hbar}\sum_k (\rho_{mk} W_{kn} - W_{mk}\rho_{kn})$$

$$\dot{\rho}_{11} = \frac{i}{\hbar}\sum_{k=1}^{2}(\rho_{1k} W_{k1} - W_{1k}\rho_{k1}) = \frac{i}{\hbar}(\rho_{11} W_{11} - W_{11}\rho_{11} + \rho_{12} W_{21} - W_{12}\rho_{21})$$

$$\dot{\rho}_{12} = \frac{i}{\hbar}(E_2 - E_1)\rho_{12} + \frac{i}{\hbar}(\rho_{11} W_{12} - W_{11}\rho_{12} + \rho_{12} W_{22} - W_{12}\rho_{22})$$

Considering the matrix for $\hat{W}$

$$\hat{W} = -\hat{\vec{\mu}}_e \cdot \vec{E} \quad \rightarrow \quad [W] = \begin{bmatrix} 0 & W_{12} \\ W_{21} & 0 \end{bmatrix} = \begin{bmatrix} 0 & W \\ W & 0 \end{bmatrix}$$

where we have supposed that W_{12} is real, so that $W_{21} = W_{12}^* = W_{12} = W$.
Simplifying the null matrix elements $W_{11} = W_{22} = 0$ we get

$$\dot{\rho}_{11} = \frac{i}{\hbar}(\rho_{12} W_{21} - W_{12}\rho_{21}) = \frac{i}{\hbar} W(\rho_{12} - \rho_{21})$$

$$\dot{\rho}_{12} = \frac{i}{\hbar}(E_2 - E_1)\rho_{12} + \frac{i}{\hbar} W(\rho_{11} - \rho_{22})$$

To find $\dot{\rho}_{22}$ we can exploit the relationship

$$\rho_{11}(t) + \rho_{22}(t) = \mathrm{Tr}(\rho) = 1$$
$$\dot{\rho}_{11}(t) + \dot{\rho}_{22}(t) = 0$$
$$\dot{\rho}_{22} = -\dot{\rho}_{11} = -\frac{i}{\hbar} W(\rho_{12} - \rho_{21})$$

The last remaining term is

$$\dot{\rho}_{21} = \dot{\rho}_{12}^* = -\frac{i}{\hbar}(E_2 - E_1)\rho_{12}^* - \frac{i}{\hbar} W(\rho_{11}^* - \rho_{22}^*) = -\frac{i}{\hbar}(E_2 - E_1)\rho_{21} - \frac{i}{\hbar} W(\rho_{11} - \rho_{22})$$

where we have used the relation $\rho_{12}^* = \rho_{21}$ and the fact that ρ_{11} and ρ_{22} are real.

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In conclusion

$$\begin{cases} \dot{\boldsymbol{\rho}}_{\mathbf{11}}(\boldsymbol{t}) = \frac{\boldsymbol{i}}{\hbar}\boldsymbol{W}(\boldsymbol{\rho}_{\mathbf{12}} - \boldsymbol{\rho}_{\mathbf{21}}) \\ \dot{\boldsymbol{\rho}}_{\mathbf{12}}(\boldsymbol{t}) = \frac{\boldsymbol{i}}{\hbar}(\boldsymbol{E}_{\mathbf{2}} - \boldsymbol{E}_{\mathbf{1}})\boldsymbol{\rho}_{\mathbf{12}} + \frac{\boldsymbol{i}}{\hbar}\boldsymbol{W}(\boldsymbol{\rho}_{\mathbf{11}} - \boldsymbol{\rho}_{\mathbf{22}}) \\ \dot{\rho}_{21}(t) = -\frac{i}{\hbar}(E_2 - E_1)\rho_{21} - \frac{i}{\hbar}W(\rho_{11} - \rho_{22}) \\ \dot{\rho}_{22}(t) = -\frac{i}{\hbar}W(\rho_{12} - \rho_{21}) \end{cases}$$

Since $\widehat{\Omega} = -\frac{\hat{\vec{\mu}}_e \cdot \vec{E}_0}{\hbar}$ we can write $\widehat{W} = -\hat{\vec{\mu}}_e \cdot \vec{E} = -\hat{\vec{\mu}}_e \cdot \vec{E}_0 \cos(\omega t) = \hbar\widehat{\Omega}\cos(\omega t)$, thus

$$W = W_{12} = \langle \varphi_1 | \widehat{W} | \varphi_2 \rangle = \hbar \langle \varphi_1 | \widehat{\Omega} | \varphi_2 \rangle \cos(\omega t) = \hbar\Omega_{12}\cos(\omega t) = \hbar\Omega\cos(\omega t)$$

$$W = \hbar\Omega\left(\frac{e^{i\omega t} + e^{-i\omega t}}{2}\right)$$

By applying a <u>rotating frame transformation</u>

$$\begin{matrix} \tilde{\rho}_{11} = \rho_{11} \\ \tilde{\rho}_{12} = \rho_{12}e^{-i\omega t} \\ \tilde{\rho}_{21} = \tilde{\rho}_{12}^* = \rho_{12}^* e^{i\omega t} = \rho_{21}e^{i\omega t} \\ \tilde{\rho}_{22} = \rho_{22} \end{matrix} \quad \leftrightarrow \quad \begin{matrix} \boldsymbol{\rho}_{\mathbf{11}} = \widetilde{\boldsymbol{\rho}}_{\mathbf{11}} \\ \boldsymbol{\rho}_{\mathbf{12}} = \widetilde{\boldsymbol{\rho}}_{\mathbf{12}}\boldsymbol{e}^{\boldsymbol{i\omega t}} \\ \boldsymbol{\rho}_{\mathbf{21}} = \widetilde{\boldsymbol{\rho}}_{\mathbf{21}}\boldsymbol{e}^{-\boldsymbol{i\omega t}} \\ \boldsymbol{\rho}_{\mathbf{22}} = \widetilde{\boldsymbol{\rho}}_{\mathbf{22}} \end{matrix}$$

We are going to work exclusively on $\tilde{\rho}_{11}$ and $\tilde{\rho}_{12}$ since the other two terms can be found from these ones.

$$\dot{\tilde{\rho}}_{11}(t) = \frac{i}{\hbar}\hbar\Omega\left(\frac{e^{i\omega t} + e^{-i\omega t}}{2}\right)\left(\tilde{\rho}_{12}e^{i\omega t} - \tilde{\rho}_{21}e^{-i\omega t}\right)$$

$$\dot{\tilde{\rho}}_{12}e^{i\omega t} + i\omega\tilde{\rho}_{12}e^{i\omega t} = \frac{i}{\hbar}(E_2 - E_1)\tilde{\rho}_{12}e^{i\omega t} + \frac{i}{\hbar}\hbar\Omega\left(\frac{e^{i\omega t} + e^{-i\omega t}}{2}\right)(\tilde{\rho}_{11} - \tilde{\rho}_{22})$$

$$\dot{\tilde{\rho}}_{12} + i\omega\tilde{\rho}_{12} = i\omega_0\tilde{\rho}_{12} + i\Omega\left(\frac{1 + e^{-i2\omega t}}{2}\right)(\tilde{\rho}_{11} - \tilde{\rho}_{22})$$

where ω_0 is the resonance frequency of the system.

Introducing the <u>rotating wave approximation</u> we neglect the terms oscillating at high frequency and get:

$$\dot{\tilde{\rho}}_{11}(t) = i\frac{\Omega}{2}(\tilde{\rho}_{12} - \tilde{\rho}_{21})$$

$$\dot{\tilde{\rho}}_{12} + i\omega\tilde{\rho}_{12} = i\omega_0\tilde{\rho}_{12} + i\frac{\Omega}{2}(\tilde{\rho}_{11} - \tilde{\rho}_{22})$$

$$\dot{\tilde{\rho}}_{12} = -i\Delta\omega\tilde{\rho}_{12} + i\frac{\Omega}{2}(\tilde{\rho}_{11} - \tilde{\rho}_{22})$$

where $\boldsymbol{\Delta\omega} = \boldsymbol{\omega} - \boldsymbol{\omega_0}$ is the <u>frequency detuning</u>.

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As we have done before, we can get the equations for the remaining terms using the relations

$$\dot{\tilde{\rho}}_{22} = -\dot{\tilde{\rho}}_{11} = -i\frac{\Omega}{2}(\tilde{\rho}_{12} - \tilde{\rho}_{21})$$

$$\dot{\tilde{\rho}}_{21} = \dot{\tilde{\rho}}_{12}^* = i\Delta\omega\tilde{\rho}_{12}^* - i\frac{\Omega}{2}(\tilde{\rho}_{11}^* - \tilde{\rho}_{22}^*) = i\Delta\omega\tilde{\rho}_{21} - i\frac{\Omega}{2}(\tilde{\rho}_{11} - \tilde{\rho}_{22})$$

In conclusion, the full set of **optical Bloch equations** after the RFT and RWA is

$$\begin{cases} \dot{\tilde{\boldsymbol{\rho}}}_{\mathbf{11}}(\boldsymbol{t}) = \boldsymbol{i}\frac{\boldsymbol{\Omega}}{\mathbf{2}}(\tilde{\boldsymbol{\rho}}_{\mathbf{12}} - \tilde{\boldsymbol{\rho}}_{\mathbf{21}}) \\ \dot{\tilde{\boldsymbol{\rho}}}_{\mathbf{12}} = -\boldsymbol{i\Delta\omega}\tilde{\boldsymbol{\rho}}_{\mathbf{12}} + \boldsymbol{i}\frac{\boldsymbol{\Omega}}{\mathbf{2}}(\tilde{\boldsymbol{\rho}}_{\mathbf{11}} - \tilde{\boldsymbol{\rho}}_{\mathbf{22}}) \\ \dot{\tilde{\rho}}_{21} = i\Delta\omega\tilde{\rho}_{21} - i\frac{\Omega}{2}(\tilde{\rho}_{11} - \tilde{\rho}_{22}) \\ \dot{\tilde{\rho}}_{22} = -i\frac{\Omega}{2}(\tilde{\rho}_{12} - \tilde{\rho}_{21}) \end{cases}$$

We are still not including the effect of spontaneous emission into this equations. There are different ways to include this phenomenon: we are going to see the phenomenological inclusion of spontaneous emission.

Phenomenological inclusion of spontaneous emission

We consider a two-state system in a <u>pure state</u>

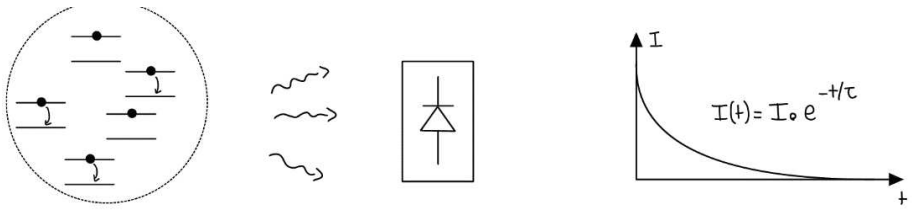
$$|\psi(t)\rangle = a_1(t)|\varphi_1\rangle + a_2(t)|\varphi_2\rangle$$

for which

$$[\rho] = \begin{bmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{bmatrix} = \begin{bmatrix} |a_1(t)|^2 & a_1 a_2^* \\ a_2 a_1^* & |a_2(t)|^2 \end{bmatrix}$$

It will be possible to see that this model can be extended to the mixed state.

First we want to include the effect of spontaneous emission in the <u>populations</u>.



Let us consider an ensemble of two-level atoms, each described by the pure state wavefunction written above and initially excited. Some atoms are going to transition and we measure the intensity of the emitted radiation as a function of time. The result of the measurement goes like

$$I(t) = I_0 e^{-\frac{t}{\tau}} = I_0 e^{-\Gamma t}$$

where τ is the lifetime of the excited state and $\Gamma = 1/\tau$.

We can therefore infer that the population of atoms is given by

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$$N_2(t) = N_2(0)e^{-\Gamma t}$$

But we know that

$$N_2(t) = |a_2(t)|^2 N_t$$

thus

$$|a_2(t)|^2 = |a_2(0)|^2 e^{-\Gamma t}$$
$$\rho_{22}(t) = \rho_{22}(0)e^{-\Gamma t}$$
$$\dot{\rho}_{22}(t) = -\Gamma\rho_{22}(0)e^{-\Gamma t} = -\Gamma\rho_{22}(t)$$
$$\dot{\tilde{\rho}}_{22}^{sp} = -\Gamma\tilde{\rho}_{22}(t)$$
$$\dot{\tilde{\rho}}_{11}^{sp} = -\dot{\tilde{\rho}}_{22}^{sp} = \Gamma\tilde{\rho}_{22}(t)$$

We now want to study the <u>coherences</u>.

The power spectral density of the previous result is a Lorentzian profile with $FWHM = \Gamma$. Therefore

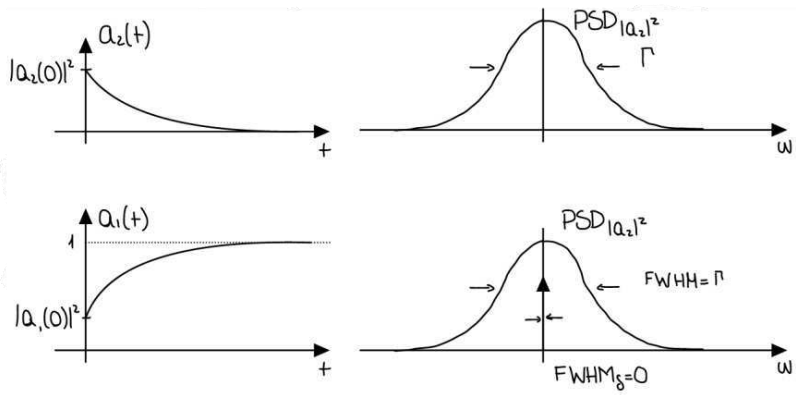
$$\dot{\tilde{\rho}}_{22}^{sp} = -\Gamma\tilde{\rho}_{22}(t) = -\mathrm{FWHM}\tilde{\rho}_{22}(t)$$
$$\frac{d}{dt}|a_2(t)|^2\Big|_{sp} = -\mathrm{FWHM}|a_2(t)|^2$$

We can generalise this result and write

$$\dot{\tilde{\rho}}_{12}^{sp} = -k\tilde{\rho}_{12}$$
$$\frac{d}{dt}[a_1(t)a_2^*(t)]\Big|_{sp} = -k[a_1(t)a_2^*(t)]$$

where k is the average between the FWHM of the power spectral densities of $|a_1(t)|^2$ and $|a_2(t)|^2$.

$$|a_2(t)|^2 = |a_2(0)|^2 e^{-\Gamma t}$$
$$|a_1(t)|^2 = 1 - |a_2(t)|^2 = 1 - |a_2(0)|^2 e^{-\Gamma t}$$
$$\mathrm{PSD}_{|a_1|^2} = \delta(\omega) + \mathrm{PSD}_{|a_2|^2}$$



Therefore, the FWHM of $\mathrm{PSD}_{|a_1|^2}$ is determined by the delta function and is zero, thus

$$k = \frac{\Gamma}{2}$$

Therefore

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$$\dot{\tilde{\rho}}_{12}^{sp} = -\frac{\Gamma}{2}\tilde{\rho}_{12}$$
$$\dot{\tilde{\rho}}_{21}^{sp} = -\frac{\Gamma}{2}\tilde{\rho}_{21}$$

Adding these spontaneous emission terms in the previous relations we get the set of **optical Bloch equations with RFT, RWA and spontaneous emission**:

$$\begin{cases} \dot{\tilde{\boldsymbol{\rho}}}_{\mathbf{11}}(\boldsymbol{t}) = \boldsymbol{i}\frac{\boldsymbol{\Omega}}{\mathbf{2}}(\tilde{\boldsymbol{\rho}}_{\mathbf{12}} - \tilde{\boldsymbol{\rho}}_{\mathbf{21}}) + \boldsymbol{\Gamma}\tilde{\boldsymbol{\rho}}_{\mathbf{22}} \\ \dot{\tilde{\boldsymbol{\rho}}}_{\mathbf{12}} = -\boldsymbol{i}\boldsymbol{\Delta\omega}\tilde{\boldsymbol{\rho}}_{\mathbf{12}} + \boldsymbol{i}\frac{\boldsymbol{\Omega}}{\mathbf{2}}(\tilde{\boldsymbol{\rho}}_{\mathbf{11}} - \tilde{\boldsymbol{\rho}}_{\mathbf{22}}) - \frac{\boldsymbol{\Gamma}}{\mathbf{2}}\tilde{\boldsymbol{\rho}}_{\mathbf{12}} \\ \dot{\tilde{\boldsymbol{\rho}}}_{\mathbf{21}} = \boldsymbol{i}\boldsymbol{\Delta\omega}\tilde{\boldsymbol{\rho}}_{\mathbf{21}} - \boldsymbol{i}\frac{\boldsymbol{\Omega}}{\mathbf{2}}(\tilde{\boldsymbol{\rho}}_{\mathbf{11}} - \tilde{\boldsymbol{\rho}}_{\mathbf{22}}) - \frac{\boldsymbol{\Gamma}}{\mathbf{2}}\tilde{\boldsymbol{\rho}}_{\mathbf{21}} \\ \dot{\tilde{\boldsymbol{\rho}}}_{\mathbf{22}} = -\boldsymbol{i}\frac{\boldsymbol{\Omega}}{\mathbf{2}}(\tilde{\boldsymbol{\rho}}_{\mathbf{12}} - \tilde{\boldsymbol{\rho}}_{\mathbf{21}}) - \boldsymbol{\Gamma}\tilde{\boldsymbol{\rho}}_{\mathbf{22}} \end{cases}$$

Stationary solutions

In stationary regime

$$0 = \dot{\tilde{\rho}}_{12} = -i\Delta\omega\tilde{\rho}_{12} + i\frac{\Omega}{2}(\tilde{\rho}_{11} - \tilde{\rho}_{22}) - \frac{\Gamma}{2}\tilde{\rho}_{12}$$

$$\tilde{\rho}_{12}\left(\frac{\Gamma}{2} + i\Delta\omega\right) = i\frac{\Omega}{2}(\tilde{\rho}_{11} - \tilde{\rho}_{22}) = i\frac{\Omega}{2}(1 - \tilde{\rho}_{22} - \tilde{\rho}_{22}) = i\frac{\Omega}{2}(1 - 2\tilde{\rho}_{22})$$

$$\tilde{\rho}_{12} = i\frac{\Omega}{2}\frac{1}{\left(\frac{\Gamma}{2} + i\Delta\omega\right)}(1 - 2\tilde{\rho}_{22}) = i\frac{\Omega}{2}\frac{\frac{\Gamma}{2} - i\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}(1 - 2\tilde{\rho}_{22})$$

$$0 = \dot{\tilde{\rho}}_{22} = -i\frac{\Omega}{2}(\tilde{\rho}_{12} - \tilde{\rho}_{21}) - \Gamma\tilde{\rho}_{22}$$

$$\tilde{\rho}_{22} = -\frac{i}{\Gamma}\frac{\Omega}{2}(\tilde{\rho}_{12} - \tilde{\rho}_{12}^*) = -\frac{i}{\Gamma}\frac{\Omega}{2}(2i\,\mathbb{Im}\{\tilde{\rho}_{12}\}) = \frac{\Omega}{\Gamma}\mathbb{Im}\{\tilde{\rho}_{12}\} = \frac{\Omega}{\Gamma}\frac{\Omega}{2}\frac{\frac{\Gamma}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}(1 - 2\tilde{\rho}_{22})$$

$$= \frac{1}{2}\frac{\frac{\Omega^2}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}(1 - 2\tilde{\rho}_{22})$$

$$\tilde{\rho}_{22}\left[1 + \frac{\frac{\Omega^2}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\right] = \frac{1}{2}\frac{\frac{\Omega^2}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}$$

$$\tilde{\boldsymbol{\rho}}_{\mathbf{22}} = \frac{1}{2}\frac{\frac{\Omega^2}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\frac{1}{\left[1 + \frac{\frac{\Omega^2}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\right]} = \frac{\mathbf{1}}{\mathbf{2}}\frac{\boldsymbol{s}}{\mathbf{1} + \boldsymbol{s}}$$

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where we have defined the saturation parameter

$$\boldsymbol{s} = \frac{\frac{\boldsymbol{\Omega^2}}{\boldsymbol{2}}}{\left(\frac{\boldsymbol{\Gamma}}{\boldsymbol{2}}\right)^{\boldsymbol{2}} + \boldsymbol{\Delta\omega^2}}$$

which is a function of $\Delta\omega$ since

$$\boldsymbol{s} = s(\Delta\omega) = \frac{\frac{\Omega^2}{2}}{\left(\frac{\Gamma}{2}\right)^2\left[1 + \left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]} = \frac{\boldsymbol{s_0}}{\boldsymbol{1} + \left(\frac{\boldsymbol{\Delta\omega}}{\boldsymbol{\Gamma/2}}\right)^{\boldsymbol{2}}}$$

where

$$\boldsymbol{s_0} = s(\Delta\omega = 0) = \frac{\frac{\Omega^2}{2}}{\left(\frac{\Gamma}{2}\right)^2} = \boldsymbol{2}\frac{\boldsymbol{\Omega^2}}{\boldsymbol{\Gamma^2}}$$

is the <u>on-resonance saturation parameter</u>.

Substituting this expression in $\tilde{\rho}_{12}$ we get

$$\tilde{\rho}_{12} = i\frac{\Omega}{2}\frac{\frac{\Gamma}{2} - i\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}(1 - 2\tilde{\rho}_{22}) = i\frac{\Omega}{2}\frac{\frac{\Gamma}{2} - i\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\left(1 - 2\frac{1}{2}\frac{s}{1+s}\right)$$

$$= i\frac{\Omega}{2}\frac{\frac{\Gamma}{2} - i\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\left(\frac{1}{1+s}\right) = i\frac{\Omega}{2}\frac{\frac{\Gamma}{2} - i\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\frac{1}{\left[1 + \frac{\frac{\Omega^2}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\right]}$$

$$= i\frac{\Omega}{2}\frac{\frac{\Gamma}{2} - i\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{\Omega^2}{2}}$$

Finally, coming back to the original time frame we find the stationary population $\rho_{22}(t) = |a_2(t)|^2$

$$\boldsymbol{\rho_{22}} = \boldsymbol{\tilde{\rho}_{22}} = \frac{\boldsymbol{1}}{\boldsymbol{2}}\frac{\boldsymbol{s}}{\boldsymbol{1+s}}$$

and the stationary coherence $\rho_{12} = a_1(t)a_2^*(t)$

$$\boldsymbol{\rho_{12}} = \boldsymbol{\tilde{\rho}_{12}e^{i\omega t}} = \boldsymbol{i}\frac{\boldsymbol{\Omega}}{\boldsymbol{2}}\frac{\frac{\boldsymbol{\Gamma}}{\boldsymbol{2}} - \boldsymbol{i\Delta\omega}}{\left(\frac{\boldsymbol{\Gamma}}{\boldsymbol{2}}\right)^{\boldsymbol{2}} + \boldsymbol{\Delta\omega^2} + \frac{\boldsymbol{\Omega^2}}{\boldsymbol{2}}}\boldsymbol{e^{i\omega t}}$$

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Florescence rate

In stationary regime

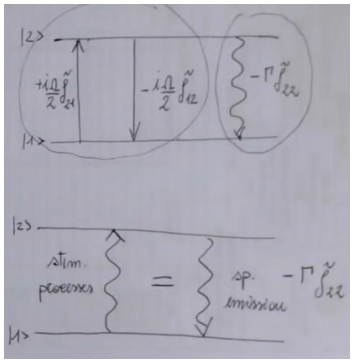
$$0 = \dot{\tilde{\rho}}_{22} = -i\frac{\Omega}{2}(\tilde{\rho}_{12} - \tilde{\rho}_{21}) - \Gamma\tilde{\rho}_{22}$$

where we have shown that the last term represents the effect of spontaneous emission. The first term, instead, represents stimulated processes.

$$\dot{\tilde{\rho}}_{22}^{sp} = -\Gamma\tilde{\rho}_{22}$$

$$\dot{\tilde{\rho}}_{22}^{st,e} = -i\frac{\Omega}{2}\tilde{\rho}_{12}$$

$$\dot{\tilde{\rho}}_{22}^{st,a} = i\frac{\Omega}{2}\tilde{\rho}_{21}$$



The rate of stimulated emission must be smaller than that of stimulated absorption, so that the net rate of stimulated process balances that of spontaneous emission.

The **total fluorescence rate**, meaning the rate of spontaneous emission for a single atom, is

$$\boldsymbol{\Gamma_p = \Gamma\tilde{\rho}_{22}}$$

$$\boldsymbol{\Gamma_p} = \Gamma\frac{1}{2}\frac{s}{1+s} = \frac{\Gamma}{2}\frac{\dfrac{s_0}{1+\left(\dfrac{\Delta\omega}{\Gamma/2}\right)^2}}{1+\dfrac{s_0}{1+\left(\dfrac{\Delta\omega}{\Gamma/2}\right)^2}} = \frac{\Gamma}{2}\frac{s_0}{1+s_0+\left(\dfrac{\Delta\omega}{\Gamma/2}\right)^2} = \frac{\Gamma}{2}\frac{s_0}{1+s_0}\frac{1}{1+\left(\dfrac{\Delta\omega}{\Gamma\sqrt{1+s_0}/2}\right)^2}$$

$$= \boldsymbol{\frac{\Gamma}{2}\frac{s_0}{1+s_0}\frac{1}{1+\left(\dfrac{\Delta\omega}{\Gamma'/2}\right)^2}}$$

Therefore, Γ_p has a Lorentzian profile with a FWHM $\boldsymbol{\Gamma' = \Gamma\sqrt{1+s_0}}$.

This phenomenon is called **power broadening** because

$$\Omega \propto E_0 \quad \rightarrow \quad \Omega^2 \propto I$$

so that s_0 can be expressed as

$$\boldsymbol{s_0} = 2\frac{\Omega^2}{\Gamma^2} = \boldsymbol{\frac{I}{I_s}}$$

with

$$\boldsymbol{I_s = \frac{\hbar\omega_0}{2\sigma\tau}} = \frac{\hbar\omega_0}{2\sigma}\Gamma$$

where $\hbar\omega_0$ is the quantum of the electromagnetic field and σ is the cross-section of the transition.

We have thus found

$$\frac{\Gamma_p}{\Gamma} = \frac{1}{2}\frac{s_0}{1+s_0}\frac{1}{1+\left(\dfrac{\Delta\omega}{\Gamma'/2}\right)^2}$$

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In the <u>low intensity regime</u>

$$I \ll I_s \quad s_0 \ll 1$$

$$\Gamma' = \Gamma\sqrt{1+s_0} \simeq \Gamma$$

$$\frac{\Gamma_p}{\Gamma} \simeq \frac{1}{2} s_0 \frac{1}{1+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2}$$

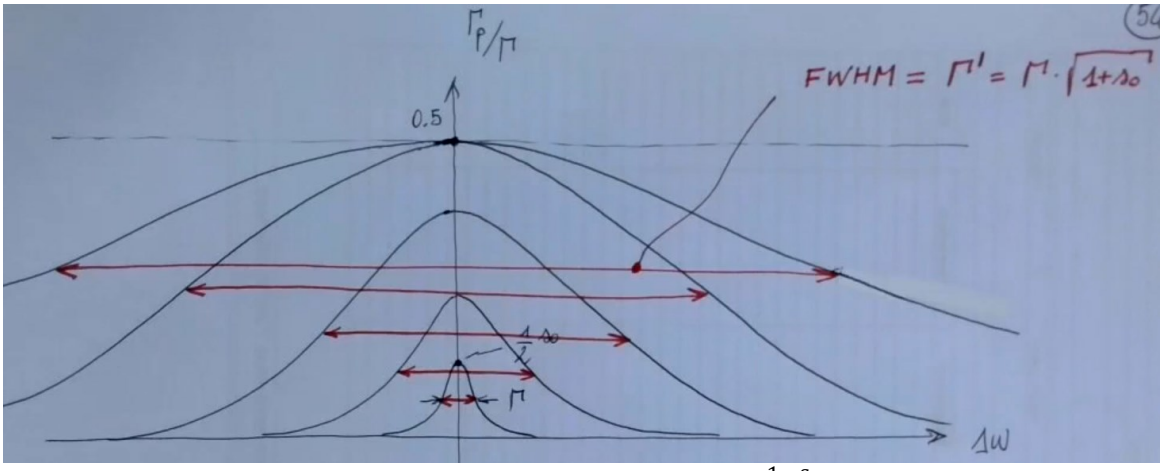
In the <u>high intensity regime</u>

$$I \gg I_s \quad s_0 \gg 1$$

$$\Gamma' = \Gamma\sqrt{1+s_0} \simeq \Gamma\sqrt{s_0}$$

$$\frac{\Gamma_p}{\Gamma} \simeq \frac{1}{2} \frac{1}{1+\left(\frac{\Delta\omega}{\Gamma\sqrt{s_0}/2}\right)^2}$$

We now want to represent Γ_p as a function of $\Delta\omega$ for different values of the intensity.



The profile is Lorentzian for all values of s_0, with height $\frac{1}{2}\frac{s_0}{1+s_0}$ and $FWHM = \Gamma' = \Gamma\sqrt{1+s_0}$: increasing s_0, both the height of the peak and the FWHM increase. There is power broadening: as the intensity of the driving field increases, the FWHM of the fluorescence rate increases too.

There is a clamping of the fluorescence rate: once the saturation of the transition is reached, the ratio is fixed for $\Delta\omega = 0$ at $\Gamma/\Gamma_p = 0.5$ and increasing the intensity of the field the FWHM increases, but the maximum is fixed.

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Radiative forces on two-level atoms

We want to study the effect of an **electromagnetic wave** on a <u>neutral atom</u>.

$$\vec{\boldsymbol{E}}(\vec{\boldsymbol{r}},\boldsymbol{t}) = \vec{\boldsymbol{\varepsilon}}\boldsymbol{E_0}(\vec{\boldsymbol{r}})\,\mathbf{cos}[\boldsymbol{\omega t} + \boldsymbol{\varphi}(\vec{\boldsymbol{r}})]$$

where $\vec{\varepsilon}$ is the unit polarisation vector, $E_0(\vec{r})$ the amplitude of the electromagnetic wave and $\varphi(\vec{r})$ the phase, which determines the shape of the wavefront.

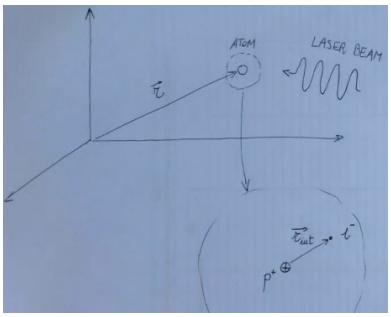
$$\vec{E}(\vec{r},t) = \vec{\varepsilon}\frac{E_0(\vec{r})}{2}e^{i\omega t + i\varphi(\vec{r})} + c.c. = \vec{\varepsilon}\frac{\tilde{E}(\vec{r})}{2}e^{i\omega t} + c.c.$$

where we have defined

$$\tilde{E}(\vec{r}) = E_0(\vec{r})e^{i\varphi(\vec{r})}$$

Description of a moving atom in a resonant laser wave

We are now going to describe the interaction of a moving atom with a resonant laser wave.



We need to define two independent position vectors:

- the **external position vector** $\vec{\boldsymbol{r}}$ describes the <u>position of the moving atom</u>
- the **internal position vector** $\vec{\boldsymbol{r}}_{\boldsymbol{int}}$ describes the position of the electron with respect to the proton, thus the <u>internal dynamic of the atomic system</u>

The two environments are separated, so we have two <u>independent subsystems</u>.

We want to include into the Hamiltonian operator the dynamic of the atom in the external system, that is with respect to the external reference frame.

On the **basis** $|\boldsymbol{\varphi_1}\rangle, |\boldsymbol{\varphi_2}\rangle$ we can write

$$\hat{H} = \hat{H}_0 + \hat{H}_I = E_1|1\rangle\langle 1| + E_2|2\rangle\langle 2| + \hat{E}_k - \hat{\vec{\mu}}_e \cdot \vec{E}$$

Choosing $E_1 = 0$, it holds $E_2 = \hbar\omega_0$.

$$\hat{H}_I = -\hat{\vec{\mu}}_e \cdot \hat{\vec{E}} = -\hat{\vec{\mu}}_e \cdot \vec{\varepsilon}\hat{E}(\vec{r},t) = -\hat{\mu}_\varepsilon \hat{E}(\vec{r},t)$$

where we have defined

$$\hat{\mu}_\varepsilon = \hat{\vec{\mu}}_e \cdot \vec{\varepsilon}$$

so that we can work with scalar quantities.

On the same basis, the matrix representation is

$$[H] = [H_0] + [H_I]$$

$$[H_0] = \begin{bmatrix} 0 & 0 \\ 0 & \hbar\omega_0 \end{bmatrix} + [E_k] = \begin{bmatrix} 0 & 0 \\ 0 & \hbar\omega_0 \end{bmatrix} + \begin{bmatrix} \left\langle 1 \middle| \frac{\hat{p}^2}{2M} \middle| 1 \right\rangle & \left\langle 1 \middle| \frac{\hat{p}^2}{2M} \middle| 2 \right\rangle \\ \left\langle 2 \middle| \frac{\hat{p}^2}{2M} \middle| 1 \right\rangle & \left\langle 2 \middle| \frac{\hat{p}^2}{2M} \middle| 2 \right\rangle \end{bmatrix}$$

In the general case

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$$E_{k,ij} = \left\langle \varphi_i(\vec{r}_{int}) \middle| \frac{\hat{p}^2}{2M} \middle| \varphi_j(\vec{r}_{int}) \right\rangle = \frac{\hat{p}^2}{2M} \langle \varphi_i | \varphi_j \rangle = \frac{\hat{p}^2}{2M} \delta_{ij}$$

since the linear momentum of the atom is only dependent on the external position $\vec{r}$ of the atom. Therefore

$$[H_0] = \begin{bmatrix} 0 & 0 \\ 0 & \hbar\omega_0 \end{bmatrix} + \frac{\hat{p}^2}{2M} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$[H_I] = \begin{bmatrix} \langle 1 | -\hat{\mu}_\varepsilon \hat{E} | 1 \rangle & \langle 1 | -\hat{\mu}_\varepsilon \hat{E} | 2 \rangle \\ \langle 2 | -\hat{\mu}_\varepsilon \hat{E} | 1 \rangle & \langle 2 | -\hat{\mu}_\varepsilon \hat{E} | 2 \rangle \end{bmatrix}$$

$$H_{I,ij} = \langle \varphi_i(\vec{r}_{int}) | -\hat{\mu}_\varepsilon \hat{E}(\vec{r},t) | \varphi_j(\vec{r}_{int}) \rangle = \langle \varphi_i(\vec{r}_{int}) | -\hat{\mu}_\varepsilon | \varphi_j(\vec{r}_{int}) \rangle E(\vec{r},t) = -\mu_{\varepsilon,ij} E(\vec{r},t)$$

since $\hat{\mu}_\varepsilon = \hat{\vec{\mu}}_e \cdot \vec{\varepsilon} = -e\hat{\vec{r}}_{int} \cdot \vec{\varepsilon}$ depends on $\vec{r}_{int}$ while $\hat{E}(\vec{r},t)$ depends exclusively on $\vec{r}$. Since $\hat{\mu}_\varepsilon$ is odd and $|\varphi_{1,2}\rangle$ have defined parity, it holds $\mu_{\varepsilon,kk} = 0$.

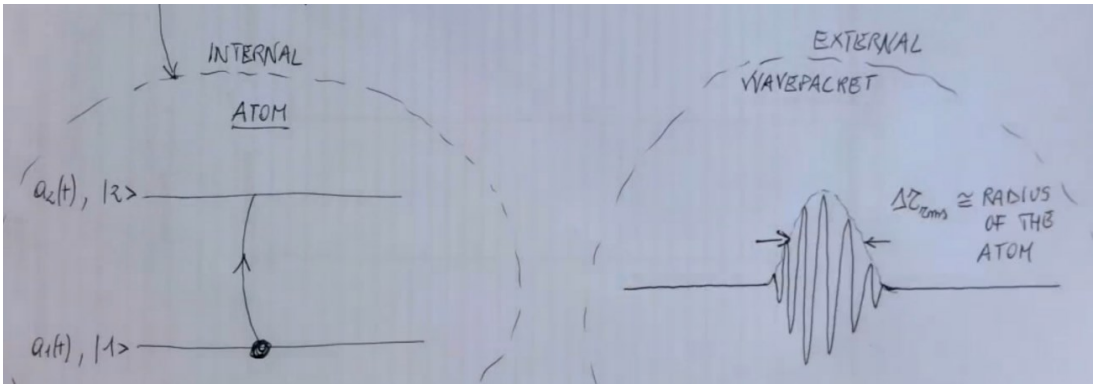
$$[H_I] = \begin{bmatrix} 0 & -\mu_{\varepsilon,12} \\ -\mu_{\varepsilon,21} & 0 \end{bmatrix} \hat{E}(\vec{r},t) = \begin{bmatrix} 0 & -\mu \\ -\mu & 0 \end{bmatrix} \hat{E}(\vec{r},t)$$

since it can be demonstrated that $\mu_{\varepsilon,12} = \mu_{\varepsilon,21} = \mu$.

State function of the atom

The overall state function of the atom is a combination, that is a tensorial product, of the internal state function and the external state function:

$$|\psi_C\rangle = |\psi_{int}\rangle \otimes |\psi_{ext}\rangle$$



The external state function can be expressed as a wave packet $\psi(\vec{r},t)$, which describes the movement of the atom. The spatial dimension of this packet is of the order of the radius of the atom.

We can also switch to a vectorial representation for the internal state of the atom.

$$|\psi_C\rangle = \begin{bmatrix} a_1(t) \\ a_2(t) \end{bmatrix} \otimes \psi(\vec{r},t)$$

Description of the position of the atom

The characteristic dimension of the atom is

$$\Delta r_{rms} = \sqrt{\overline{\left(\vec{r} - \langle \hat{\vec{r}} \rangle\right)^2}}$$

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We can describe the position of the atom as

$$\vec{r}_{at} = \langle \hat{\vec{r}} \rangle = \langle \psi | \hat{\vec{r}} | \psi \rangle = \int \vec{r} |\psi(\vec{r}, t)|^2 dr^3$$

which can be rewritten as

$$\vec{r}_{at} = \langle \hat{r}_x \rangle \vec{u}_x + \langle \hat{r}_y \rangle \vec{u}_y + \langle \hat{r}_z \rangle \vec{u}_z = x_{at}\vec{u}_x + y_{at}\vec{u}_y + z_{at}\vec{u}_z$$

where we have defined the i component of the *external* position

$$\langle \hat{r}_i \rangle = \int r_i |\psi(\vec{r}, t)|^2 dr^3 , \qquad i = x, y, z$$

Remember that given an observable A the corresponding quantum operator $\hat{A}$ can only return its eigenvalues a_n.

Description of the velocity of the atom without the laser beam

Let us consider the velocity of the atom when the laser is turned off.

$$\vec{v}_{at} = \frac{d}{dt}\vec{r}_{at} = \left\langle \frac{d}{dt}\hat{\vec{r}} \right\rangle = \left\langle \frac{d}{dt}\hat{r}_x \right\rangle \vec{u}_x + \left\langle \frac{d}{dt}\hat{r}_y \right\rangle \vec{u}_y + \left\langle \frac{d}{dt}\hat{r}_z \right\rangle \vec{u}_z = v_{at,x}\vec{u}_x + v_{at,y}\vec{u}_y + v_{at,z}\vec{u}_z$$

where

$$v_{at,i} = \left\langle \frac{d}{dt}\hat{r}_i \right\rangle = \left\langle \psi \middle| \frac{d}{dt}\hat{r}_i \middle| \psi \right\rangle$$

The time evolution of $\hat{r}_i$ is given by

$$\frac{d}{dt}\hat{r}_i = -\frac{i}{\hbar}[\hat{r}_i, \hat{H}_0]$$

where we have used the uncoupled Hamiltonian since the laser is off.

Therefore

$$v_{at,i} = \left\langle \psi \middle| -\frac{i}{\hbar}[\hat{r}_i, \hat{H}_0] \middle| \psi \right\rangle = -\frac{i}{\hbar}\langle [\hat{r}_i, \hat{H}_0] \rangle = -\frac{i}{\hbar}\langle [\hat{r}_i, \hbar\omega_0 |2\rangle\langle 2| + E_{k,i}] \rangle$$
$$= -\frac{i}{\hbar}\left\langle \left[\hat{r}_i, \hbar\omega_0 |2\rangle\langle 2| + \frac{\hat{p}_i^2}{2M}\right] \right\rangle = -\frac{i}{\hbar}\langle [\hat{r}_i, \hbar\omega_0 |2\rangle\langle 2|] \rangle - \frac{i}{\hbar}\left\langle \left[\hat{r}_i, \frac{\hat{p}_i^2}{2M}\right] \right\rangle$$

However, it holds $[\hat{r}_i, \hbar\omega_0 |2\rangle\langle 2|] = 0$ since the two operators act on independent subsystems, external and internal respectively.

It can be proved[1] that

$$v_{at,i} = -\frac{i}{\hbar}\left\langle \frac{1}{2M}[\hat{r}_i, \hat{p}_i^2] \right\rangle = -\frac{i}{\hbar}\left\langle \frac{\hat{p}_i}{M}[\hat{r}_i, \hat{p}_i] \right\rangle = -\frac{i}{\hbar}\left\langle \frac{\hat{p}_i}{M} i\hbar \right\rangle = \frac{\langle \hat{p}_i \rangle}{M} = \frac{p_i}{M}$$

Therefore when the laser is turned off

$$\vec{v}_{at} = \frac{\vec{p}}{M}$$

---

[1]

$$[\hat{r}_i, \hat{p}_i^2] = \hat{r}_i\hat{p}_i^2 - \hat{p}_i^2\hat{r}_i = \hat{r}_i\hat{p}_i\hat{p}_i - \hat{p}_i\hat{p}_i\hat{r}_i = ([\hat{r}_i, \hat{p}_i] + \hat{p}_i\hat{r}_i)\hat{p}_i - \hat{p}_i\hat{p}_i\hat{r}_i = [\hat{r}_i, \hat{p}_i]\hat{p}_i + \hat{p}_i\hat{r}_i\hat{p}_i - \hat{p}_i\hat{p}_i\hat{r}_i$$
$$= [\hat{r}_i, \hat{p}_i]\hat{p}_i + \hat{p}_i(\hat{r}_i\hat{p}_i - \hat{p}_i\hat{r}_i) = [\hat{r}_i, \hat{p}_i]\hat{p}_i + \hat{p}_i[\hat{r}_i, \hat{p}_i]$$

Since $[\hat{r}_i, \hat{p}_i] = i\hbar$ it also holds $[\hat{r}_i, \hat{p}_i]\hat{p}_i = \hat{p}_i[\hat{r}_i, \hat{p}_i]$, so that $[\hat{r}_i, \hat{p}_i^2] = 2\hat{p}_i[\hat{r}_i, \hat{p}_i]$.

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Description of the atom when the laser is turned on

We are now going to describe the atom when the laser is turned on and we are <u>near resonance</u>.

$$\vec{F} = \frac{d}{dt}\vec{p}$$

$$\left\langle \hat{\vec{F}} \right\rangle = \left\langle \frac{d}{dt}\hat{\vec{p}} \right\rangle = \left\langle -\frac{i}{\hbar}\left[\hat{\vec{p}}, \hat{H}\right] \right\rangle = -\frac{i}{\hbar}\left\langle \left[\hat{\vec{p}}, \hbar\omega_0 |2\rangle\langle 2| + \frac{\hat{p}^2}{2M} - \hat{\mu}_\varepsilon \hat{E}\right] \right\rangle$$

$$= -\frac{i}{\hbar}\left\langle \left[\hat{\vec{p}}, \hbar\omega_0 |2\rangle\langle 2|\right] \right\rangle - \frac{i}{\hbar}\left\langle \left[\hat{\vec{p}}, \frac{\hat{p}^2}{2M}\right] \right\rangle - \frac{i}{\hbar}\left\langle \left[\hat{\vec{p}}, -\hat{\mu}_\varepsilon \hat{E}\right] \right\rangle$$

The first term is equal to zero because the operators act on different subsystems, while the second is equal to zero since

$$\left[\hat{\vec{p}}, \frac{\hat{p}^2}{2M}\right] |\psi_C\rangle = \hat{\vec{p}}\frac{\hat{p}^2}{2M}|\psi_{int}\rangle|\psi_r\rangle - \frac{\hat{p}^2}{2M}\hat{\vec{p}}|\psi_{int}\rangle|\psi_r\rangle$$

$$\hat{\vec{p}}\frac{\hat{p}^2}{2M}|\psi_{int}\rangle|\psi_r\rangle = -i\hbar\vec{\nabla}\frac{-i\hbar\vec{\nabla}\cdot\left(-i\hbar\vec{\nabla}\right)}{2M}|\psi_{int}\rangle|\psi_r\rangle = |\psi_{int}\rangle\frac{i\hbar^3}{2M}\vec{\nabla}\nabla^2\psi_r(\vec{r},t)$$

$$= |\psi_{int}\rangle\frac{i\hbar^3}{2M}\left(\frac{\partial}{\partial x}\vec{u}_x + \frac{\partial}{\partial y}\vec{u}_y\right)\left(\frac{\partial^2}{\partial x^2}\psi_r + \frac{\partial^2}{\partial y^2}\psi_r\right)$$

$$= |\psi_{int}\rangle\frac{i\hbar^3}{2M}\left(\frac{\partial^3}{\partial x^3}\psi_r\vec{u}_x + \frac{\partial}{\partial x}\frac{\partial^2}{\partial y^2}\psi_r\vec{u}_x + \frac{\partial}{\partial y}\frac{\partial^2}{\partial x^2}\psi_r\vec{u}_y + \frac{\partial^3}{\partial y^3}\psi_r\vec{u}_y\right)$$

$$\frac{\hat{p}^2}{2M}\hat{\vec{p}}|\psi_{int}\rangle|\psi_r\rangle = \frac{-i\hbar\vec{\nabla}\cdot\left(-i\hbar\vec{\nabla}\right)}{2M}\left(-i\hbar\vec{\nabla}\right)|\psi_{int}\rangle|\psi_r\rangle = |\psi_{int}\rangle\frac{i\hbar^3}{2M}\nabla^2\vec{\nabla}\psi_r(\vec{r},t)$$

$$= |\psi_{int}\rangle\frac{i\hbar^3}{2M}\left(\frac{\partial^2}{\partial x^2}\psi_r + \frac{\partial^2}{\partial y^2}\psi_r\right)\left(\frac{\partial}{\partial x}\vec{u}_x + \frac{\partial}{\partial y}\vec{u}_y\right)$$

$$= |\psi_{int}\rangle\frac{i\hbar^3}{2M}\left(\frac{\partial^3}{\partial x^3}\psi_r\vec{u}_x + \frac{\partial^2}{\partial y^2}\frac{\partial}{\partial x}\psi_r\vec{u}_x + \frac{\partial^2}{\partial x^2}\frac{\partial}{\partial y}\psi_r\vec{u}_y + \frac{\partial^3}{\partial y^3}\psi_r\vec{u}_y\right)$$

Since $\frac{\partial}{\partial x}\frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial y^2}\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}\frac{\partial^2}{\partial x^2} = \frac{\partial^2}{\partial x^2}\frac{\partial}{\partial y}$, the two expressions are the same, thus

$$\left[\hat{\vec{p}}, \frac{\hat{p}^2}{2M}\right] = 0$$

Therefore, the expression simplifies to

$$\left\langle \hat{\vec{F}} \right\rangle = -\frac{i}{\hbar}\left\langle \left[\hat{\vec{p}}, -\hat{\mu}_\varepsilon \hat{E}(\vec{r},t)\right] \right\rangle$$

Notice that $\hat{\vec{p}}$ acts on the external state and so does $\hat{E}(\vec{r},t)$, while $\hat{\mu}_\varepsilon$ acts on the internal state. Therefore, this expectation value is responsible for the mixing between the two states.

$$\left\langle \hat{\vec{F}} \right\rangle = -\frac{i}{\hbar}\left\langle -\hat{\mu}_\varepsilon\left[\hat{\vec{p}}, \hat{E}(\vec{r},t)\right] \right\rangle$$

$$\left[\hat{\vec{p}}, \hat{E}\right]|\psi_{int}\rangle|\psi_r\rangle = |\psi_{int}\rangle\left[\hat{\vec{p}}, \hat{E}\right]|\psi_r\rangle = |\psi_{int}\rangle\left[\hat{\vec{p}}\left(\hat{E}\psi_r\right) - \hat{E}\hat{\vec{p}}\psi_r\right]$$

$$= |\psi_{int}\rangle\left[\left(-i\hbar\vec{\nabla}\right)\left(\hat{E}\psi_r\right) - \hat{E}\left(-i\hbar\vec{\nabla}\right)\psi_r\right] =$$

$$= -i\hbar|\psi_{int}\rangle\left[\left(\psi_r\vec{\nabla}\hat{E} + \hat{E}\vec{\nabla}\psi_r\right) - \hat{E}\vec{\nabla}\psi_r\right] = -i\hbar|\psi_{int}\rangle\psi_r\vec{\nabla}\hat{E}$$

$$= \left(-i\hbar\vec{\nabla}\hat{E}\right)|\psi_{int}\rangle|\psi_r\rangle$$

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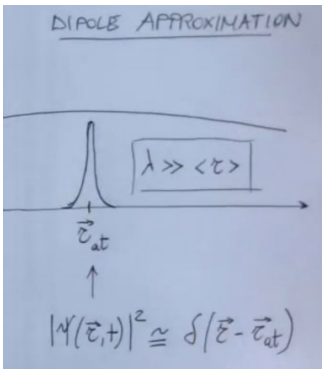
$$\Rightarrow \quad \left[\hat{\vec{p}}, \hat{E}\right] = -i\hbar\vec{\nabla}\hat{E}$$

$$\left\langle\hat{\vec{F}}\right\rangle = -\frac{i}{\hbar}\left\langle-\hat{\mu}_\varepsilon\left(-i\hbar\vec{\nabla}\hat{E}\right)\right\rangle = \left\langle\hat{\mu}_\varepsilon\vec{\nabla}\hat{E}\right\rangle = \langle\psi_r|\langle\psi_{int}|\hat{\mu}_\varepsilon\vec{\nabla}\hat{E}|\psi_{int}\rangle|\psi_r\rangle$$

Notice that the two operators act on different subspaces: $\hat{\mu}_\varepsilon$ is acting on the internal state function, while $\vec{\nabla}\hat{E}$ is acting on the external state function.

$$\left\langle\hat{\vec{\boldsymbol{F}}}\right\rangle = \langle\psi_r|\langle\psi_{int}|\hat{\mu}_\varepsilon\vec{\nabla}\hat{E}|\psi_{int}\rangle|\psi_r\rangle = \left\langle\psi_r\middle|\langle\psi_{int}|\hat{\mu}_\varepsilon|\psi_{int}\rangle\vec{\nabla}\hat{E}\middle|\psi_r\right\rangle = \langle\hat{\mu}_\varepsilon\rangle\left\langle\psi_r\middle|\vec{\nabla}\hat{E}\middle|\psi_r\right\rangle$$
$$= \langle\hat{\boldsymbol{\mu}}_{\boldsymbol{\varepsilon}}\rangle\langle\vec{\boldsymbol{\nabla}}\hat{\boldsymbol{E}}\rangle$$

Notice that in the dipole approximation the wave function is different from zero only in close proximity to the atom, so that



$$\left\langle\vec{\nabla}\hat{E}\right\rangle = \left\langle\psi_r\middle|\vec{\nabla}\hat{E}\middle|\psi_r\right\rangle = \int\vec{\nabla}\hat{E}(\vec{r},t)|\psi(\vec{r},t)|^2 dr^3$$
$$\simeq \int\vec{\nabla}\hat{E}(\vec{r},t)\,\delta(\vec{r}-\vec{r}_{at})\,dr^3 = \vec{\nabla}\hat{E}(\vec{r}_{at},t)$$

where

$$\vec{\nabla}\hat{E}(\vec{r},t) = \vec{\nabla}\left(\frac{E_0(\vec{r})}{2}e^{i\varphi(\vec{r})+i\omega t} + c.c\right)$$
$$= \frac{\vec{\nabla}E_0 e^{i\varphi+i\omega t} + E_0\left(i\vec{\nabla}\varphi\right)e^{i\varphi+i\omega t}}{2} + c.c.$$
$$= \frac{\vec{\nabla}E_0 + iE_0\vec{\nabla}\varphi}{2}e^{i\varphi+i\omega t} + c.c.$$

To find $\langle\hat{\mu}_\varepsilon\rangle$ we can use the relation $\left\langle\hat{A}\right\rangle = \mathrm{Tr}\{[\rho][A]\}$.

$$\begin{bmatrix}\rho_{11} & \rho_{12}\\ \rho_{21} & \rho_{22}\end{bmatrix}\begin{bmatrix}0 & \mu_{\varepsilon,12}\\ \mu_{\varepsilon,21} & 0\end{bmatrix} = \begin{bmatrix}\rho_{12}\mu_{\varepsilon,21} & \cdots\\ \cdots & \rho_{21}\mu_{\varepsilon,12}\end{bmatrix}$$

where

$$\rho_{12} = i\frac{\Omega_{12}}{2}\frac{\frac{\Gamma}{2} - i\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}e^{i\omega t} = \frac{\Omega_{12}}{2}\frac{\Delta\omega + i\frac{\Gamma}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}e^{i\omega t}$$

is the <u>stationary solution</u> of the optical Block equation.

Notice that here we are using the square modulus of Ω_{12}: while we have previously assumed that Ω is real, we can generalise to the case in which it is a complex number. Indeed, given the interaction Hamiltonian operator in the form

$$\hat{H}_I = -\hat{\mu}_\varepsilon\hat{E}(\vec{r},t) = -\hat{\mu}_\varepsilon\frac{E_0(\vec{r})}{2}e^{i\varphi(\vec{r})+i\omega t} + c.c. = -\frac{\hat{\mu}_\varepsilon\tilde{E}(\vec{r})}{2}e^{i\omega t} + c.c.$$

we can define a **complex Rabi frequency operator**:

$$\hat{\boldsymbol{\Omega}} = -\frac{\hat{\boldsymbol{\mu}}_{\boldsymbol{\varepsilon}}\tilde{\boldsymbol{E}}(\vec{\boldsymbol{r}})}{\hbar}$$

to write the interaction Hamiltonian in the form

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$$\boldsymbol{\hat{H}_I = \frac{\hbar\hat{\Omega}}{2} e^{i\omega t} + c.c.}$$

Therefore we can write

$$\Omega_{12} = -\frac{\mu_{\varepsilon,12}\tilde{E}(\vec{r})}{\hbar}$$

$$\langle\hat{\mu}_\varepsilon\rangle = \mathrm{Tr}\{[\rho][\mu_\varepsilon]\} = \rho_{12}\mu_{\varepsilon,21} + c.c. = -\frac{\mu_{\varepsilon,12}}{2\hbar}\tilde{E}(\vec{r})\frac{\Delta\omega + i\frac{\Gamma}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}e^{i\omega t}\mu_{\varepsilon,21} + c.c.$$

$$= -\frac{|\mu_{\varepsilon,12}|^2}{2\hbar}\frac{\Delta\omega + i\frac{\Gamma}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}\tilde{E}(\vec{r})e^{i\omega t} + c.c.$$

Finally,

$$\left\langle\hat{\vec{F}}\right\rangle = \langle\hat{\mu}_\varepsilon\rangle\left\langle\vec{\nabla}\hat{E}\right\rangle = \langle\hat{\mu}_\varepsilon\rangle\vec{\nabla}\hat{E}(\vec{r}_{at}, t)$$

$$= \left\{-\frac{|\mu_{\varepsilon,12}|^2}{2\hbar}\frac{\Delta\omega + i\frac{\Gamma}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}E_0 e^{i\varphi + i\omega t} + c.c.\right\}\left\{\frac{\vec{\nabla}E_0 + iE_0\vec{\nabla}\varphi}{2}e^{i\varphi + i\omega t} + c.c.\right\}$$

As usual, we consider only the slow-varying field and <u>neglect the high frequency terms</u>.

$$\left\langle\hat{\vec{F}}\right\rangle \simeq -\frac{|\mu_{\varepsilon,12}|^2}{4\hbar}\frac{\Delta\omega + i\frac{\Gamma}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}E_0\left[\vec{\nabla}E_0 - iE_0\vec{\nabla}\varphi\right] + c.c. = 2\,\mathbb{R}\mathrm{e}\{\ldots\} =$$

$$= -\frac{|\mu_{\varepsilon,12}|^2}{2\hbar}\frac{\Delta\omega E_0\vec{\nabla}E_0 + \frac{\Gamma}{2}E_0^2\vec{\nabla}\varphi}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}$$

$$= -\frac{|\mu_{\varepsilon,12}|^2}{2\hbar}\frac{\Delta\omega E_0(\vec{r})\boldsymbol{\vec{\nabla}E_0(\vec{r})}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}} - \frac{|\mu_{\varepsilon,12}|^2}{2\hbar}\frac{\frac{\Gamma}{2}E_0^2(\vec{r})\boldsymbol{\vec{\nabla}\varphi(\vec{r})}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}$$

$$= \vec{F}_{DIP} + \vec{F}_{RRP}$$

When the laser is interacting with the atom, the expectation value of the force is the sum of two separate contributions:

- the **dipole force** $\boldsymbol{\vec{F}_{DIP}}$ depends on the gradient of the amplitude of the field and is typically used to confine (trap) atoms or other entities;
- the **resonant radiation pressure force** $\boldsymbol{\vec{F}_{RRP}}$ depends on the gradient of its phase function and is used to cool down (laser cooling), deflect or more in general manipulate the atom (atom optics).

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Resonant radiation pressure force

A laser beam can be used <u>to slow down the atoms</u>.

As electric field

$$\vec{E}(\vec{r},t) = \vec{\varepsilon}E(\vec{r},t) = \vec{\varepsilon}E_0(\vec{r})\cos[\omega t + \varphi(\vec{r})]$$

we use a **plane wave**, so

$$\begin{cases} E_0(\vec{r}) = E_0 = const. \\ \varphi(\vec{r}) = -\vec{k}\cdot\vec{r} \end{cases}$$

$$\vec{E}(\vec{r},t) = \vec{\varepsilon}E_0\cos\left[\omega t - \vec{k}\cdot\vec{r}\right] = \vec{\varepsilon}\frac{E_0 e^{-i\vec{k}\cdot\vec{r}}}{2}e^{i\omega t} + c.c.$$

In this case

$$\vec{\nabla}E_0(\vec{r}) = 0 \quad \rightarrow \quad \vec{F}_{DIP} \propto \vec{\nabla}E_0(\vec{r}) = 0$$

$$\begin{aligned}\vec{\nabla}\varphi(\vec{r}) &= \left[\frac{\partial}{\partial x}\vec{u}_x + \frac{\partial}{\partial y}\vec{u}_y + \frac{\partial}{\partial z}\vec{u}_z\right]\left[-\left(k_x\vec{u}_x + k_y\vec{u}_y + k_z\vec{u}_z\right)\left(x\vec{u}_x + y\vec{u}_y + z\vec{u}_z\right)\right] \\ &= -\left[\frac{\partial}{\partial x}\vec{u}_x + \frac{\partial}{\partial y}\vec{u}_y + \frac{\partial}{\partial z}\vec{u}_z\right]\left(k_x x + k_y y + k_z z\right) = -k_x\vec{u}_x - k_y\vec{u}_y - k_z\vec{u}_z \\ &= -\vec{k}\end{aligned}$$

$$\vec{F}_{RRP} = -\frac{\left|\mu_{\varepsilon,12}\right|^2}{2\hbar}\frac{\frac{\Gamma}{2}E_0^2(\vec{r})\vec{\nabla}\varphi(\vec{r})}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}} = -\frac{\left|\mu_{\varepsilon,12}\right|^2}{2\hbar}\frac{\frac{\Gamma}{2}E_0^2(-\vec{k})}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}$$

We recall that

$$|\Omega_{12}|^2 = \frac{\left|\mu_{\varepsilon,12}\right|^2 E_0^2}{\hbar^2}$$

so that we can write

$$\begin{aligned}\vec{\boldsymbol{F}}_{\boldsymbol{RRP}} &= \hbar\vec{k}\frac{|\Omega_{12}|^2}{2}\frac{\frac{\Gamma}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}} = \hbar\vec{k}\frac{\Gamma}{2}\frac{\frac{|\Omega_{12}|^2}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\frac{1}{\left[1 + \frac{\frac{|\Omega_{12}|^2}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\right]} \\ &= \hbar\vec{\boldsymbol{k}}\frac{\boldsymbol{\Gamma}}{\boldsymbol{2}}\frac{\boldsymbol{s}}{\boldsymbol{1+s}} = \vec{\boldsymbol{p}}_{\boldsymbol{ph}}\boldsymbol{\Gamma}_{\boldsymbol{p}}\end{aligned}$$

where we have used the saturation parameter s and the total fluorescence rate Γ_p and we have recognised the momentum of the photon. The <u>force</u> is related to the change of the momentum of the atom due to the action of the lasering: accordingly, it <u>corresponds to the momentum of a photon multiplied by the rate of fluorescence</u>.

With the same calculations done when discussing the fluorescence rate, we find that the force has the following Lorentzian profile:

$$\vec{F}_{RRP} = \hbar\vec{k}\frac{\Gamma}{2}\frac{s}{1+s} = \hbar\vec{k}\frac{\Gamma}{2}\frac{s_0}{1+s_0}\frac{1}{1 + \left(\frac{\Delta\omega}{\Gamma'/2}\right)^2}$$

where we have introduced

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$$\Gamma' = \Gamma\sqrt{1+s_0}$$

which describes the power broadening.

In the **low intensity** case

$$s_0 = \frac{I}{I_s} \simeq 0$$

$$\Gamma' = \Gamma\sqrt{1+s_0} \simeq \Gamma$$

thus there is no power broadening.

$$\vec{F}_{RRP} = \hbar\vec{k}\frac{\Gamma}{2}s_0\frac{1}{1+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2}$$

In the **high intensity** case

$$s_0 = \frac{I}{I_s} \gg 1$$

$$\Gamma' = \Gamma\sqrt{1+s_0} \simeq \Gamma\sqrt{s_0}$$

thus there is a strong power broadening effect.

$$\vec{F}_{RRP} = \hbar\vec{k}\frac{\Gamma}{2}\frac{1}{1+\left(\frac{\Delta\omega}{\Gamma'/2}\right)^2}$$

In the **high intensity** case and **on resonance** the force is maximum

$$\vec{F}_{RRP}(\Delta\omega = 0) = \hbar\vec{k}\frac{\Gamma}{2} = \vec{F}_{\max}$$

$$Ma_{\max} = \left|\vec{F}_{\max}\right| = \hbar\left|\vec{k}\right|\frac{\Gamma}{2}$$

$$a_{\max} = \frac{\hbar k}{M}\frac{\Gamma}{2} = v_R\frac{\Gamma}{2}$$

where we have introduce the **recoil velocity** v_R <u>induced on the atom by absorption of one photon</u>:

$$\boldsymbol{v_R} := \frac{p}{M} = \frac{\hbar\boldsymbol{k}}{\boldsymbol{M}}$$

This way, the <u>maximum deceleration</u> is expressed as the recoil velocity induced by the absorption of a photon times the maximum rate of absorption.

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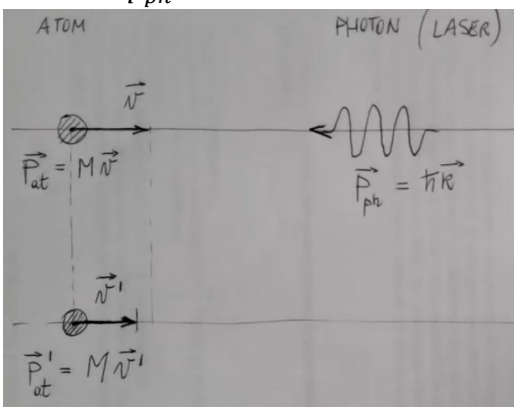
To understand the physical meaning of the recoil velocity, let us consider an atom moving with velocity $\vec{v}$ and linear momentum $\vec{p}_{at}$ and a photon moving in the opposite direction with momentum $\vec{p}_{ph} = \hbar\vec{k}$.



The overall momentum of the system is

$$\vec{p} = \vec{p}_{at} + \vec{p}_{ph} = M\vec{v} + \hbar\vec{k}$$

After the absorption of the photon, the velocity and the momentum of the atom are lower

$$M\vec{v} + \hbar\vec{k} = \vec{p} = \vec{p}'_{at} = M\vec{v}'$$

$$|v - v'| = \frac{\hbar k}{M} = v_R$$

Absorbing the photon changes the momentum by $\hbar k$ and accordingly the velocity by a quantity v_R.

Let us consider the case of a sodium atom (Na):

$$\lambda_0 = 589\ nm$$

$$M = 23\ a.m.u. = 23 \cdot 1.66 \cdot 10^{-27}\ kg \simeq 38 \cdot 10^{-27}\ kg$$

$$\left.\frac{\Gamma}{2\pi}\right|_{Na\ @\lambda_0} \simeq 10\ MHz$$

In this case we get

$$v_R = \frac{\hbar k}{M} = \frac{\hbar}{M}\frac{2\pi}{\lambda_0} = \frac{h}{M\lambda_0} = \frac{6.626 \cdot 10^{-34}\ J \cdot s}{38 \cdot 10^{-27}\ kg \cdot 589\ nm} \simeq 0.025\ m/s$$

$$a_{\max} = v_R\frac{\Gamma}{2} = 0.025\ m/s \cdot \frac{2\pi \cdot 10^7\ Hz}{2} \simeq 10^6\ m/s^2 \simeq 10^5\ g$$

We have thus obtained a huge deceleration.

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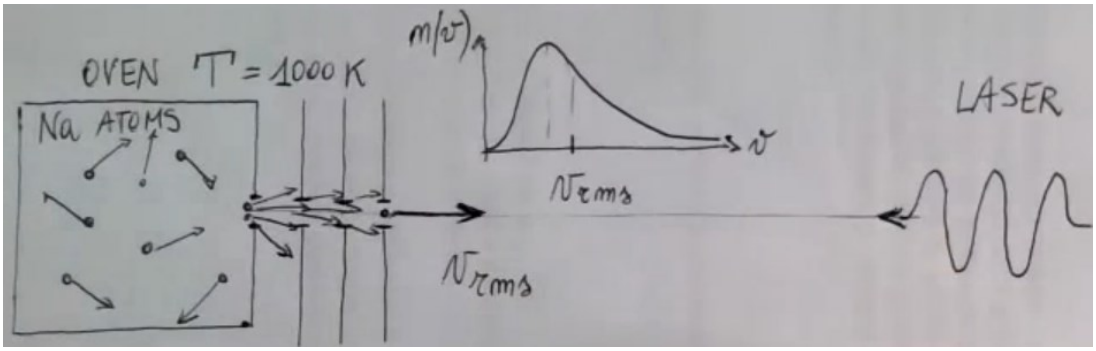

Let us consider the sodium atom moving at thermal velocity: atoms evaporate from an oven of temperature $T = 1000\ K$ and are filtered to select a certain propagation direction. We get a certain Maxwell-Boltzmann distribution of the velocities and we specifically tailor our system as to slow down atoms moving at the root mean square velocity $v_{rms} = \sqrt{\langle v^2 \rangle}$. Considering the atoms non-interacting, the average kinetic energy is

$$\langle E_k \rangle = \frac{1}{2} M \langle v^2 \rangle = \frac{1}{2} M v_{rms}^2$$

Since each atom has 3 degrees of freedom, the energy is also

$$\langle E_k \rangle = \frac{3}{2} kT$$

thus

$$v_{rms} = \sqrt{\frac{3kT}{M}} \simeq 1000 \frac{m}{s}$$

We want to calculate the <u>minimum stopping distance</u>, meaning the minimum distance to bring atoms to a halt.

$$a(t) = a_0 = -a_{\max}$$

$$v(t) = v(0) + a_0 t = v_{rms} - a_{\max} t$$

$$z(t) = z(0) + v(0)t + \frac{1}{2} a_0 t^2 = v_{rms} t - \frac{1}{2} a_{\max} t^2$$

$$0 = v(t_{\min}) = v_{rms} - a_{\max} t \quad \rightarrow \quad t_{\min} = \frac{v_{rms}}{a_{\max}}$$

$$L_{\min} = z(t_{\min}) = v_{rms} \frac{v_{rms}}{a_{\max}} - \frac{1}{2} a_{\max} \left(\frac{v_{rms}}{a_{\max}} \right)^2 = \frac{1}{2} \frac{v_{rms}^2}{a_{\max}} \simeq 0.5\ m$$

This is a very useful result since this distance is practicable in a lab.

Let us <u>estimate the power of the laser required</u> to achieve the maximum deceleration. Let us consider a strong saturation parameter, i.e. $s_0 = \frac{I}{I_s} = 10$. Consider a two-level system and a wave with $I \gg I_s$. In these conditions the stimulated processes are much stronger than the spontaneous one, so that in stationary conditions

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$$0 = \frac{dN_2}{dt} = -WN_2 + WN_1 - \frac{N_2}{\tau} \simeq -WN_2 + WN_1$$

$$\Rightarrow \quad N_1 \simeq N_2 \simeq \frac{N_t}{2}$$

We can thus consider the system as a series of discs of surface σ and the saturation condition, meaning the intensity required to keep the system saturated, becomes having one photon on each disc in each time interval τ, which corresponds to the time after which the atom excited by the previous photon has relaxed.

$$\hbar\omega_0 \frac{1}{\tau} = P_{\text{on single atom}} = I_s \sigma$$

Since we have seen that in a stationary condition we have to excite only half of the atoms, the required saturation intensity is

$$I_s = \frac{\hbar\omega_0}{2\sigma\tau}$$

In the case of the sodium atom the transition cross section is $\sigma_{Na} = 165.7 \cdot 10^{-15}\, m^2$ and the lifetime is $\tau = 16\, ns$, thus $\boldsymbol{I_s \simeq 6.6\, mW/cm^2}$.

Assuming a laser beam cross section of $A = 1\, cm^2$, the required intensity is

$$I > 10 I_s = 66\, mW/cm^2 \quad \Rightarrow \quad P = 66\, mW$$

Therefore the laser sources needed to slow down thermal atoms are easily available.

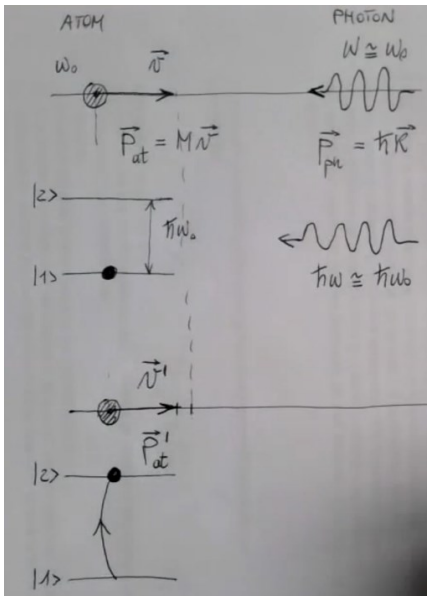
Intuitive physical picture of $\vec{F}_{RRP}$

Consider an atom moving with velocity $\vec{v}$ and interacting with a photon in the form of a plane electromagnetic wave defined by ω and $\vec{k}$.

The result we have obtained,

$$\begin{cases} \vec{F}_{RRP} = \hbar\vec{k}\frac{\Gamma}{2}\frac{s}{1+s} = \hbar\vec{k}\Gamma_p \\ \vec{F}_{DIP} = 0 \end{cases}$$

can be explained in terms of fluorescent cycles.



The atom can be seen as a two-level system with energy separation $\hbar\omega_0$. <u>Near resonance</u>, the photon has an energy $\hbar\omega \simeq \hbar\omega_0$.

Before the interaction, the overall momentum is

$$\vec{p}_{tot} = \vec{p}_{at} + \vec{p}_{ph} = M\vec{v} + \hbar\vec{k}$$

The absorbed photons are coming always from the same direction, thus the velocity of the atom decreases, while the atom gets excited.

$$\vec{p}'_{at} = \vec{p}_{tot} = M\vec{v} + \hbar\vec{k} = \vec{p}_{at} + \hbar\vec{k}$$

Then, there is a scattering event. Scattering is characterised by isotropy, thus it is uniformly distributed in all directions. The atom undergoes spontaneous emission,

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emitting a photon with momentum $\vec{p}_{\text{scat}} = \hbar\vec{k}_{\text{scat}}$. This is a scattering event of the original photon, since it is isotropically reemitted by spontaneous emission.

The recoil changes the momentum of the atom:

$$\vec{p}''_{at} + \hbar\vec{k}_{\text{scat}} = \vec{p}_{tot} = \vec{p}'_{at}$$
$$\vec{p}''_{at} = \vec{p}'_{at} - \hbar\vec{k}_{\text{scat}} = \vec{p}_{at} + \hbar\vec{k} - \hbar\vec{k}_{\text{scat}}$$

At the end of the process, therefore, the atom is back in the ground state. This <u>can be considered both as a fluorescence cycle or as a scattering cycle</u>.

The change in the momentum of the atom over a cycle is thus

$$\Delta\vec{p}_{at}|_{\text{cycle}} = \vec{p}''_{at} - \vec{p}_{at} = \hbar\vec{k} - \hbar\vec{k}_{\text{scat}}$$
$$\langle\Delta\vec{p}_{at}\rangle|_{\text{cycle}} = \langle\hbar\vec{k}\rangle - \langle\hbar\vec{k}_{\text{scat}}\rangle = \hbar\vec{k}$$

where the average value of the scattering momentum is zero due to isotropy, while $\langle\hbar\vec{k}\rangle = \hbar\vec{k}$ since it is constant.

The force is thus

$$\vec{F}_{RRP} = \frac{d\vec{p}_{at}}{dt} = \langle\Delta\vec{p}_{at}\rangle|_{\text{cycle}}\Gamma_p$$

where the fluorescence rate can also be seen as a scattering rate.

Consider the optical Bloch equation

$$\dot{\tilde{\rho}}_{22} = -i\frac{\Omega}{2}(\tilde{\rho}_{12} - \tilde{\rho}_{21}) - \Gamma\tilde{\rho}_{22} = 0$$

Using the <u>stationary solution</u> of the optical Bloch equation for $\tilde{\rho}_{12}$ and $\tilde{\rho}_{22}$

$$\tilde{\rho}_{12} = i\frac{\Omega}{2}\frac{\frac{\Gamma}{2} - i\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{\Omega^2}{2}}$$
$$\tilde{\rho}_{22} = \frac{1}{2}\frac{s}{1+s}$$

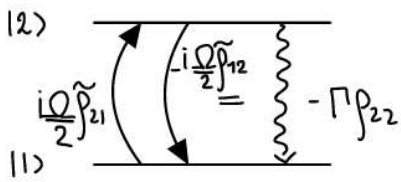
the rate due to stimulated emission and absorption can be written as

$$\dot{\tilde{\rho}}_{22}\Big|_{st} = -i\frac{\Omega}{2}(\tilde{\rho}_{12} - \tilde{\rho}_{21}) = -i\frac{\Omega}{2}(\tilde{\rho}_{12} - \tilde{\rho}^*_{12}) = -i\frac{\Omega}{2}2i\,\mathbb{Im}\{\tilde{\rho}_{12}\}$$
$$= \Omega\cdot\frac{\Omega}{2}\frac{\Gamma}{2}\frac{1}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{\Omega^2}{2}} = \frac{\Gamma}{2}\frac{s}{1+s}$$

and the rate due to spontaneous emission can be written as

$$\dot{\tilde{\rho}}_{22}\Big|_{sp} = -\Gamma\tilde{\rho}_{22} = -\Gamma\left(\frac{1}{2}\frac{s}{1+s}\right) = -\dot{\tilde{\rho}}_{22}\Big|_{st}$$

As expected, the overall stimulated processes and the spontaneous process balance each other. This dynamic describes the fluorescence or scattering cycle.



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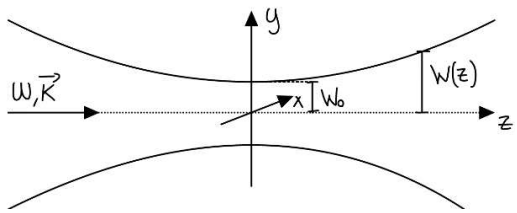
Dipole force



Consider a laser beam focused by a lens, with the reference system centred in the waist of the beam. At the waist ($z = 0$) of the laser beam

$$\vec{E}(x,y,z,t) = \vec{\varepsilon}E_0(x,y)\cos[\omega t + \varphi(\vec{r})] = \vec{\varepsilon}E_0' e^{-\frac{x^2+y^2}{2w_0^2}}\cos\left[\omega t - \vec{k}\cdot z\vec{u}_z\right]$$
$$= \vec{\varepsilon}\frac{E_0'}{2}e^{-\frac{x^2+y^2}{2w_0^2}}e^{-i\vec{k}\cdot z\vec{u}_z}e^{i\omega t} + c.c. = \vec{\varepsilon}\frac{\tilde{E}(\vec{r})}{2}e^{i\omega t} + c.c.$$

where we have defined $E_0' = E_0(0,0)$ and

$$\tilde{E}(\vec{r}) = E_0(x,y)e^{-i\vec{k}\cdot z\vec{u}_z} = E_0' e^{-\frac{x^2+y^2}{2w_0^2}}e^{-i\vec{k}\cdot z\vec{u}_z}$$

We expect to have both the resonant radiation pressure and dipole force, since we have both phase and amplitude modulation in space.

$$\vec{F}_{RRP} = \hbar\vec{k}\frac{\Gamma}{2}\frac{s}{1+s}$$

Using the relation

$$\vec{\nabla}(E_0^2) = \vec{\nabla}E_0\cdot E_0 + E_0\vec{\nabla}E_0 = 2E_0\vec{\nabla}E_0$$

we can write

$$\vec{F}_{DIP} = -\frac{\left|\mu_{\varepsilon,12}\right|^2}{2\hbar}\frac{\Delta\omega E_0(\vec{r})\vec{\nabla}E_0(\vec{r})}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}} = -\frac{\left|\mu_{\varepsilon,12}\right|^2}{2\hbar}\frac{\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}\frac{\vec{\nabla}(E_0^2)}{2}$$

Recall that to write

$$\hat{H}_I = -\hat{\mu}_\varepsilon\hat{E}(\vec{r},t) = -\hat{\mu}_\varepsilon\frac{\tilde{E}(\vec{r})}{2}e^{i\omega t} + c.c. = \frac{\hbar\hat{\Omega}}{2}e^{i\omega t} + c.c.$$

we need to define

$$\hat{\Omega} = -\frac{\hat{\mu}_\varepsilon\tilde{E}(\vec{r})}{\hbar}$$

Therefore

$$\Omega_{12} = -\frac{\mu_{\varepsilon,12}\tilde{E}(\vec{r})}{\hbar}$$

$$|\Omega_{12}|^2 = \frac{\left|\mu_{\varepsilon,12}\right|^2\left|\tilde{E}(\vec{r})\right|^2}{\hbar^2} = \frac{\left|\mu_{\varepsilon,12}\right|^2}{\hbar^2}E_0'^2\, e^{-\frac{x^2+y^2}{w_0^2}} = \frac{\left|\mu_{\varepsilon,12}\right|^2}{\hbar^2}E_0^2(x,y) \propto I(x,y)$$

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$$\vec{F}_{DIP} = -\frac{|\mu_{\varepsilon,12}|^2}{2\hbar}\frac{\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}\frac{\vec{\nabla}(E_0^2)}{2}$$

$$= -\frac{|\mu_{\varepsilon,12}|^2}{2\hbar}\frac{\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}\frac{\hbar^2}{|\mu_{\varepsilon,12}|^2}\frac{\vec{\nabla}(|\Omega_{12}|^2)}{2}$$

$$= -\frac{\hbar}{2}\frac{\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}\frac{\vec{\nabla}(|\Omega_{12}|^2)}{2} \propto -\frac{\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega_{12}|^2}{2}}\vec{\nabla}\big(I(x,y)\big)$$

where

$$I(x,y) = I_0 e^{-\frac{x^2+y^2}{w_0^2}}$$

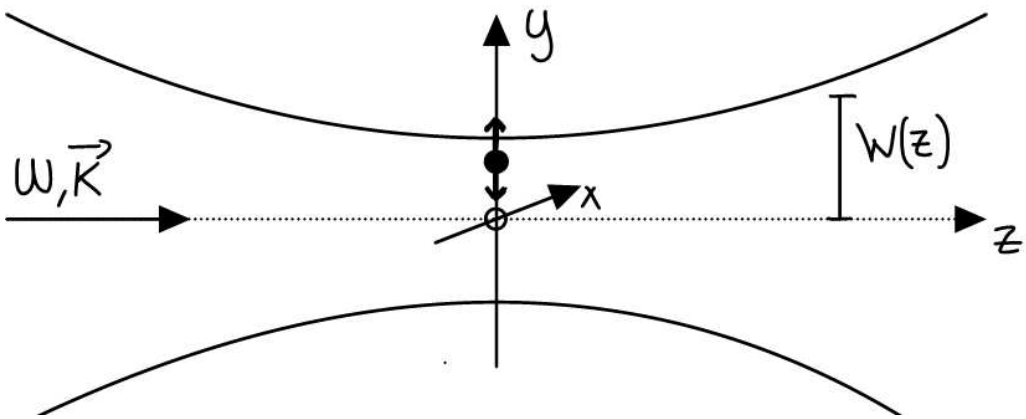
Geometrical interpretation of the gradient

$$dI(x,y) = \vec{\nabla}I(x,y)\cdot d\vec{s} = |\vec{\nabla}I||d\vec{s}|\cos\theta$$

The differential is maximum along the direction of the gradient, where $\theta = 0$ so that $\cos\theta = 1$. The gradient is therefore the direction of greatest variation of a function.

$$\boldsymbol{\vec{F}_{DIP} \propto -\Delta\omega\vec{\nabla}\big(I(x,y)\big)}$$

Therefore, when $\Delta\omega > 0$ the force tends to the direction which minimises the intensity, while when $\Delta\omega < 0$ the force tends to the direction which maximises the intensity.



In the $\Delta\omega < 0$ case, called red detuning, the force on the atom is always pointing towards the centre of the beam waist, where the intensity is maximum.

We can introduce the potential energy associated to the dipole force $\vec{F}_{dip}$.

$$\boldsymbol{E_p(\vec{r}) = \frac{\hbar}{2}\Delta\omega\ln\left[1 + \frac{\frac{|\Omega|^2}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\right]}$$

which describes a potential well for the atom in the case $\Delta\omega < 0$.

Let us show that this expression is correct:

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$$-\vec{\nabla}E_p(\vec{r}) = -\frac{\hbar}{2}\Delta\omega\vec{\nabla}\left[\ln\left(1+\frac{\frac{|\Omega|^2}{2}}{\left(\frac{\Gamma}{2}\right)^2+\Delta\omega^2}\right)\right]$$

$$= -\frac{\hbar}{2}\Delta\omega\frac{1}{1+\frac{\frac{|\Omega|^2}{2}}{\left(\frac{\Gamma}{2}\right)^2+\Delta\omega^2}}\frac{1}{\left(\frac{\Gamma}{2}\right)^2+\Delta\omega^2}\vec{\nabla}\left(\frac{|\Omega|^2}{2}\right)$$

$$= -\frac{\hbar}{2}\frac{\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2+\Delta\omega^2+\frac{|\Omega|^2}{2}}\vec{\nabla}\left(\frac{|\Omega|^2}{2}\right) = \vec{F}_{DIP}$$

In the general three-dimensional case $z \neq 0$ it holds

$$w(z) = w_0\left[1+\left(\frac{z}{z_R}\right)^2\right]^{1/2}$$

$$\vec{E}(x,y,z) = \vec{\varepsilon}\frac{E_0'}{2}e^{-\frac{x^2+y^2}{2w^2(z)}}e^{-ikz}e^{i\omega t} + c.c.$$

$$I(x,y,z) \propto \left|\vec{E}(x,y,z)\right|^2 = {E_0'}^2 e^{-\frac{x^2+y^2}{w^2(z)}}$$

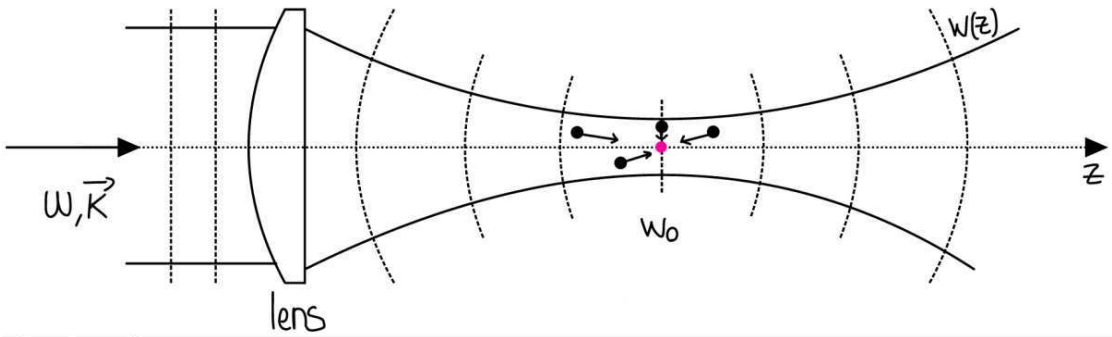
$$\vec{F}_{DIP} \propto -\Delta\omega\vec{\nabla}\big(I(\vec{r})\big)$$

In the case of red detuning, $\Delta\omega < 0$, the force has the same direction as $\vec{\nabla}\big(I(\vec{r})\big)$.

Practical implementation: laser trap

A large collimated laser beam, almost a plane wave, with $\omega < \omega_0$, impinges on a focusing lens.

The dipole force on the atom is always directed to the point of highest intensity of the beam, which is the centre of the beam waist.



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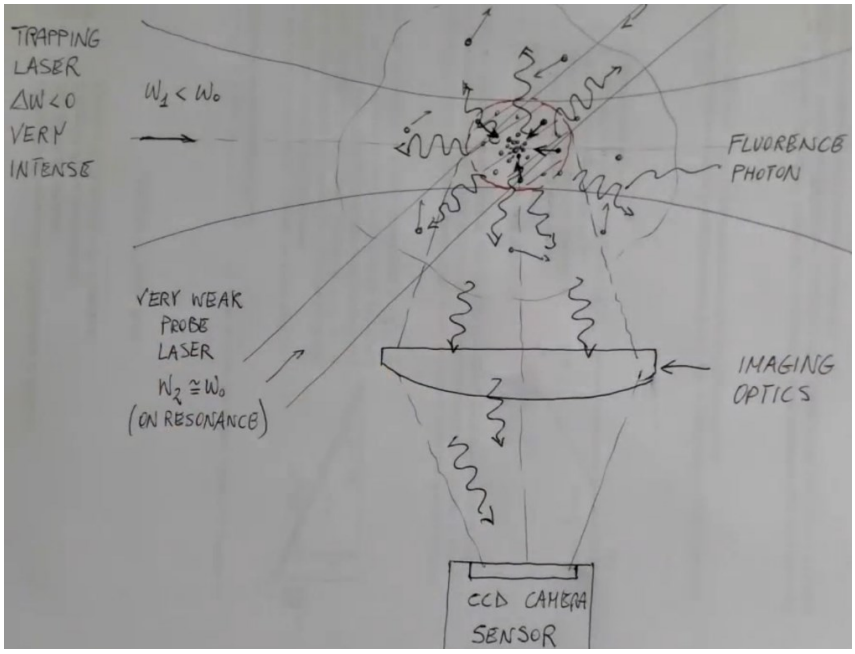
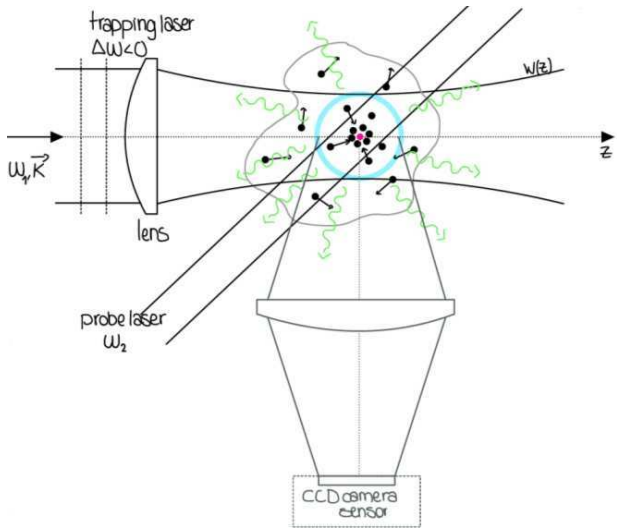
Optical tweezer



Consider a cloud of atoms moving in all directions and a trapping laser consisting of a very intense laser beam with frequency $\omega_1 < \omega_0$, so that $\Delta\omega < 0$. In the region around the centre of the laser beam the intensity is strong and causes a force on the atoms that brings them to the centre. Thus, the density of atoms at the centre of the trap increases. We can use an additional weak laser beam, called <u>probe laser</u>, coming from a different direction at a frequency $\omega_2 \simeq \omega_0$ to detect the atoms: we excite them to the higher energy level and measure the isotropic fluoresce that follows. The measurement uses an objective to collect the photons, followed by a CCD camera sensor, which will show a bright spot in the region around the centre of the beam, in which the density of atoms is highest. The presence of atoms in that region is in this way detected.



Notice that:

1) we can trap atoms at <u>a few hundreds of mK</u>;
2) optical tweezers are very common;
3) if we are able to move the focus of the beam we can move the cloud of trapped atoms from one place to another.

Physical intuitive picture of $\vec{F}_{DIP}$

Consider once again an atom in a laser beam.

$$\vec{E}(x,y,z) = \vec{\varepsilon}E(x,y,z) \quad \vec{\varepsilon} = \vec{u}_y$$

$$\hat{H}_I = -\vec{\mu}_e \cdot \vec{E}$$

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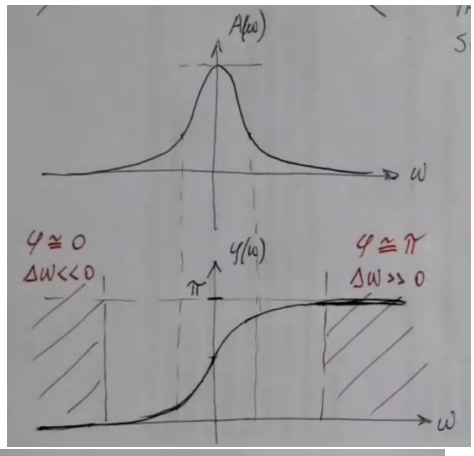
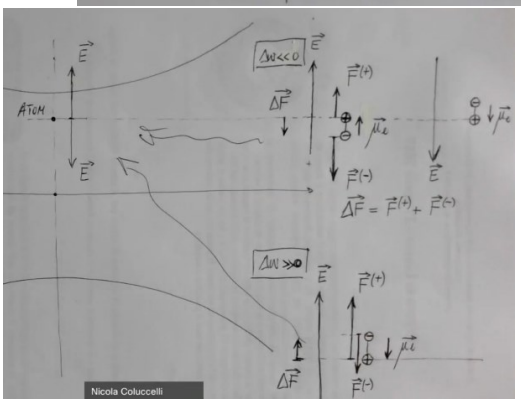
If we consider the atom as a classical oscillator subject to a driving force, we can model the electric dipole moment as a function of frequency with resonance at $\omega = \omega_0$:

$$\vec{\mu}_e(\omega) = \vec{\mu}_{e0} A(\omega) e^{i\varphi(\omega)}$$

Since it is resonant, the amplitude follows a Lorentzian profile. When the frequency of the driving field is low, the system can keep up and is in phase ($\varphi = 0$). When the frequency of the driving field is much higher than that of the oscillator, it can't keep up and oscillates out of phase with the driving field ($\varphi = \pi$).

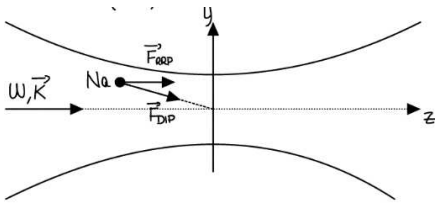


The force is stronger for the charge closer to the centre of the beam. In the case $\Delta\omega \ll 0$, the electric dipole moment is always in phase with the electric field, thus the net force is directed towards the centre of the beam. In the case $\Delta\omega \gg 0$ the electric dipole is out of phase with the electric field, thus the net force is directed away from the centre.



Practical implementation with sodium atoms

Consider the transition of sodium atoms at $\lambda_0 \simeq 589\ nm$. As a laser consider $Nd{:}YAG$ with $\lambda \simeq 1064\ nm$.



The force acting on an atom is:

$$\vec{F} = \vec{F}_{DIP} + \vec{F}_{RRP} = -\frac{\hbar}{2}\frac{\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega|^2}{2}}\vec{\nabla}\left(\frac{|\Omega|^2}{2}\right) + \hbar\vec{k}\frac{\Gamma}{2}\frac{s}{1+s}$$

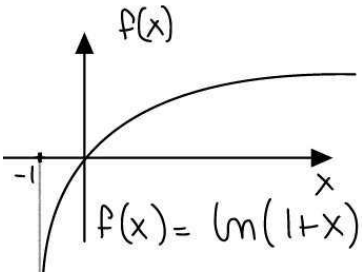
The frequency detuning is

$$\Delta\omega = \omega - \omega_0 = 2\pi(\nu - \nu_0) = 2\pi c\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right) \simeq -4\pi \cdot 10^{14}\ rad/s \ll 0$$

We recall that the saturation intensity is $I_s = 6.6\ mW/cm^2$.

The potential well has the expression

$$E_p(\vec{r}) = \frac{\hbar}{2}\Delta\omega \ln\left[1 + \frac{\frac{|\Omega|^2}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\right]$$



For $r \gg w_0$ it holds $I(r) \simeq 0$, thus $|\Omega|^2 \propto I(r) = 0$ which implies $E_p(r) = 0$. For $r < w_0$ it holds $I(r) \neq 0$, thus $|\Omega|^2 \propto I(r) \neq 0$ which, keeping in mind that $\Delta\omega < 0$, implies $E_p(r) < 0$.

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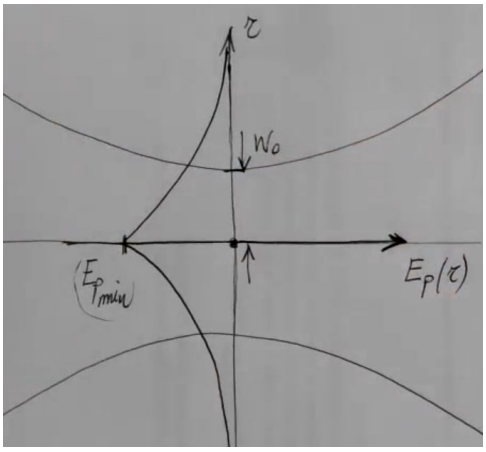
On axis, that is for $r = 0$, the intensity is maximum and the energy will reach a minimum value $E_p^{\min} < 0$. In $r = 0$ it holds:



$$I(0) = I_{\max} \quad \rightarrow \quad |\Omega_{12}|^2(0) = |\Omega_{12}|^2_{\max}$$

$$2\frac{|\Omega_{12}|^2}{\Gamma^2} = s_0 = \frac{I(r)}{I_s}$$

$$\Rightarrow \quad |\Omega_{12}|^2_{\max} = s_0|_{\max}\frac{\Gamma^2}{2} = \frac{I(0)}{I_s}\frac{\Gamma^2}{2} = \frac{I_{\max}}{I_s}\frac{\Gamma^2}{2}$$

Considering a laser with $P_{laser} \simeq 1\ mW$ and spot size $w_0 = 10\ \mu m$, the peak intensity is

$$I_{\max} = \frac{P}{A_{eff}} = \frac{P}{\pi r_{\text{eff}}^2} = \frac{P}{\pi\left(w_0/\sqrt{2}\right)^2} = \frac{2P}{\pi w_0^2}$$
$$\simeq 6.6 \cdot 10^{10}\ W/m^2$$

Therefore

$$s_0|_{\max} = \frac{I_{\max}}{I_s} \simeq 10^8$$

$$|\Omega_{12}|^2_{\max} = s_0|_{\max}\frac{\Gamma^2}{2} \simeq 18 \cdot 10^{22}\ rad^2/s^2$$

$$E_p^{\min} = E_p(0) = \frac{\hbar}{2}\Delta\omega \ln\left[1 + \frac{\frac{|\Omega_{12}|^2_{\max}}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2}\right]$$

Notice that

$$\frac{\frac{|\Omega_{12}|^2_{\max}}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2} \simeq 10^{-7} \ll 1$$

thus we can use the relation $\ln(1 + x) \simeq x$ to estimate the depth of the optical trap

$$E_p^{\min} \simeq \frac{\hbar}{2}\Delta\omega\frac{\frac{|\Omega_{12}|^2_{\max}}{2}}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2} \simeq -2.2 \cdot 10^{-26}\ J$$

Atoms can be trapped provided that they have very low temperature.

The maximum temperature can be estimated as

$$\frac{3}{2}kT = \langle E_k \rangle = \frac{1}{2}M\langle v^2 \rangle = \frac{1}{2}Mv_{rms}^2$$

$$T = \frac{2}{3}\frac{\langle E_k \rangle}{k} = \frac{2}{3}\frac{|E_p(0)|}{k} \simeq \mathbf{1\ mK}$$

Therefore, if we put our optical trap at the exit of an oven, we will trap only the atoms coming out at a speed corresponding to a temperature lower than $1\ mK$.

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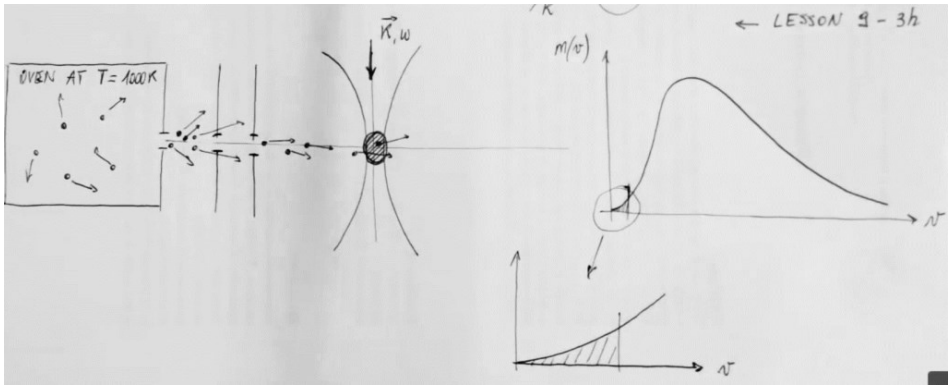
Let us consider an oven at $T = 1000\ K$ (we need high temperatures to obtain a substantial evaporation of sodium atoms). As atoms pass through the trap, if their kinetic energy is lower than the depth of the optical trap they get trapped.

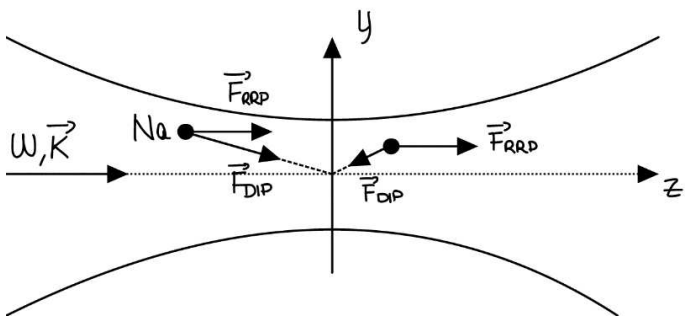


Given the Maxwell-Boltzmann distribution of velocity, only the atoms in a small low velocity interval will get trapped. This device is very simple, but has a **low efficiency**. It is used often in laboratories.

The overall force is

$$\vec{F} = \vec{F}_{DIP} + \vec{F}_{RRP} = -\frac{\hbar}{2}\frac{\Delta\omega}{\left(\frac{\Gamma}{2}\right)^2 + \Delta\omega^2 + \frac{|\Omega|^2}{2}}\vec{\nabla}\left(\frac{|\Omega|^2}{2}\right) + \hbar\vec{k}\frac{\Gamma}{2}\frac{s}{1+s}$$

In the case of a very large red detuning ($\Delta\omega \ll 0$), the situation is the one shown in figure.



$$\vec{F}_{RRP} = \hbar\vec{k}\frac{\Gamma}{2}\frac{s}{1+s} = \hbar\vec{k}\Gamma_p$$

$$\Gamma_p = \frac{\Gamma}{2}\frac{s}{1+s} \quad \Gamma_{\max} = \frac{\Gamma}{2}\frac{s_{\max}}{1+s_{\max}}$$

Since in our example the detuning is very large, the $s_{\max}$ is very small:

$$s_{\max} = \frac{s_{0,\max}}{1+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2} \simeq \frac{10^8}{1+\left(\frac{-4\pi\cdot 10^{14}\ rad/s}{2\pi\cdot 10^7/2}\right)^2} \simeq \frac{1}{16}10^{-6} \ll 1$$

Therefore

$$\Gamma_{\max} = \frac{\Gamma}{2}\frac{s_{\max}}{1+s_{\max}} \simeq \frac{\Gamma}{2}s_{\max} \simeq 2\ s^{-1}$$

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We have thus demonstrated that the maximum scattering rate is in the order of two cycles per second, which is very low. The resonant radiation pressure force, therefore, is negligible.

If the detuning were low, than the saturation parameter would be high and $\vec{F}_{RRP}$ would not be negligible.

Atom mirrors

Consider now <u>blue detuning</u> ($\Delta\omega > 0$). Consider a laser beam impinging on the interface between glass and air at an angle big enough to have total internal reflection.

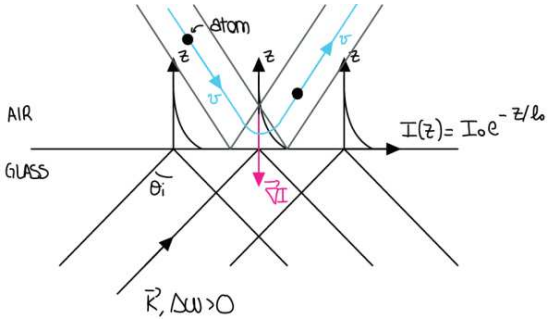
In air the field is evanescent and there is no propagation, thus the intensity follows the law

$$I(z) = I_0 e^{-\frac{z}{L}}$$

Since

$$\vec{F}_{DIP} \propto -\Delta\omega \vec{\nabla} I$$

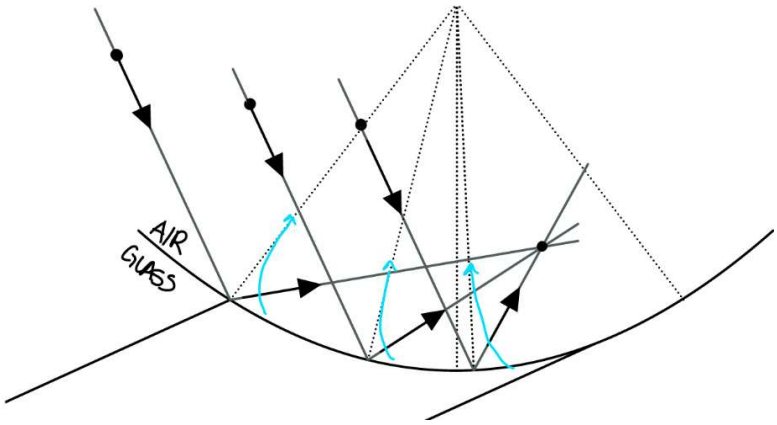
and $\Delta\omega > 0$, the force is opposite to $\vec{\nabla} I$.



The effect of this field on a beam of atoms coming from air is then a repulsion, which reverses the velocity along z. We have obtained a mirror for the atoms.

Focusing atom mirror

If we have a curved interface between glass and air, with a laser beam coming from the glass side, the interface behaves as a focusing mirror for atoms coming from the air side.



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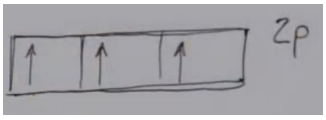
Energy level structure of alkali atoms

Alkali atoms are very important for many of the applications we are going to see. They are:

Lithium	Li	$[He]2s$
Sodium	Na	$[Ne]3s$
Potassium	K	$[Ar]4s$
Rubidium	Rb	$[Kr]5s$
Caesium	Cs	$[Xe]6s$

Their structure consists of a closed (filled) shell and a <u>single valence electron</u>.

The closed shell has $\vec{L} = 0$ and $\vec{S} = 0$. For example, Na atoms have configuration $[1s^2 2s^2 2p^6]3s$. The s orbitals have spherical symmetry, thus the associated charge distribution $\propto \left|\psi_{1s,2s}(r,\theta,\varphi)\right|^2$ has spherical symmetry and $\vec{l} = 0$. The six p orbitals give each $l = 1$. We recall that



$$\psi_{nlm}(r,\theta,\varphi) = R_{nl}(r)Y_{lm}(\theta,\varphi)$$

Let us start by considering three of the electrons:

$$\begin{aligned} l_1 &= 1 \quad m_1 = 1 \quad \rightarrow \quad \psi_{2,1,-1} = R_{2,1}Y_{1,-1} \\ l_2 &= 1 \quad m_2 = 0 \quad \rightarrow \quad \psi_{2,1,0} = R_{2,1}Y_{1,0} \\ l_3 &= 1 \quad m_3 = -1 \quad \rightarrow \quad \psi_{2,1,1} = R_{2,1}Y_{1,1} \end{aligned}$$

$$|\psi|^2 = \left|R_{2,1}\right|^2\left|Y_{1,-1}\right|^2 + \left|R_{2,1}\right|^2\left|Y_{1,0}\right|^2 + \left|R_{2,1}\right|^2\left|Y_{1,1}\right|^2 = \left|R_{2,1}\right|^2\left(\left|Y_{1,-1}\right|^2 + \left|Y_{1,0}\right|^2 + \left|Y_{1,1}\right|^2\right)$$

However, the sum in the brackets is a spherical distribution, thus the angular momentum is zero. The same exact result is obtained when considering the remaining three electrons, therefore the overall angular momentum is $\vec{L} = 0$.

Moreover, since all orbitals in the closed shell are filled, all electrons are in pairs and $\vec{S} = 0$.

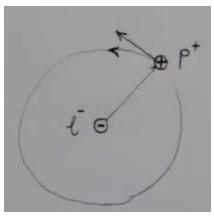
The total angular momentum and spin of the atom are therefore determined by the outer electron:

$$\vec{L} = \vec{l}_{\text{outer } e^-}$$
$$\vec{S} = \vec{s}_{\text{outer } e^-}$$

Spin-orbit interaction

We need to consider the fine structure of alkaline atoms (LS coupling).

Let us consider hydrogen: we can see the system as a proton rotating around the electron. The magnetic field generated by this current interacts with the magnetic moment associated to the spin angular momentum of the electron.



The energy of the interaction of a magnetic field with a (generic) magnetic dipole moment is:

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$$\boldsymbol{\Delta E = -\vec{\mu}_m \cdot \vec{B}}$$

In our case the magnetic dipole moment is that associated to the spin of the electron, thus

$$\Delta E_{LS} = -\vec{\mu}_s \cdot \vec{B}$$

The magnetic field generated by the rotating proton is

$$\vec{B}(z) = \frac{\mu_0}{2} \frac{R^2 I}{[z^2 + R^2]^{3/2}} \vec{u}_z$$

$$\vec{B}(0) = \frac{\mu_0}{2} \frac{R^2 I}{R^3} = \frac{\mu_0 I}{2R} \vec{u}_z$$

Since

$$\vec{L} = \vec{r} \times \vec{p} = Rmv\, \vec{u}_z$$

we can write

$$\vec{\mu}_m = IA\vec{u}_z = \frac{q}{T} A\vec{u}_z = \frac{q}{2\pi R/v} \pi R^2 \vec{u}_z = \frac{qRv}{2} \vec{u}_z = \frac{q}{2m} \vec{L}$$

$$\vec{B}(0) = \frac{\mu_0 I}{2R} \vec{u}_z = \frac{\mu_0}{2RA} \vec{\mu}_m = \frac{\mu_0}{2RA} \frac{q}{2m} \vec{L} \propto \vec{L}$$

In analogy with $\vec{\mu}_m$, the magnetic spin momentum $\vec{\mu}_s$ can be written as

$$\vec{\mu}_s = g_s \frac{q}{2m} \vec{S} = -g_s \frac{e}{2m} \vec{S}$$

where g_s is called giromagnetic factor.

Therefore

$$\Delta E_{LS} = -\vec{\mu}_s \cdot \vec{B} = a_{LS} \vec{S} \cdot \vec{L}$$

where a_{LS} depends on the specific values of $\vec{L}$ and $\vec{S}$.

Fine structure (LS coupling)

When there are both $\vec{L}$ and $\vec{S}$, then LS coupling takes place.

$$\vec{J} = \vec{L} + \vec{S}$$

$$\left|\vec{J}\right|^2 = j(j+1)\hbar^2, \qquad |l - s| \leq j \leq l + s$$

$$J_z = m_j \hbar, \qquad m_j = -j, \dots, j$$

The energy level structure of alkaline atoms can be written using the <u>Russell-Saunders notation</u>

$$n\ {}^{2s+1}L_j$$

For the sodium atom:

$$n = 3, \qquad l = 0,1,2, \qquad s = 1/2$$

thus the possible combinations are:

1) $n = 3 \quad l = 0 \quad s = 1/2 \quad \rightarrow \quad j = 1/2$

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$$3\ ^2S_{1/2}$$

2) $n = 3 \quad l = 1 \quad s = 1/2 \quad \rightarrow \quad 1/2 \leq j \leq 3/2$

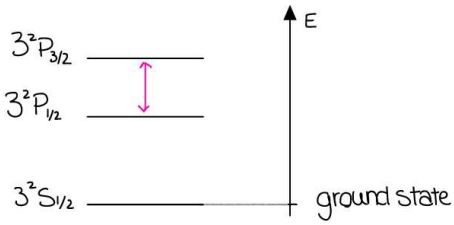
$$3\ ^2P_{1/2} \quad 3\ ^2P_{3/2}$$

3) $n = 3 \quad l = 2 \quad s = 1/2 \quad \rightarrow \quad 3/2 \leq j \leq 5/2$

$$3\ ^2D_{3/2} \quad 3\ ^2D_{5/2}$$

The three lowest energy levels are presented in figure.

We can easily show for example that the energy of $3\ ^2P_{1/2}$ is lower than that of $3\ ^2P_{3/2}$. The coupling energy is $\Delta E_{LS} = a_{LS}\vec{S} \cdot \vec{B}$: when $j = 1/2$ it means that $\vec{S}$ and $\vec{B}$ are antiparallel, thus the energy contribution is negative; when $j = 3/2$ they are parallel and their contribution is positive.



Hyperfine structure (JI coupling)

We can also take into account the <u>coupling of the *overall* angular momentum with</u> the **nuclear spin angular momentum $\vec{I}$**.

The nucleus of an atom may have a spin angular momentum $\vec{I}$. In the case of <u>sodium</u>, for example, <u>$I = 3/2$</u>.

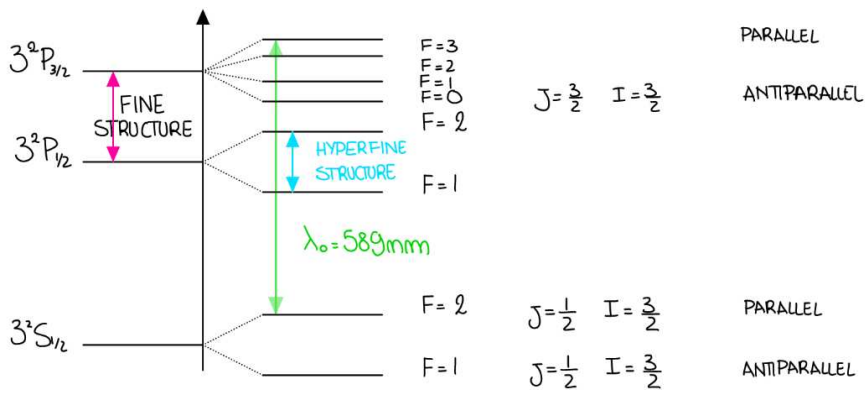
$$\vec{F} = \vec{I} + \vec{J}$$

$$\left|\vec{F}\right|^2 = F(F+1)\hbar^2, \qquad |J - I| \leq F \leq J + I$$

$$F_z = M_F\hbar, \qquad M_F = -F, \dots, F$$

$$\Delta E_{HFS} = b_{JI}\vec{I} \cdot \vec{J}$$

With this level of precision, we can show in the figure the cycling transition used for laser cooling of sodium atoms.



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Zeeman effect

The Zeeman effect is a further removal of degeneracy induced by the presence of a magnetic field applied to the atom: for different values of M_F there are different values of the projection of the overall angular momentum along the direction defined by the magnetic field. The splitting of the energy levels due to the Zeeman effect is

$$\Delta E_{\text{Zeeman}} = -\vec{\mu}_m \cdot \vec{B}$$

where $\vec{B} = B_z \vec{u}_z$, while $\vec{\mu}_m$ can be expressed using a generalisation to $\vec{F}$ of the relation used for $\vec{L}$ and $\vec{S}$:

$$\vec{\mu}_m = g_F \frac{q}{2m} \vec{F} = -g_F \frac{e}{2m} \vec{F}$$

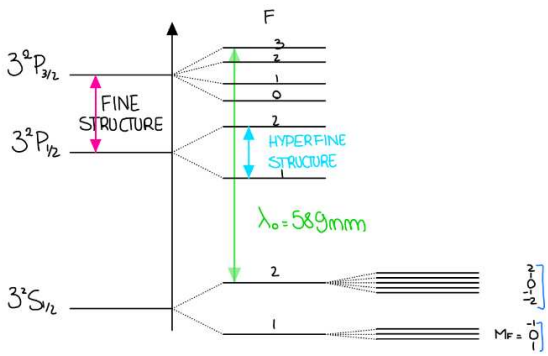
Therefore

$$\boldsymbol{\Delta E_{\text{Zeeman}}} = -\vec{\mu}_m \cdot \vec{B} = g_F \frac{e}{2m} \vec{F} \cdot \vec{B} = g_F \frac{e}{2m} (\hbar M_F) B_z = \boldsymbol{\mu_B g_F M_F B_z}$$

where we have introduced the Bohr magneton

$$\mu_B = \frac{e\hbar}{2m}$$

Since M_F spans between $\pm F$, there are <u>$2F + 1$ different levels</u> for each unperturbed state.



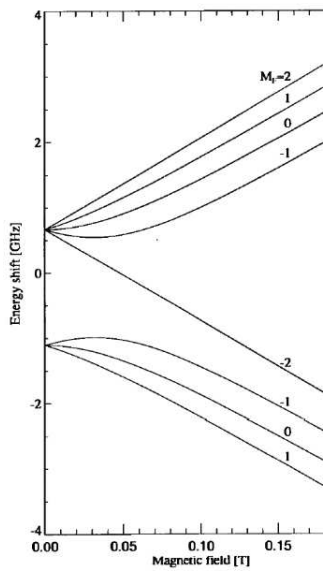
As we can see in the figures, the energy ordering of these levels in not fixed, since it depends on the sign of the Landé g-factor

$$g_F = g_j \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}$$

where

$$g_j = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{J(J+1)}$$

The figure shows the energy levels of the ground state of sodium, $3\ ^2S_{1/2}$. We can see that the levels split linearly with the amplitude of the magnetic field when the field is low, while the behaviour becomes non-linear when the field increases. These results are consistent with both calculations and measurements.



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Selection rules and polarisation of light

We are going to consider the selection rules in the context of the experiments of laser cooling, trapping and atom optics. We also want to provide a way to accurately represent the transitions and the selection rules.

Consider a <u>two-level system</u>:

$$\hat{H}_0|a\rangle = E_a|a\rangle$$
$$\hat{H}_0|b\rangle = E_b|b\rangle$$

The perturbation is given by the electric dipole interaction

$$\hat{H}_I = -\hat{\vec{\mu}}_e \cdot \vec{E}(\vec{r},t)$$

Since in general

$$\langle a|\hat{H}_I|b\rangle \neq 0$$

the transition between $|a\rangle$ and $|b\rangle$ is possible, but we still need to check if it is allowed.

In the <u>electric dipole approximation</u>

$$\hat{H}_I = -\hat{\vec{\mu}}_e \cdot \vec{E}(\vec{r},t) \simeq -\hat{\vec{\mu}}_e \cdot \vec{E}(0,t)$$

Under this assumption, and using the more compact notation $\vec{E}(t) = \vec{E}(0,t)$, the matrix element is

$$\langle a|\hat{H}_I|b\rangle = -\langle a|\hat{\vec{\mu}}_e \cdot \vec{E}(t)|b\rangle = -\langle a|\hat{\vec{\mu}}_e|b\rangle\vec{E}(t) = -\vec{\mu}_{ab}\vec{E}(t)$$

For the hydrogen atom, since there is only one electron, it holds $\hat{\vec{\mu}}_e = -e\hat{\vec{r}}$, thus

$$\vec{\mu}_{ab} = \langle a|\hat{\vec{\mu}}_e|b\rangle = -e\langle a|\hat{\vec{r}}|b\rangle = -e\vec{r}_{ab} = -e\int \phi_a^*(\vec{r})\vec{r}\phi_b(\vec{r})d\vec{r}$$

If $|a\rangle$ and $|b\rangle$ have the same parity, then the transition is forbidden, while if they have opposite parity it is possible, but subject to additional selection rules.

For the problem we are dealing with it will be useful to express the unit polarisation vector using the spherical basis

$$\vec{\boldsymbol{u}}_- = \frac{\mathbf{1}}{\sqrt{\mathbf{2}}}(\vec{\boldsymbol{u}}_x - \boldsymbol{i}\vec{\boldsymbol{u}}_y) = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-i\\0\end{bmatrix} \qquad \vec{u}_x = \frac{1}{\sqrt{2}}(-\vec{u}_+ + \vec{u}_-) = \frac{1}{\sqrt{2}}\begin{bmatrix}-1\\1\\0\end{bmatrix}$$

$$\vec{\boldsymbol{u}}_{\mathbf{0}} = \vec{\boldsymbol{u}}_{\boldsymbol{z}} = \begin{bmatrix}0\\0\\1\end{bmatrix} \qquad \leftrightarrow \qquad \vec{u}_0 = \vec{u}_z = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

$$\vec{\boldsymbol{u}}_+ = \frac{-\mathbf{1}}{\sqrt{\mathbf{2}}}(\vec{\boldsymbol{u}}_x + \boldsymbol{i}\vec{\boldsymbol{u}}_y) = \frac{1}{\sqrt{2}}\begin{bmatrix}-1\\-i\\0\end{bmatrix} \qquad \vec{u}_y = \frac{1}{\sqrt{2}}(i\vec{u}_+ + i\vec{u}_-) = \frac{1}{\sqrt{2}}\begin{bmatrix}i\\i\\0\end{bmatrix}$$

Indeed, the electric field can be expressed as

$$\vec{E}(\vec{r},t) \simeq \vec{E}(0,t) = \vec{\varepsilon}E_0\cos(\omega t) = \vec{\varepsilon}\frac{E_0}{2}e^{i\omega t} + c.c.$$

where $\vec{\varepsilon} = \vec{u}_0 = \vec{u}_z$ for linearly polarised light, $\vec{\varepsilon} = \vec{u}_+$ for right-handed circularly polarised light and $\vec{\varepsilon} = \vec{u}_-$ for left-handed circularly polarised light.

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Linearly polarised light

Hydrogen atom

In the case of linearly polarised light interacting with an hydrogen atom

$$\langle a|\hat{H}_I|b\rangle = \langle a|-\hat{\vec{\mu}}_e \cdot \vec{E}(t)|b\rangle = -\langle a|\hat{\vec{\mu}}_e \cdot \vec{\varepsilon}|b\rangle E(t) = -\langle a|-e\hat{\vec{r}} \cdot \vec{\varepsilon}|b\rangle E(t) = e\langle a|\hat{z}|b\rangle E(t)$$
$$= e z_{ab} E_0 \cos(\omega t)$$

where we have also found that

$$\mu_{\varepsilon,ab} = \langle a|\hat{\vec{\mu}}_e \cdot \vec{\varepsilon}|b\rangle = -e z_{ab}$$

It holds

$$z_{ab} = \langle a|\hat{z}|b\rangle = \int \phi_a^*(\vec{r}) z \phi_b(\vec{r}) d\vec{r}$$

where for the hydrogen atom

$$\phi_a(\vec{r}) = \phi_{nlm}(\vec{r}) = R_{nl}(r) Y_{lm}(\theta, \varphi)$$

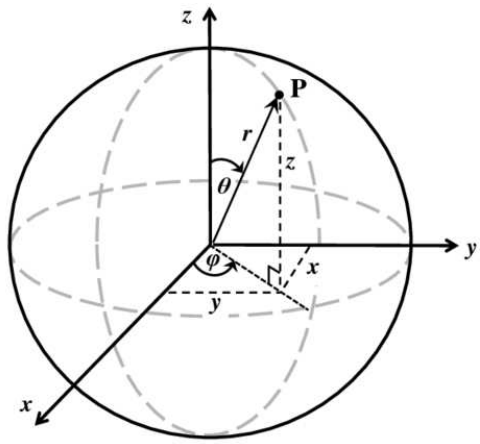
Assume that $|a\rangle = |n = 1, l = 0, m = 0\rangle = |100\rangle$ is the ground state and $|b\rangle = |nlm\rangle$ is an excited state. Notice that

$$Y_{00}(\theta, \varphi) = \frac{1}{\sqrt{4\pi}}$$

$$Y_{10}(\theta, \varphi) = \sqrt{\frac{3}{4\pi}} \cos\theta$$

$$Y_{11}(\theta, \varphi) = -\sqrt{\frac{3}{8\pi}} \sin\theta \, e^{i\varphi}$$

$$Y_{1,-1}(\theta, \varphi) = \sqrt{\frac{3}{8\pi}} \sin\theta \, e^{-i\varphi}$$



therefore

$$z = r\cos\theta = r\sqrt{\frac{4\pi}{3}} Y_{10}(\theta, \varphi)$$

We can thus calculate z_{ab} as

$$z_{ab} = \langle a|\hat{z}|b\rangle = \int R_{10}^*(r) Y_{00}^*(\theta, \varphi) z R_{nl}(r) Y_{lm}(\theta, \varphi) r^2 \sin\theta \, dr d\theta d\varphi$$
$$= \int R_{10}^*(r) \frac{1}{\sqrt{4\pi}} \left[r \sqrt{\frac{4\pi}{3}} Y_{10}(\theta, \varphi) \right] R_{nl}(r) Y_{lm}(\theta, \varphi) r^2 \sin\theta \, dr d\theta d\varphi$$
$$= \frac{1}{\sqrt{3}} \int R_{10}^*(r) r Y_{10}(\theta, \varphi) R_{nl}(r) Y_{lm}(\theta, \varphi) r^2 \sin\theta \, dr d\theta d\varphi$$
$$= \frac{1}{\sqrt{3}} \int R_{10}^*(r) r^3 R_{nl}(r) dr \int Y_{\mathbf{10}}(\theta, \varphi) Y_{\boldsymbol{lm}}(\theta, \varphi) \sin\theta \, d\theta d\varphi = \frac{1}{\sqrt{3}} \mathcal{I}_{rad} \mathcal{I}_{ang}$$

There is <u>no general selection rule acting on the radial integral</u> $\mathcal{I}_{rad}$, which is in general different from zero. The <u>orthogonality relations</u> of the spherical harmonics

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impose some conditions on the angular integral $\mathcal{I}_{ang}$, which determine some selection rules:

$$\boldsymbol{\mathcal{I}_{ang} = \delta_{1l}\delta_{0m}}$$

For linearly polarised electric field, therefore, the transition is allowed only if $l = 1$ and at the same time $m = 0$:

$$|a\rangle = |100\rangle \quad \rightarrow \quad |b\rangle = |n10\rangle$$

$$\Delta n = n_f - n_i \text{ free}, \qquad \Delta l = l_f - l_i = +1, \qquad \Delta m = m_f - m_i = 0$$

In the general case

$$|a\rangle = |n_i l_i m_i\rangle \quad |b\rangle = |n_f l_f m_f\rangle$$

with an electric field linearly polarised along the quantisation axis we get the following selection rules:

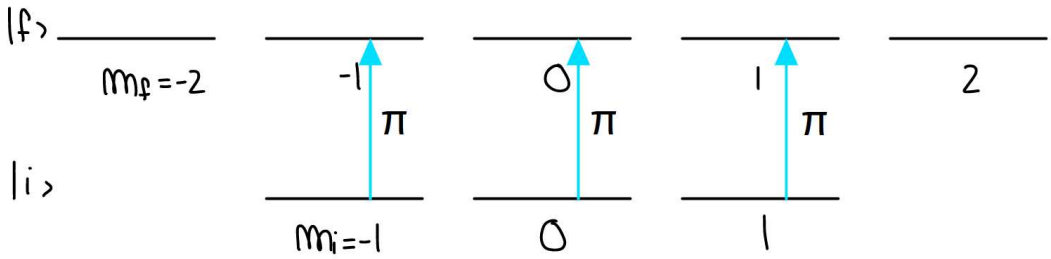
$$\boldsymbol{\Delta n \text{ free}, \qquad \Delta l = \begin{cases} +1 \text{ (absorption)} \\ -1 \text{ (emission)} \end{cases}, \qquad \Delta m = 0}$$

These are called $\boldsymbol{\pi}$ **transitions**. As an example, the allowed π absorption transitions for a system with states

$$|i\rangle = |n_i; l_i = 1; m_i = 0, \pm 1\rangle$$

$$|f\rangle = |n_f; l_f = 2; m_f = 0, \pm 1, \pm 2\rangle$$

are reported in figure.



We can analyse our results in light of the <u>conservation law</u> of angular momentum.

Let us consider light polarised along the z direction, and propagating in the xy plane, and an atom in the state $|a\rangle = |100\rangle$. The atom's angular momentum is

$$\left|\vec{L}_i\right|^2 = l_i(l_i + 1)\hbar^2 = 0 \quad L_{zi} = m_i\hbar = 0$$

After an absorption process, the atom is in a state $|b\rangle = |n10\rangle$ and has angular momentum

$$\left|\vec{L}_f\right|^2 = l_f(l_f + 1)\hbar^2 = 2\hbar^2 \quad L_{zf} = m_f\hbar = 0$$

The law of conservation of angular momentum implies that the absorbed photon had an angular momentum

$$\boldsymbol{\left|\vec{L}_{ph}\right|^2 = 2\hbar^2 \quad L_{z,ph} = 0}$$

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Multi-electron atoms

We also need to consider the case of multi-electron atoms. As before, we need to consider the interaction Hamiltonian

$$\hat{H}_I = -\hat{\vec{\mu}}_e \cdot \vec{E}(\vec{r}, t)$$

To see if a transition is allowed we must evaluate the matrix element

$$\langle a|\hat{H}_I|b\rangle = -\langle a|\hat{\vec{\mu}}_e \cdot \vec{E}(t)|b\rangle = -\langle a|\hat{\vec{\mu}}_e \cdot \vec{\varepsilon}|b\rangle E(t) = -\mu_{\varepsilon,ab} E(t)$$

The relation $\hat{\vec{\mu}} = -e\hat{\vec{r}}$ doesn't hold in this case and the matrix element $\mu_{\varepsilon,ab}$ cannot be calculated in closed form, therefore it must be computed numerically.

We define the **transition** (or line) **strength** as the square modulus of the projection of $\vec{\mu}_{ab}$ on the polarisation direction $\vec{\varepsilon}$:

$$S = |\mu_{\varepsilon,ab}|^2$$

It is important to consider the **LS coupling**.

$$\vec{J} = \vec{L} + \vec{S}$$

For π transitions we get a generalisation of the selection rules found before:

$$\Delta J = \begin{cases} \pm 1 \ (\Delta S = 0) \\ 0 \ \ (\Delta S = \pm 1) \end{cases}, \qquad \Delta L = \begin{cases} +1 \ (\text{absorption}) \\ -1 \ \ (\text{emission}) \end{cases}, \qquad \Delta m_J = 0$$

Indeed, since $\vec{J} = \vec{L} + \vec{S}$, if $\vec{S}$ doesn't switch then the change in $\vec{J}$ is determined by the one in $\vec{L}$. Therefore, the transitions with $\Delta J = \pm 1$ correspond to those with $\Delta L = \pm 1$ respectively (absorption and emission respectively). The transition with $\Delta J = 0$, instead, also requires a change in spin, $\Delta S = \pm 1$, therefore it is weak.

We then need to consider the **JI coupling**.

$$\vec{F} = \vec{J} + \vec{I} = \vec{L} + \vec{S} + \vec{I}$$

For π transitions we get:

$$\boldsymbol{\Delta F = \pm 1, 0}, \qquad \boldsymbol{\Delta J = \pm 1, 0}, \qquad \boldsymbol{\Delta L = \begin{cases} +1 \ (\text{absorption}) \\ -1 \ \ (\text{emission}) \end{cases}}, \qquad \boldsymbol{\Delta M_F = 0}$$

where the transition with $\Delta J = 0$ is weak. ~~Since I is fixed, the transitions with $\Delta J = \pm 1,0$ correspond to $\Delta F = \pm 1,0$ respectively~~.

The ΔM_F selection rule comes into play only if there is Zeeman splitting.

Example: cycling transition of sodium atoms

Consider the transition used for laser cooling of sodium atoms ($I = 3/2$):

$$3\,^2S_{1/2}\ (F = 2) \quad \leftrightarrow \quad 3\,^2P_{3/2}\ (F = 3)$$

$$L_i = 0, \qquad J_i = 1/2, \qquad F_i = 2 \quad \leftrightarrow \quad L_f = 1, \qquad J_f = 3/2, \qquad F_f = 3$$

$$\Delta L = +1, \qquad \Delta J = +1, \qquad \Delta F = +1$$

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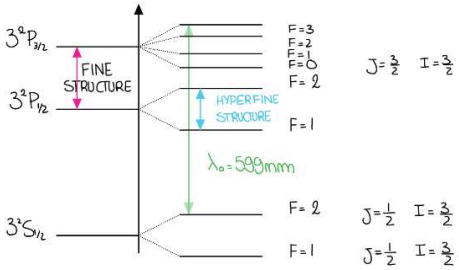
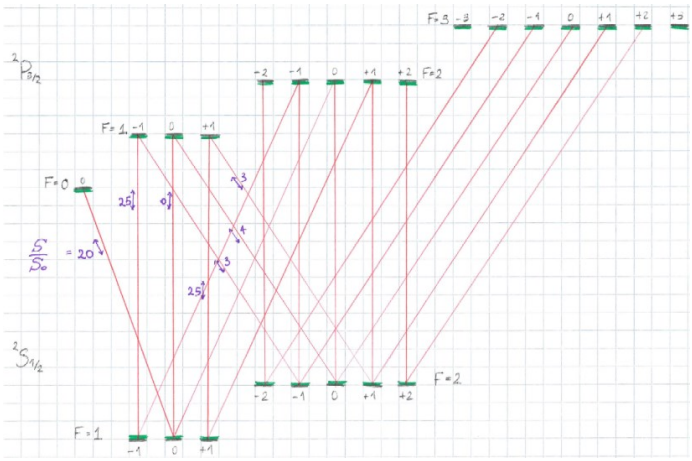
We can see that it respects the selection rules for π transitions.



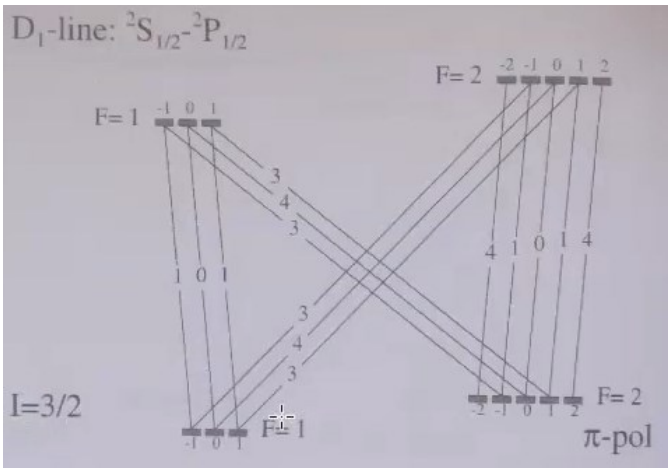
To study the transition strengths $S = |\mu_{\varepsilon,ab}|$ in the case of π polarised light, let us consider the following transitions of sodium atoms:

$$3\ ^2S_{1/2} \quad \rightarrow \quad 3\ ^2P_{3/2}$$

In the figure, the π transitions are represented along with their normalised transition strength S/S_0, where S_0 is the transition strength of a specified reference transition.



The following figure shows another example of this representation:



Circularly polarised light

Hydrogen atom

We are working under the electric dipole approximation, therefore the electric field of <u>circularly polarised right-handed light</u> (+) is

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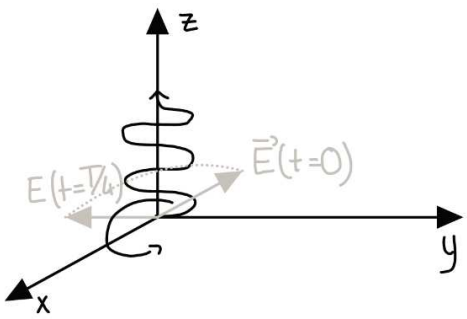
$$\vec{E}(\vec{r},t) \simeq \vec{E}(0,t) = \vec{\varepsilon}E(t) = \mathbb{R}e\{\vec{\varepsilon}E_0 e^{-i\omega t}\} = \mathbb{R}e\{\vec{u}_+ E_0 e^{-i\omega t}\}$$
$$= \mathbb{R}e\left\{\left(-\frac{1}{\sqrt{2}}\vec{u}_x - \frac{i}{\sqrt{2}}\vec{u}_y\right)E_0 e^{-i\omega t}\right\}$$
$$= \mathbb{R}e\left\{-\frac{1}{\sqrt{2}}\vec{u}_x E_0 e^{-i\omega t} - \frac{i}{\sqrt{2}}\vec{u}_y E_0 e^{-i\omega t}\right\}$$
$$= -\frac{1}{\sqrt{2}}\vec{u}_x E_0 \cos(\omega t) - \frac{1}{\sqrt{2}}\vec{u}_y E_0 \sin(\omega t)$$
$$= \vec{u}_+ \frac{E_0}{2} e^{-i\omega t} + c.c.$$



Therefore the interaction Hamiltonian is

$$\hat{H}_I = -\hat{\vec{\mu}}_e \cdot \vec{E} = -\hat{\vec{\mu}}_e \cdot \vec{u}_+ \frac{E_0}{2} e^{-i\omega t} + c.c. = \frac{1}{2}\hat{\tilde{H}}_I + c.c.$$

where we have defined

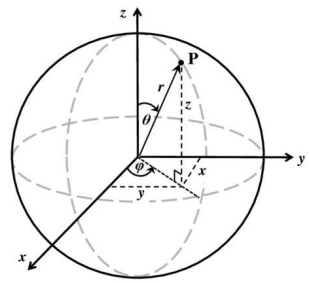
$$\hat{\tilde{H}}_I = -\hat{\vec{\mu}}_e \cdot \vec{u}_+ E_0 e^{-i\omega t}$$

To see if a transition is allowed we must evaluate the matrix element

$$\left\langle a\middle|\hat{\tilde{H}}_I\middle|b\right\rangle = \langle a|-\hat{\vec{\mu}}_e \cdot \vec{u}_+|b\rangle E_0 e^{-i\omega t} = -\vec{\mu}_{ab} \cdot \vec{u}_+ E_0 e^{-i\omega t} = -\mu_{\varepsilon,ab} E_0 e^{-i\omega t}$$

$$\mu_{\varepsilon,ab} = \langle a|\vec{\mu}_e \cdot \vec{u}_+|b\rangle = \left\langle a\middle|-e\hat{\vec{r}} \cdot \left(-\frac{1}{\sqrt{2}}\vec{u}_x - \frac{i}{\sqrt{2}}\vec{u}_y\right)\middle|b\right\rangle = e\left\langle a\middle|\frac{x+iy}{\sqrt{2}}\middle|b\right\rangle$$

Let's consider for example the ground state $|a\rangle = |1,0,0\rangle$ and an excited state $|b\rangle = |n,l,m\rangle$. We notice that



$$Y_{00}(\theta,\varphi) = \frac{1}{\sqrt{4\pi}}$$

$$Y_{11}(\theta,\varphi) = -\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{i\varphi}$$

$$\frac{x+iy}{\sqrt{2}} = \frac{r\sin\theta\cos\varphi + ir\sin\theta\sin\varphi}{\sqrt{2}} = \frac{r\sin\theta}{\sqrt{2}}e^{i\varphi} = -\frac{r}{\sqrt{2}}\sqrt{\frac{8\pi}{3}}Y_{11}(\theta,\varphi) = -r\sqrt{\frac{4\pi}{3}}Y_{11}$$

thus

$$\left\langle a\middle|\frac{x+iy}{\sqrt{2}}\middle|b\right\rangle = \int \phi_a(\vec{r})\left(\frac{x+iy}{\sqrt{2}}\right)\phi_b(\vec{r})d\vec{r}$$
$$= \int R_{10}^*(r)\frac{1}{\sqrt{4\pi}}\left(-r\sqrt{\frac{4\pi}{3}}Y_{11}\right)R_{nl}(r)Y_{lm}(\theta,\varphi)r^2\sin\theta\, dr d\theta d\varphi =$$
$$= -\frac{1}{\sqrt{3}}\int R_{10}^*(r)r^3 R_{nl}(r)dr \int Y_{\mathbf{11}}(\theta,\varphi)Y_{\boldsymbol{lm}}(\theta,\varphi)\sin\theta\, d\theta d\varphi$$
$$= -\frac{1}{\sqrt{3}}\mathcal{I}_{rad}\mathcal{I}_{ang}$$

There is <u>no general selection rule acting on the radial integral</u> $\mathcal{I}_{rad}$, which is in general different from zero. The <u>orthogonality relations</u> of the spherical harmonics

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impose some conditions on the angular integral $\mathcal{I}_{ang}$, which determine some selection rules:

$$\boldsymbol{\mathcal{I}_{ang} = \delta_{1l}\delta_{1m}}$$

For circularly polarised and right-handed electric field, therefore, the transition is allowed only if $l = 1$ and at the same time $m = 1$:

$$|a\rangle = |100\rangle \quad \rightarrow \quad |b\rangle = |n11\rangle$$

$$\Delta n \text{ free}, \qquad \Delta l = +1, \qquad \Delta m = +1$$

In the general case

$$|a\rangle = |n_i l_i m_i\rangle \quad |b\rangle = |n_f l_f m_f\rangle$$

with an electric field circularly polarised and right-handed we get the following selection rules:

$$\boldsymbol{\Delta n \text{ free}, \qquad \Delta l = \begin{cases} +1 \text{ (absorption)} \\ -1 \text{ (emission)} \end{cases}, \qquad \Delta m = +1}$$

These are called $\boldsymbol{\sigma^+}$ **transitions**.

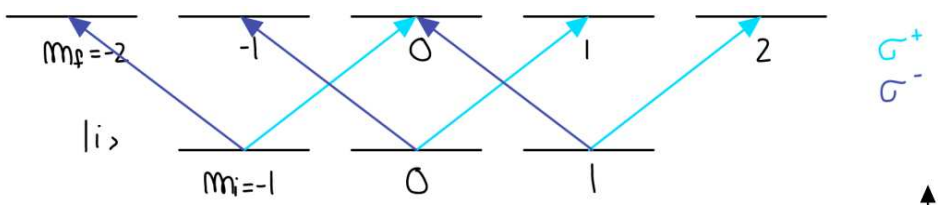
For σ^- transitions, the selection rules are the same, except for $\Delta m = -1$.

As an example, the allowed σ^+ and σ^- absorption transitions for a system with states

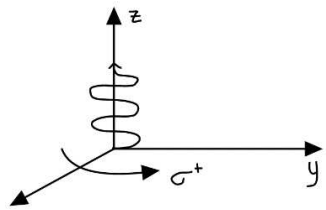
$$|i\rangle = |n_i, l_i = 1, m_i = 0, \pm 1\rangle$$

$$|f\rangle = |n_f, l_f = 2, m_f = 0, \pm 1, \pm 2\rangle$$

are reported in figure.



We can once again analyse our results in light of the <u>conservation law</u> of angular momentum.



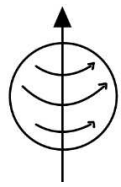
Let us consider right-handed light polarised in the xy plane and propagating in the z direction, and an atom in the state $|a\rangle = |100\rangle$. The atom's angular momentum is

$$\left|\vec{L}_i\right|^2 = l_i(l_i + 1)\hbar^2 = 0 \quad L_{zi} = m_i\hbar = 0$$

After an absorption process, the atom is in a state $|b\rangle = |n11\rangle$ and has angular momentum

$$\left|\vec{L}_f\right|^2 = l_f(l_f + 1)\hbar^2 = 2\hbar^2 \quad L_{zf} = m_f\hbar = \hbar$$

The law of conservation of angular momentum implies that the absorbed photon had an angular momentum



$$\left|\vec{L}_{ph}\right|^2 = 2\hbar^2 \quad L_{z,ph} = \hbar$$

as if the photon were spinning on itself along the z axis.

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Multi-electron atoms

Let us consider circularly polarised light and a multi-electron atom. As always, we need to consider the interaction Hamiltonian

$$\hat{H}_I = -\hat{\vec{\mu}}_e \cdot \vec{E}(\vec{r}, t)$$

To see if a transition is allowed we must evaluate the matrix element

$$\langle a|\hat{H}_I|b\rangle = -\langle a|\hat{\vec{\mu}}_e \cdot \vec{E}(t)|b\rangle = -\langle a|\hat{\vec{\mu}}_e \cdot \vec{\varepsilon}|b\rangle E(t) = -\langle a|\hat{\vec{\mu}}_e \cdot \vec{u}_\pm|b\rangle E(t) = -\mu_{\varepsilon,ab} E(t)$$

The relation $\hat{\vec{\mu}} = -e\hat{\vec{r}}$ doesn't hold in this case and the matrix element $\mu_{\varepsilon,ab}$ cannot be calculated in closed form, therefore it must be computed numerically.

The transition (or line) strength is defined as:

$$S = |\mu_{\varepsilon,ab}|^2$$

It is important to consider the **LS coupling**.

$$\vec{J} = \vec{L} + \vec{S}$$

We get a generalisation of the selection rules found before:

$$\Delta J = \pm 1, 0, \qquad \Delta L = \begin{cases} +1, & \text{absorption} \\ -1, & \text{emission} \end{cases}, \qquad \Delta m_J = \begin{cases} +1, & \text{right-handed} \\ -1, & \text{left-handed} \end{cases}$$

We then need to consider the **JI coupling**:

$$\boldsymbol{\Delta F = \pm 1, 0}, \qquad \boldsymbol{\Delta J = \pm 1, 0}, \qquad \boldsymbol{\Delta L = \begin{cases} +1, & \text{absorption} \\ -1, & \text{emission} \end{cases}}, \qquad \boldsymbol{\Delta M_F = \begin{cases} +1, & \text{right-handed} \\ -1, & \text{left-handed} \end{cases}}$$

Observation valid for $\boldsymbol{\pi}$ and $\boldsymbol{\sigma}$ transitions: light shift

Previously, studying the light shift we have found that the change in the energies of the excited and ground states are:

$$\Delta E_{e,g} = \pm \frac{\hbar |\Omega_{ab}|^2}{4\Delta\omega}$$

This relationship applies to both π and σ transitions.

Since to write

$$\hat{H}_I = -\hat{\vec{\mu}}_e \cdot \vec{\varepsilon} E(t) = -\hat{\vec{\mu}}_e \cdot \frac{\vec{\varepsilon} E_0}{2} e^{-i\omega t} + c.c. = \frac{\hbar\hat{\Omega}}{2} e^{-i\omega t} + c.c. =$$

we defined

$$\hat{\Omega} = -\frac{\hat{\vec{\mu}}_e \cdot \vec{\varepsilon} E_0}{\hbar} = -\frac{\hat{\mu}_\varepsilon E_0}{\hbar}$$

therefore

$$\Omega_{ab} = \langle a|\hat{\Omega}|b\rangle = \left\langle a\left| -\frac{\hat{\mu}_\varepsilon E_0}{\hbar} \right| b\right\rangle = -\frac{\mu_{\varepsilon,ab} E_0}{\hbar}$$

$$|\Omega_{ab}|^2 = \frac{|\mu_{\varepsilon,ab}|^2 |E_0|^2}{\hbar^2} \propto |\mu_{\varepsilon,ab}|^2 = S$$

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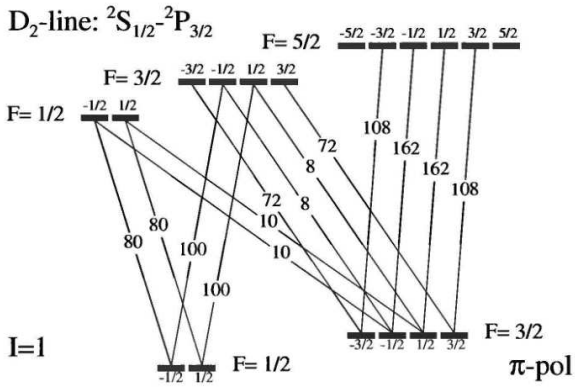
This means that the transition strength gives us an indication of the magnitude of Ω_{ab}. Therefore, the light shift is proportional to the transition strength:

$$\Delta E_{e,g} = \pm \frac{\hbar |\Omega_{ab}|^2}{4\Delta\omega} \propto S$$

The transition strength reported in representations like the one in figure is then an indicator of the strength of the light shift.



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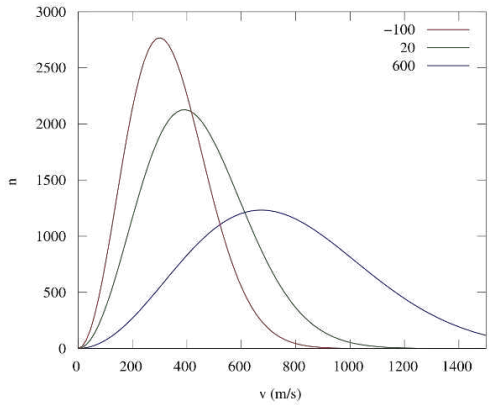
Velocity distribution of atomic beams

If we consider a population of atoms at a certain temperature, we can describe the velocity distribution

$$n(v) = \frac{dN_{v,v+dv}}{dv}, \qquad dN_{v,v+dv} = n(v)dv$$

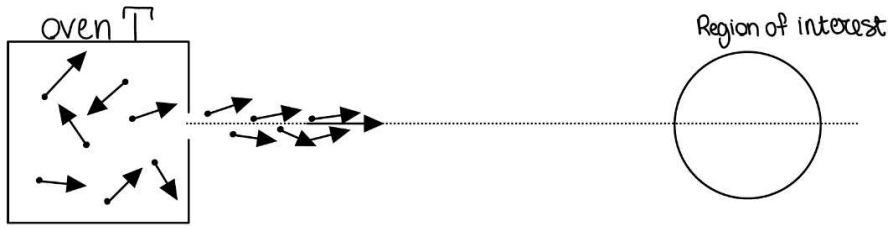


It is possible to demonstrate that for atoms <u>in an oven</u> of temperature T the velocity distribution follows the **Maxwell-Boltzmann distribution**

$$\boldsymbol{n(v) \propto v^2 e^{-\frac{v^2}{2\tilde{v}^2}}}$$

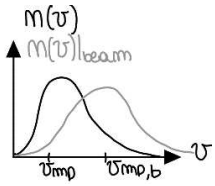
$$\boldsymbol{\tilde{v} = \sqrt{\frac{kT}{M}}}$$

which is the product of a parabolic term and a gaussian one.



We need to consider how the distribution of velocities evolves <u>after the atom beam propagates</u> from the output of the oven to the region of interest.

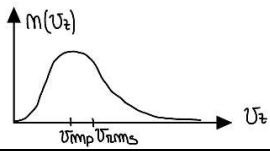
$$\boldsymbol{n(v) \propto v^3 e^{-\frac{v^2}{2\tilde{v}^2}} \propto v \cdot n_{oven}(v)}$$



This can be explained by considering that slower atoms have a lower chance of reaching the region of interest.

Velocity distribution parameters

It is useful to introduce the most probable velocity v_{mp}, the average velocity v_{ave} and the root mean square velocity $v_{rms} = \sqrt{\langle v^2 \rangle}$.



	$\boldsymbol{n(v)}$	$\boldsymbol{v_{mp}}$	$\boldsymbol{v_{ave}}$	$\boldsymbol{v_{rms}}$
3D gas (oven)	$\sqrt{\frac{2}{\pi}}\frac{v^2}{\tilde{v}^3}e^{-\frac{v^2}{2\tilde{v}^2}}$	$\sqrt{2}\tilde{v}$	$\sqrt{\frac{8}{\pi}}\tilde{v}$	$\boldsymbol{\sqrt{3}\tilde{v}}$
Atomic beam	$\frac{v_z^3}{2\tilde{v}^4}e^{-\frac{v_z^2}{2\tilde{v}^2}}$	$\sqrt{3}\tilde{v}$	$\sqrt{\frac{9\pi}{8}}\tilde{v}$	$\boldsymbol{2\tilde{v}}$

The root mean square velocity is very important because it is linked to the average kinetic energy:

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$$\langle E_k \rangle = \frac{1}{2} M \langle v^2 \rangle = \frac{1}{2} M v_{rms}^2$$

For a 3D gas it is easy to calculate the average kinetic energy and thus v_{rms}:

$$\frac{1}{2} M v_{rms}^2 = \langle E_k \rangle = \frac{3}{2} kT \quad \rightarrow \quad v_{rms} = \sqrt{3 \frac{kT}{M}} = \sqrt{3} \tilde{v}$$

Doppler broadening

Let us consider an atom with a certain velocity $\vec{v}$ and a resonance frequency ω_0 and a photon propagating in the opposite direction and with a wavevector $\vec{k}$ and frequency ω. The frequency of the photon seen from the atom is higher than ω, while it would be lower if the atom and the photon were moving in the same direction.

In a more quantitative way, because of the Doppler effect the frequency of the photon seen by the atom is

$$\boldsymbol{\omega' = \omega - \vec{k} \cdot \vec{v} = \omega + k v_z}$$

The resonance condition becomes

$$\omega + k v_z = \omega' = \omega_0 \quad \Rightarrow \quad \omega = \omega_0 - k v_z$$

We can define a <u>modified resonance frequency</u> due to the Doppler effect

$$\boldsymbol{\omega_0' = \omega_0 - k v_z = \omega_0 + \vec{k} \cdot \vec{v}}$$

such that the resonance condition can be expressed in the external reference system as

$$\omega = \omega_0'$$

Let us consider the atoms exiting an oven and an area of interest for an experiment. There isn't just one single velocity in the atom beam, but a distribution. Then, referring to the formula of ω_0', it can be seen that the presence of a distribution of velocities leads to a distribution of atomic resonance frequencies:

$$n(v_z) \propto v_z^3 e^{-\frac{v_z^2}{2\tilde{v}^2}} \quad \leftrightarrow \quad n(\omega_0')$$

This leads to the Doppler broadening, which is the broadening, due to Doppler effect, of the line corresponding to a certain transition. This mechanics leads to homogeneous broadening.

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Techniques of beam deceleration

Introduction to the detuning problem

We have seen that

$$\vec{F}_{RRP} = \hbar\vec{k}\frac{\Gamma}{2}\frac{s}{1+s} = \hbar\vec{k}\frac{\Gamma}{2}\frac{s_0}{1+s_0+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2}$$

We can include the Doppler shift in the detuning using the modified resonance frequency

$$\Delta\omega \quad \rightarrow \quad \Delta\omega' = \omega' - \omega_0 = \left(\omega - \vec{k}\cdot\vec{v}\right) - \omega_0 = \omega - \omega_0 + kv_z = \Delta\omega + kv_z$$

This way

$$\vec{F}_{RRP} = \hbar\vec{k}\frac{\Gamma}{2}\frac{s_0}{1+s_0+\left(\frac{\Delta\omega + kv_z}{\Gamma/2}\right)^2}$$

To maximise $\vec{F}_{RRP}$ and have maximum deceleration we need to be in <u>saturation</u> condition

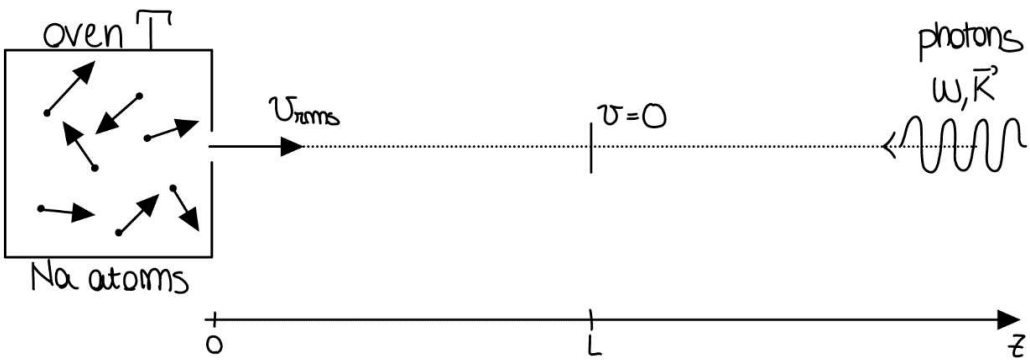
$$s_0 = \frac{I}{I_s} \gg 1$$

and to be near enough to resonance, that is having a detuning $\Delta\omega$ much smaller than the full width at half maximum Γ of the line:

$$\Delta\omega + kv_z = \Delta\omega' \ll \Gamma$$

Under these conditions

$$Ma_{\max} = \left|\vec{F}_{RRP}\right|_{\max} \simeq \hbar k\frac{\Gamma}{2}$$



Let us consider sodium atoms coming from an oven at $T = 700\ K$ and photons coming from a laser source. In the position of interest, the atoms have a well defined distribution of velocities

$$n(v_z) \propto v_z^3 e^{-\frac{v_z^2}{2\tilde{v}^2}}$$

We assume that the relevant velocity is v_{rms} since it is linked to the energy of the atom beam. Therefore

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$$\begin{cases} a(t) = -a_{\max} \\ v(t) = v_{rms} - a_{\max} t \\ z(t) = z_0 + v_{rms} t - \frac{1}{2} a_{\max} t^2 \end{cases}$$

The minimum time required to stop an atom is given by

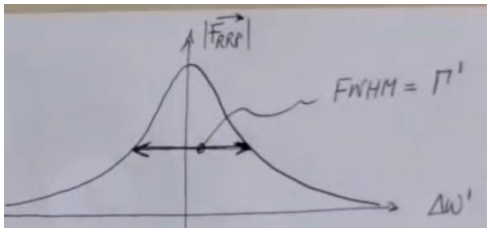
$$0 = v(t_{\min}) = v_{rms} - a_{\max} t_{\min} \quad \Rightarrow \quad t_{\min} = \frac{v_{rms}}{a_{\max}}$$

while the space travelled by the atom before stopping (defining $z_0 = 0$ at the oven) is

$$\boldsymbol{L} = z(t_{\min}) = v_{rms} t_{\min} - \frac{1}{2} a_{\max} t_{\min}^2 = \boldsymbol{\frac{1}{2} \frac{v_{rms}^2}{a_{\max}}}$$

<u>Without taking additional measures, there is actually no chance of stopping the atom</u>.

The resonant radiation pressure force as a function of the detuning $\Delta\omega'$ is Lorentzian



$$\vec{F}_{RRP} = \hbar \vec{k} \frac{\Gamma}{2} \frac{s_0}{1 + s_0 + \left(\frac{\Delta\omega'}{\Gamma/2}\right)^2} = \hbar \vec{k} \frac{\Gamma}{2} \frac{s_0}{1 + s_0} \frac{1}{1 + \left(\frac{\Delta\omega'}{\Gamma'/2}\right)^2},$$

$$\Gamma' = \Gamma \sqrt{1 + s_0}$$

Given

$$\Delta\omega' = \omega' - \omega_0 = \omega + k v_{rms} - \omega_0$$

to have resonance *at the output of the oven* we need a frequency

$$0 = \Delta\tilde{\omega}' = \tilde{\omega} + k v_{rms} - \omega_0$$

$$\tilde{\omega} = \omega_0 - k v_{rms}$$

After a certain propagation, due to deceleration of the atom caused by the laser beam, we have a change in the velocity and thus a change in the detuning:

$$\Delta\omega' = \omega' - \omega_0 = \tilde{\omega} + k(v_{rms} - \Delta v_z) - \omega_0 = \Delta\tilde{\omega}' - k\Delta v_z = -k\Delta v_z$$

where we have considered that we had fixed $\omega = \tilde{\omega}$.

For the deceleration to work we need

$$k\Delta v_z = |\Delta\omega'| < \Gamma'$$

For sodium atoms this means

$$\Delta v_z < \frac{\Gamma'}{k} \simeq \frac{\Gamma}{k} \simeq \frac{2\pi \cdot 10^7 m/s}{2\pi/(0.6 \cdot 10^{-6})} = 6\ m/s$$

while at $T = 700\ K$

$$v_{rms} = 2\tilde{v} = 2\sqrt{\frac{kT}{M}} \simeq 1000\ m/s$$

Thus, because of the Doppler shift, after we slow down the atom by a small amount it goes out of resonance and doesn't interact anymore. This problem is solved using Zeeman cooling.

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Zeeman cooling

Zeeman cooling keeps the atoms resonant with the laser beam during the deceleration using a <u>non-uniform magnetic field to induce a Zeeman effect that will compensate for the Doppler shift</u>.

To study this technique we need to determine $v = v(z)$. We start by finding $t(z)$:

$$\begin{cases} a(t) = -a_{\max} \\ v(t) = v_{rms} - a_{\max} t \\ z(t) = v_{rms} t - \frac{1}{2} a_{\max} t^2 \end{cases}$$

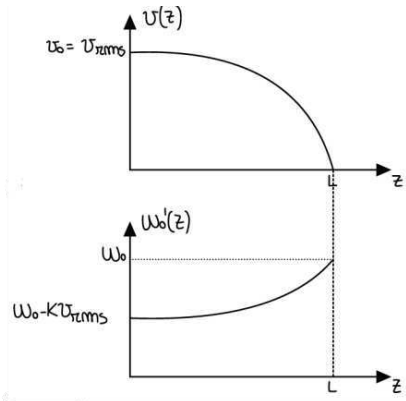
$$\frac{1}{2} a_{\max} t^2 - v_{rms} t + z = 0$$

$$t_{12}(z) = \frac{v_{rms} \pm \sqrt{v_{rms}^2 - 2a_{\max} z}}{a_{\max}}$$

Since at $t = 0$ the atom is in $z = 0$, the acceptable solution is the one with the minus.

$$t(z) = \frac{v_{rms} - \sqrt{v_{rms}^2 - 2a_{\max} z}}{a_{\max}}$$

$$\begin{aligned} \boldsymbol{v(z)} = v\big(t(z)\big) &= v_{rms} - a_{\max} t(z) \\ &= v_{rms} - \left(v_{rms} - \sqrt{v_{rms}^2 - 2a_{\max} z}\right) \\ &= \sqrt{v_{rms}^2 - 2a_{\max} z} = v_{rms} \sqrt{1 - \frac{2a_{\max} z}{v_{rms}^2}} \\ &= \boldsymbol{v_{rms} \sqrt{1 - \frac{z}{L}}} \end{aligned}$$



where L is the minimum stopping distance previously defined.

The resonance frequency as a function of z is then

$$\omega_0'(z) = \omega_0 - k v_z(z) = \omega_0 - k v_{rms} \sqrt{1 - \frac{z}{L}}$$

In the external reference system the resonance frequency changes while the frequency of our beam is fixed.

For multi-electron atoms having $\vec{I} \neq 0$ (for example sodium atoms have $I = 3/2$) we have to consider JI coupling. Considering atoms in a state $|J, I, F, M_F\rangle$ inside a magnetic field, we get Zeeman splitting by a quantity

$$\Delta E_{Zeeman} = \mu_B g_F M_F B, \qquad M_F = -F, \dots, 0, \dots, +F$$

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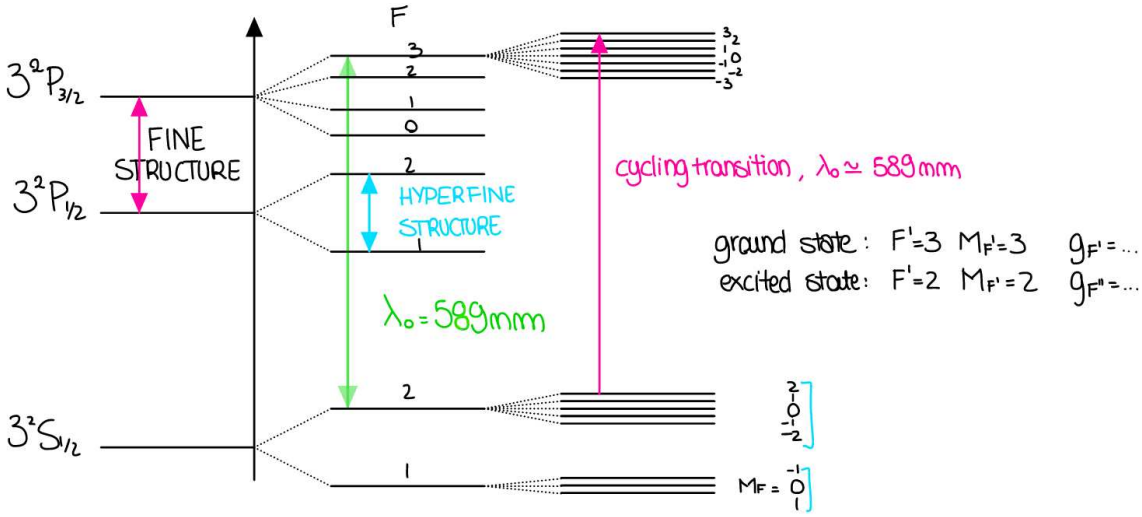
Let us consider the case of sodium atoms: the cycling transition at $\lambda_0 \simeq 589\ nm$ is between a ground state $F' = 2, M_F' = +2$ and an excited state $F'' = 3, M_F'' = +3$.

$$\Delta E_{Zeeman} = \mu_B g_F M_F B$$
$$\Delta E_Z' = \mu_B g_{F'} M_F' B$$
$$\Delta E_Z'' = \mu_B g_{F''} M_F'' B$$



The overall <u>energy shift of the transition</u> is then in this case

$$\Delta E_Z = \Delta E_Z'' - \Delta E_Z' = \mu_B (g_{F''} M_F'' - g_{F'} M_F') B \simeq \mu_B \cdot 1 \cdot B = \mu_B B$$

The change of frequency induced by the Zeeman shift is then

$$\Delta\omega_Z = \frac{\Delta E_z}{\hbar} = \frac{\mu_B B}{\hbar}$$

To compensate for the Doppler shift it must hold

$$\omega_0 = \omega_0'' = \omega_0' + \Delta\omega_Z = \omega_0 - \Delta\omega_D + \Delta\omega_Z = \omega_0 - k v_Z + \frac{\mu_B B}{\hbar}$$

$$k v_Z(z) = \frac{\mu_B B(z)}{\hbar}$$

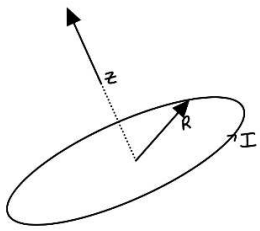
$$B(z) = \frac{\hbar k}{\mu_B} v_z(z) = \frac{\hbar k}{\mu_B} v_{rms} \sqrt{1 - \frac{z}{L}} = B_0 \sqrt{1 - \frac{z}{L}}$$

where

$$B_0 = B(0) = \frac{\hbar k}{\mu_B} v_{rms}$$

We need therefore a non-uniform magnetic field.

We can exploit the field generated by a ring carrying an electric current:



$$\vec{B}(z) = \frac{\mu_0}{2} \frac{R^2 N I}{[z^2 + R^2]^{3/2}} \vec{u}_z$$

$$\vec{B}(0) = \frac{\mu_0}{2} \frac{N I}{R} \vec{u}_z$$

where N is the number of loops made by the current along the ring.

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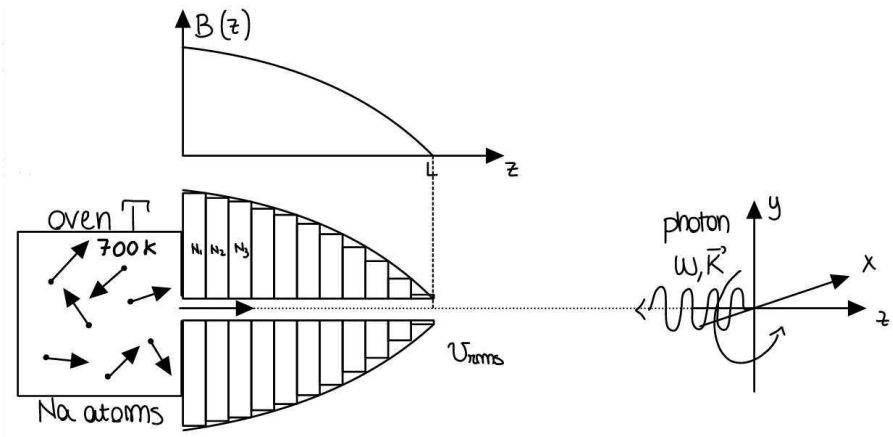
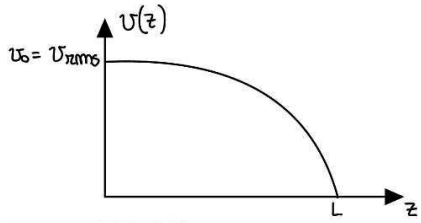
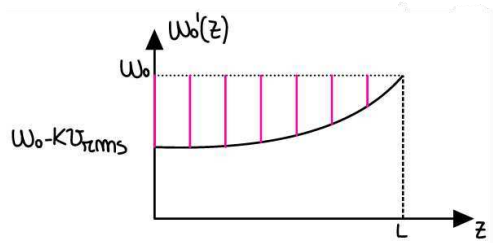
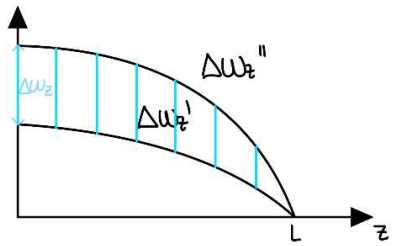
To approximate the required field, we use a certain number of coils all injected with the same current, but having each a different number of loops. This device is called a Zeeman slower or Zeeman cooler.



For our cycling transition the change ΔM_F is $+1$, therefore to stimulate this transition we need $\boldsymbol{\sigma^+}$ **polarised light**.

	$v(z) = v_{rms}\sqrt{1-\frac{z}{L}}$ $B(z) = B_0\sqrt{1-\frac{z}{L}}$
	$\omega_0' = \omega_0 - \Delta\omega_D = \omega_0 - kv_z(z)$
	$\Delta\omega_Z' = \frac{\Delta E_Z'}{\hbar} = \frac{1}{\hbar}\mu_B g_{F'} M_F' B$ $\Delta\omega_Z'' = \frac{\Delta E_Z''}{\hbar} = \frac{1}{\hbar}\mu_B g_{F''} M_F'' B$ $g_{F''}M_F'' \gg g_{F'}M_F' \quad \Delta\omega_Z'' - \Delta\omega_Z' \propto B(z) > 0$

Therefore, as the Doppler effect decreases during propagation, so does the Zeeman shift and the two compensate each other in all positions.

In practical implementations the device is cut just shorter than the stopping distance L, therefore there is a <u>small residual velocity</u> $v_{res} \sim cm/s$. This is because we need to leave the atom with a residual velocity for it to move, very slowly, to the area where the next phase of the experiment takes place.

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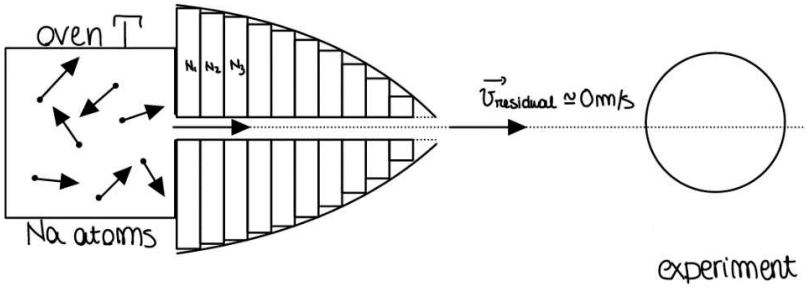
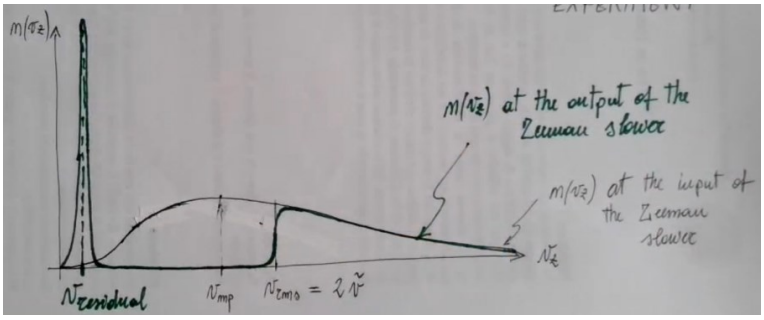


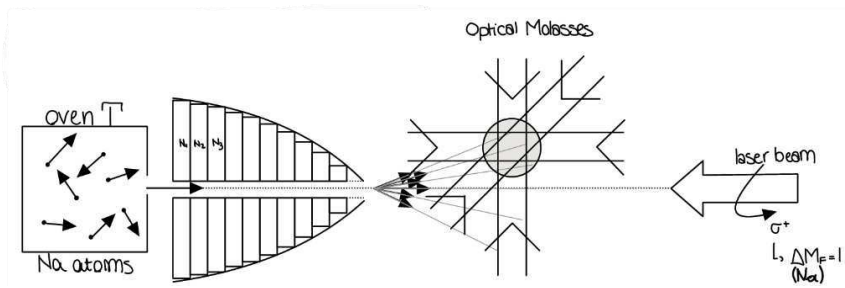
In the figure are represented the velocity distributions at the input and at the output of the Zeeman slower. Since the device has been optimised for v_{rms}, the atoms with a larger velocity are never at resonance and don't slow down: the Zeeman shift can't compensate the Doppler shift of these atoms in any section of the device. Atoms at $v < v_{rms}$ get in resonance at some point in the device, where the Zeeman shift gets lower and compensates exactly the Doppler one. From there on, they remain in resonance with the device, thus coming out at v_{res}: the area of $v < v_{rms}$ at the input corresponds therefore to the area of the peak in v_{res} at the output.



Optical molasses

The optical molasses is a device to cool atoms, usually employed as a <u>second cooling stage after a Zeeman device</u>. Indeed, the Zeeman slower is effective at reducing the kinetic energy of atoms only along one axis. We need to also reduce the kinetic energy of the atoms along the transverse directions.

In figure is reported the scheme of a typical arrangement for atom cooling, which consists of a Zeeman slower followed by an optical molasses.



At the output of the Zeeman slower there is some dispersion in the direction of the velocities. An optical molasses consists of the <u>intersection of three orthogonal couples of counter-propagating laser beams</u>. It is placed outside of the cooling laser beam and it is <u>intercepted only by the atoms moving at a specific angle</u> with respect to the axis of the system, which therefore get trapped.

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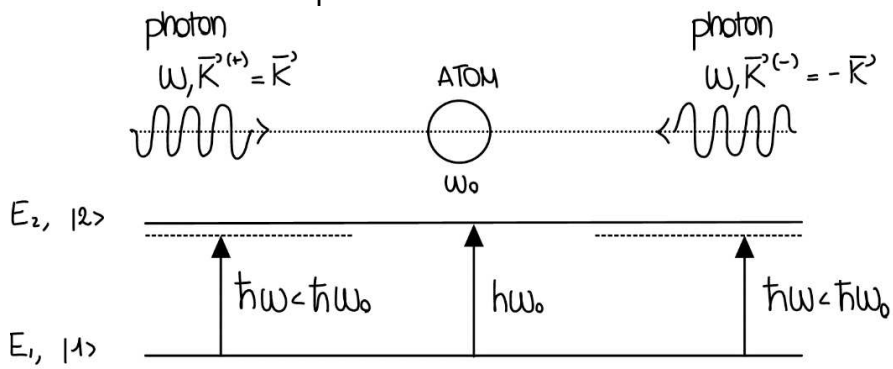
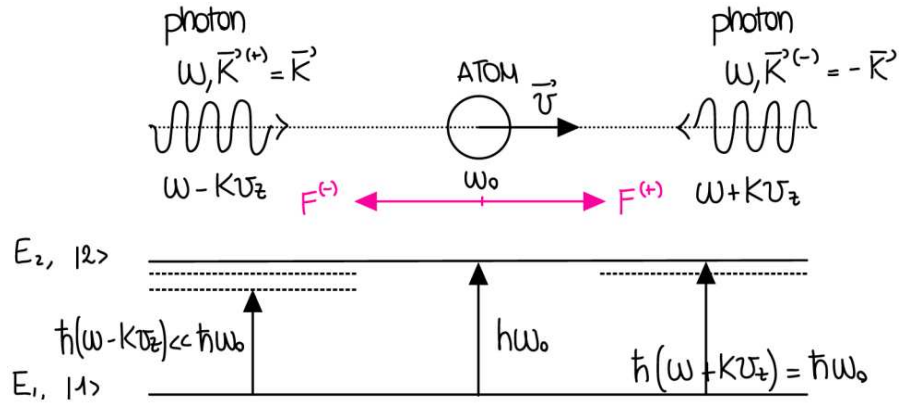
1D optical molasses

Let us assume that we are working with <u>red detuning</u> ($\Delta\omega < 0$). Two photons with the same frequency impinge on the atom from opposite directions, having opposite $\vec{k}$.

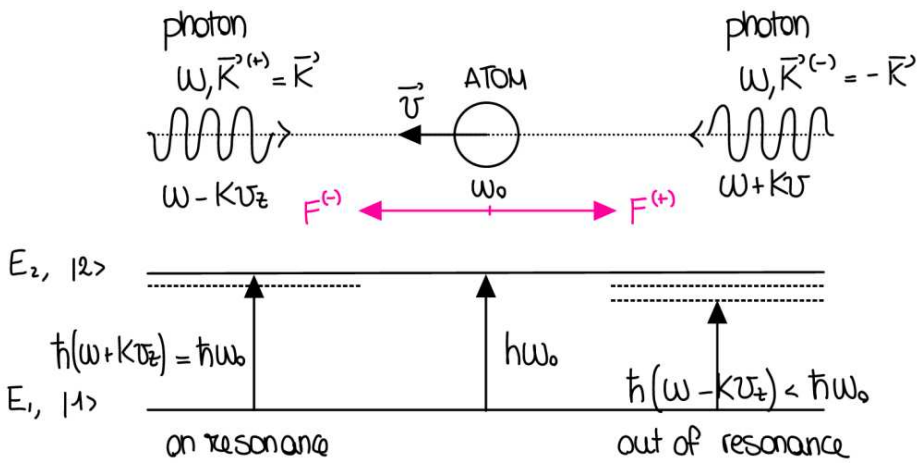
Let us first consider the case in which the atom is at rest: both lasers are out of resonance and there is no absorption.



When the atom is moving with a positive velocity, $\vec{v} = v_z\vec{u}_z$, the two photons are seen with different frequencies, $\omega \pm kv_z$. Therefore under these conditions the counter-propagating photon is on resonance and the atom is slowed down.



When the atom is moving with a negative velocity, $\vec{v} = -v_z\vec{u}_z$, the two photons are once again seen with different frequencies, $\omega \pm kv_z$. As in the previous case, the counter-propagating photon is on resonance and the atom is decelerated.



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Damping coefficient

We now want to quantitively study this device and calculate the damping coefficient.

For an atom at rest

$$\vec{F}_{RRP}(\Delta\omega') = \hbar\vec{k}\frac{\Gamma}{2}\frac{s_0}{1+s_0+\left(\frac{\Delta\omega'}{\Gamma/2}\right)^2}$$

Considering both photons and the Doppler shift, in an optical molasses the force on an <u>atom with positive velocity</u> is

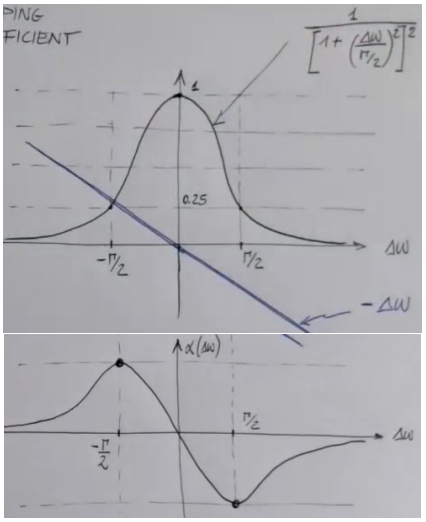
$$\begin{aligned}\vec{F}_{OM} = \vec{F}^+ + \vec{F}^- &= \hbar\vec{k}^+\frac{\Gamma}{2}\frac{s_0}{1+s_0+\left(\frac{\Delta\omega - kv_z}{\Gamma/2}\right)^2} + \hbar\vec{k}^-\frac{\Gamma}{2}\frac{s_0}{1+s_0+\left(\frac{\Delta\omega + kv_z}{\Gamma/2}\right)^2}\\ &= \hbar k\frac{\Gamma}{2}\frac{s_0}{1+s_0+\left(\frac{\Delta\omega - kv_z}{\Gamma/2}\right)^2}\vec{u}_z - \hbar k\frac{\Gamma}{2}\frac{s_0}{1+s_0+\left(\frac{\Delta\omega + kv_z}{\Gamma/2}\right)^2}\vec{u}_z\\ &= F_{RRP}(\Delta\omega - kv_z)\vec{u}_z - F_{RRP}(\Delta\omega + kv_z)\vec{u}_z\end{aligned}$$

Under the assumption $|kv_z| \ll \Delta\omega$ we can write

$$\begin{aligned}\vec{F}_{OM} &\simeq \left[F_{RRP}(\Delta\omega) + (-kv_z)\left.\frac{\partial F_{RRP}}{\partial\Delta\omega'}\right|_{\Delta\omega}\right]\vec{u}_z - \left[F_{RRP}(\Delta\omega) + (kv_z)\left.\frac{\partial F_{RRP}}{\partial\Delta\omega'}\right|_{\Delta\omega}\right]\vec{u}_z\\ &= -2kv_z\left.\frac{\partial F_{RRP}}{\partial\Delta\omega'}\right|_{\Delta\omega}\vec{u}_z = -2kv_z\hbar k\frac{\Gamma}{2}s_0\frac{-1}{\left[1+s_0+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]^2}2\left(\frac{\Delta\omega}{\Gamma/2}\right)\frac{1}{\Gamma/2}\vec{u}_z\\ &= \frac{8\hbar k^2 s_0}{\Gamma}\frac{\Delta\omega}{\left[1+s_0+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]^2}v_z\vec{u}_z = -\alpha(\Delta\omega)\vec{v}_z\end{aligned}$$

where we have defined the damping coefficient

$$\boldsymbol{\alpha(\Delta\omega) = -\frac{8\hbar k^2 s_0}{\Gamma}\frac{\Delta\omega}{\left[1+s_0+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]^2}}$$



Let us consider for simplicity the case $s_0 \ll 1$:

$$\alpha(\Delta\omega) \propto -\Delta\omega\cdot\frac{1}{\left[1+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]^2}$$

We have found that

$$\vec{F}_{OM} = -\alpha(\Delta\omega)\vec{v}_z$$

which describes a friction force. Thus, the force in optical molasses is of the same kind as the force acting on a particle moving in a fluid with a defined viscosity, that is, a particle moving with a molasses.

The graph of $\alpha(\Delta\omega)$ shows a quite intuitive result: there is an effective reduction of the velocity for red detuning, while for blue detuning the atom will be accelerated. Notice that the **damping is maximum for $\boldsymbol{\Delta\omega = -\Gamma/2}$**.

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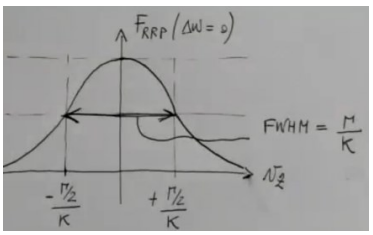
Capture range

We are going to compute the capture range in the $s_0 \ll 1$ case (for simplicity) and under assumption that there is red detuning with the maximum damping, $\Delta\omega = -\Gamma/2$.

$$\vec{F}_{OM} = \vec{F}^+ + \vec{F}^- \simeq \hbar\vec{k}^+ \frac{\Gamma}{2} \frac{s_0}{1 + \left(\frac{\Delta\omega - kv_z}{\Gamma/2}\right)^2} + \hbar\vec{k}^- \frac{\Gamma}{2} \frac{s_0}{1 + \left(\frac{\Delta\omega + kv_z}{\Gamma/2}\right)^2}$$

This time we want to study the force as a function of the velocity of the atom instead of as a function of the detuning (which is now fixed).



If there were no detuning

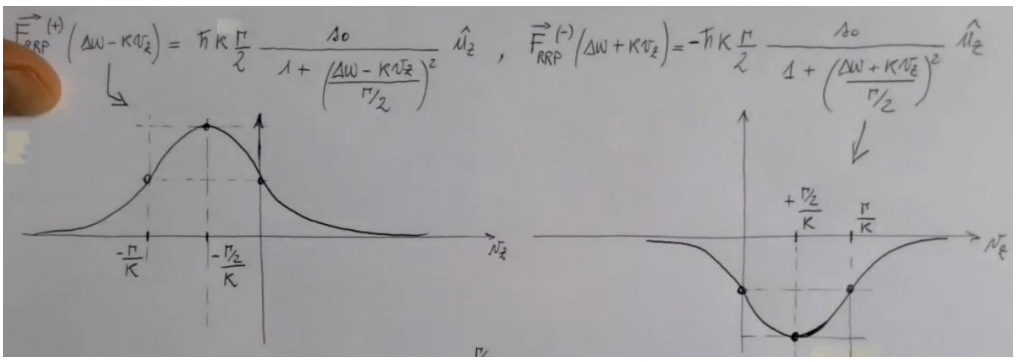
$$F_{RRP}(\Delta\omega = 0) = F_{RRP}(\pm kv_z) = \hbar k \frac{\Gamma}{2} \frac{s_0}{1 + \left(\frac{kv_z}{\Gamma/2}\right)^2}$$

where the FWHM is Γ/k.

Since there is a detuning, the FWHM is unchanged, but the position of the peak is

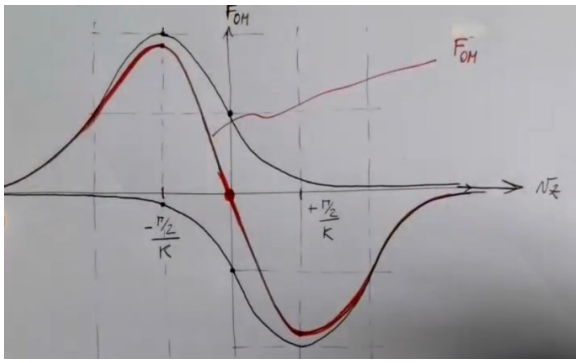
$$\Delta\omega \pm kv_z = 0$$

$$v_z = \mp\frac{\Delta\omega}{k} = \pm\frac{\Gamma/2}{k}$$



The optical molasses force is the sum of the two, therefore

$$\vec{F}_{OM} = \vec{F}^+ + \vec{F}^- \simeq F^+_{RRP}\vec{u}_z - \vec{F}^-_{RRP}\vec{u}_z = [F_{RRP}(\Delta\omega - kv_z) - F_{RRP}(\Delta\omega + kv_z)]\vec{u}_z$$



As expected, the force is always opposite to the direction of movement of the atom, thus slowing it down. However, the force is strong only for slow atoms: the **capture range** is defined as the velocity range

$$|\boldsymbol{v_z}| < \frac{\boldsymbol{\Gamma}}{\boldsymbol{2k}}$$

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3D optical molasses

Let us consider six counter-propagating laser beams.

$$\vec{F}_{OM} = -\alpha v_x \vec{u}_x - \alpha v_y \vec{u}_y - \alpha v_z \vec{u}_z = -\alpha \vec{v}$$

$$M\frac{d\vec{v}}{dt} = M\vec{a} = \vec{F}_{OM} = -\alpha \vec{v}$$

The equation of motion of an atom inside the molasses is then

$$\frac{d\vec{v}}{dt} = -\frac{\alpha}{M}\vec{v}$$

The <u>stationary solution</u> is

$$0 = \frac{d\vec{v}}{dt} = -\frac{\alpha}{M}\vec{v} \quad \Rightarrow \quad \vec{v} = 0$$

It looks like the atom is completely stopped by the optical molasses, but actually <u>we are considering expectation values</u>, since the radiation reaction force was computed as such. If we consider the stochastic quantum nature of the force acting on the atom we will find a dispersion of the force around its expectation value which will add a chaotic component of motion to the atoms and thus a "heating effect".

Fluctuations and heating

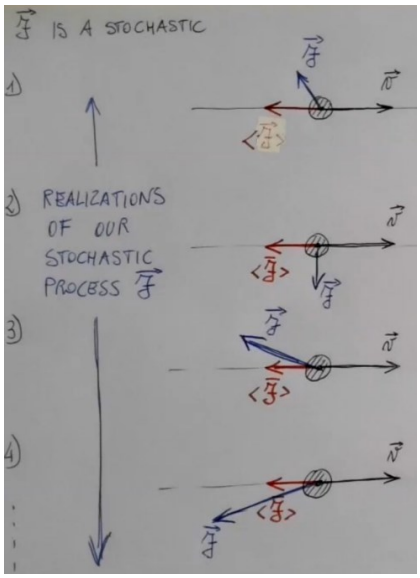
Let us now take into account the quantum nature of the force, which is at the origin of the fluctuations in the force.

Let us consider an atom described by $\psi(x,t)$: we are interested in measuring the position of the atom. The dispersion of the results of the measurements is

$$\Delta x^2 = \int (x - \langle x \rangle)^2 |\psi(x,t)|^2 \, dx$$

Since the observable x is a stochastic process, so are $v = dx/dt$, $a = dv/dt$ and, most importantly, $\vec{\mathcal{F}} = M\vec{a}$. With $\vec{\mathcal{F}}$ we are denoting the stochastic quantum force.

Assume we are able to replicate our system and make many measurements, obtaining many realisations of the stochastic process. The expectation value of the force is always the same, but because of the fluctuations the realisation of the force changes from replica to replica.



The quantum force acting at time t on an atom within the cloud of atoms is

$$\boldsymbol{\vec{\mathcal{F}}(t) = \vec{F}_{RRP} + \vec{f}(t)}$$

where $\vec{F}_{RRP} = \langle \vec{\mathcal{F}}(t) \rangle$, therefore $\vec{f}(t)$, called <u>Langevin force</u>, must have

$$\langle \vec{f}(t) \rangle = 0$$

We can write the force as a sum of two contributions

$$\vec{\mathcal{F}}(t) = \vec{\mathcal{F}}_{\text{abs}}(t) + \vec{\mathcal{F}}_{\text{sca}}(t)$$

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Absorption force

Since the laser beam can be considered as a train of photons moving in the opposite direction with respect to the atom, each with a momentum $\hbar\vec{k}$, the force due to each photon is

$$\vec{\mathcal{F}}_{\text{abs},j}(t) = \hbar\vec{k}\delta(t - t_j)$$

so that

$$\Delta\vec{p}_j = \int \vec{\mathcal{F}}_{\text{abs},j}(t)dt = \int \hbar\vec{k}\delta(t - t_j)dt = \hbar\vec{k}$$

The total absorption force is

$$\vec{\mathcal{F}}_{\text{abs}}(t) = \vec{F}_{RRP}(t) + \vec{f}_{\text{abs}}(t)$$

and we know that

$$\langle\vec{\mathcal{F}}_{\text{abs}}(t)\rangle = \vec{F}_{RRP} = \hbar\vec{k}\Gamma_p = \hbar\vec{k}\frac{\Gamma}{2}\frac{s}{1+s}$$

where Γ_p is the scattering rate.

In the complete expression of the absorption force we want to account for the distribution of instants in which the absorption events occur:

$$\boldsymbol{\vec{\mathcal{F}}_{\text{abs}}(t) = \sum_{j=1}^{n} \hbar\vec{k}\delta(t - t_j)}$$

where n is the number of absorption events in a time interval $t_0 \mapsto t$. The average number of these events in this time interval $t_0 \mapsto t$ is

$$N = \Gamma_p \cdot (t - t_0)$$

The distribution of n in Poisson statistics is

$$P(n) = \frac{N^n}{n!}e^{-N}$$

where $P(n)$ is the probability of having n absorption events in the time interval $t_0 \mapsto t$.

$$\mu_n = \langle n\rangle = \sum_n np(n) = N$$

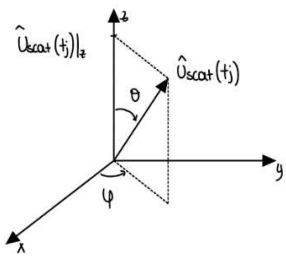
$$\sigma_n^2 = \langle(n - \mu_n)^2\rangle = \langle(n - \langle n\rangle)^2\rangle = N$$

$$\sigma_n = \sqrt{N}$$

Scattering force

The scattering force is given by

$$\boldsymbol{\vec{\mathcal{F}}_{\text{sca}}(t) = \vec{f}_{\text{sca}}(t) = \sum_{j=1}^{n} \hbar k\delta(t - t_j)\vec{u}_{\text{sca}}(t_j)}$$



where $\vec{u}_{\text{sca}}(t_j)$ is the stochastic scattering versor, thus

1) $\langle\vec{u}_{\text{sca}}(t_i)\vec{u}_{\text{sca}}(t_j)\rangle = \delta_{ij}$
2) $\langle\vec{u}_{\text{sca}}(t_j)\rangle = 0$

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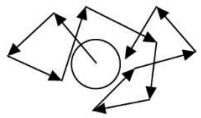
We need to study the distribution of these orientations. We want to calculate the average of the square of the projection of $\vec{u}_{\text{sca}}(t_j)$ along the x, y and z axes



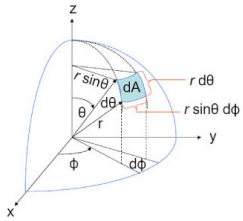
$$\left\langle \vec{u}_{\text{sca}}(t_j)\big|_{\alpha}^{2} \right\rangle$$

It is reasonable to assume that $\vec{u}_{\text{sca}}(t_j)$ is isotropically distributed.

$$\vec{u}_{\text{sca}}(t_j)\big|_{z} = \cos\theta$$

We need to find the probability $p(\theta,\varphi)$ of finding $\vec{u}_{\text{sca}}(t_j)$ at an angle (θ,φ).

Consider the infinitesimal solid angle $d\Omega$ and the corresponding surface dS:



$$\frac{d\Omega}{dS} = \frac{4\pi}{4\pi R^2}$$

$$d\Omega = \frac{dS}{R^2} = \frac{(R\sin\theta\, d\varphi)Rd\theta}{R^2} = \sin\theta\, d\theta d\varphi$$

Since because of isotropy $p(\Omega) = \text{A} = \text{const.}$, from the normalisation condition we get

$$1 = \int_{4\pi} p(\Omega)d\Omega = A\int_{4\pi} d\Omega = A\cdot 4\pi \quad \Rightarrow \quad p(\Omega) = A = \frac{1}{4\pi}$$

$$p(\theta,\varphi)d\theta d\varphi = \frac{1}{4\pi}d\Omega = \frac{1}{4\pi}\sin\theta\, d\theta d\varphi$$

$$p(\theta,\varphi) = \frac{\sin\theta}{4\pi}$$

We can now calculate the quantity we are looking for:

$$\left\langle \vec{u}_{\text{sca}}(t_j)\big|_{z}^{2} \right\rangle = \langle\cos^2\theta\rangle = \int_{\varphi=0}^{2\pi}\int_{\theta=0}^{\pi}\cos^2\theta\, p(\theta,\varphi)d\theta d\varphi = 2\pi\int_0^{\pi}\cos^2\theta\,\frac{\sin\theta}{4\pi}d\theta$$

$$= \frac{1}{2}\left[-\frac{1}{3}\cos^3\theta\right]_0^{\pi} = \frac{1}{3}$$

Since there is isotropy, this result holds for any axis:

$$\left\langle \vec{u}_{\text{sca}}(t_j)\big|_{\alpha}^{2} \right\rangle, \qquad \alpha = x, y, z$$

Fluctuations with a single laser beam (1D case)

For a complete arrangement we need six laser beams. For now we will consider only the fluctuations due to a **single laser beam**.

For an atom illuminated by a laser beam:

$$\boldsymbol{\vec{p}(t) = \vec{p}(t_0) + \hbar\vec{k}n + \sum_{j=1}^{n}\hbar k\vec{u}_{\text{sca}}(t_j)}$$

where n is the number of fluorescence cycles within a time interval $t_0 \mapsto t$.

Let us first consider only $\vec{\mathcal{F}}_{\text{abs}}(t)$:

$$\vec{p}(t) = \vec{p}(t_0) + \hbar\vec{k}n$$

$$\langle\vec{p}(t)\rangle = \langle\vec{p}(t_0)\rangle + \langle\hbar\vec{k}n\rangle = \vec{p}(t_0) + \hbar\vec{k}\langle n\rangle = \vec{p}(t_0) + \hbar\vec{k}N$$

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The variance of the linear momentum is

$$\sigma_{\vec{p}}^2 = \langle[\vec{p}(t) - \langle\vec{p}(t)\rangle]^2\rangle = \left\langle\left[\left(\vec{p}(t_0) + \hbar\vec{k}n\right) - \left(\vec{p}(t_0) + \hbar\vec{k}N\right)\right]^2\right\rangle = \left\langle\left[\hbar\vec{k}(n-N)\right]^2\right\rangle$$
$$= \left(\hbar\vec{k}\right)^2\langle(n-N)^2\rangle = \left(\hbar\vec{k}\right)^2\sigma_n^2 = \hbar^2k^2N = \hbar^2k^2\Gamma_p\cdot(t-t_0)$$

Let us now consider only $\vec{\mathcal{F}}_{\text{sca}}(t)$:

$$\vec{p}(t) = \vec{p}(t_0) + \sum_{j=1}^{n}\hbar k\vec{u}_{\text{sca}}(t_j)$$

$$\langle\vec{p}(t)\rangle = \langle\vec{p}(t_0)\rangle + \left\langle\sum\nolimits_{j=1}^{n}\hbar k\vec{u}_{\text{sca}}(t_j)\right\rangle = \vec{p}(t_0) + \sum_{j=1}^{n}\hbar k\langle\vec{u}_{\text{sca}}(t_j)\rangle = \vec{p}(t_0)$$

$$\sigma_{\vec{p}}^2 = \langle[\vec{p}(t) - \langle\vec{p}(t)\rangle]^2\rangle = \left\langle\left[\sum\nolimits_{j=1}^{n}\hbar k\vec{u}_{\text{sca}}(t_j)\right]^2\right\rangle = \left\langle\sum\nolimits_{j=1}^{n}\hbar k\vec{u}_{\text{sca}}(t_j)\cdot\sum\nolimits_{i=1}^{n}\hbar k\vec{u}_{\text{sca}}(t_i)\right\rangle$$
$$= \hbar^2k^2\left(\left\langle\sum\nolimits_{j=1}^{n}\vec{u}_{\text{sca}}(t_j)\cdot\vec{u}_{\text{sca}}(t_j)\right\rangle + \left\langle\sum\nolimits_{i\neq j=1}^{n}\vec{u}_{\text{sca}}(t_j)\cdot\vec{u}_{\text{sca}}(t_i)\right\rangle\right) =$$
$$= \hbar^2k^2\left(\left\langle\sum\nolimits_{j=1}^{n}1\right\rangle + \sum\nolimits_{i\neq j=1}^{n}\langle\vec{u}_{\text{sca}}(t_j)\cdot\vec{u}_{\text{sca}}(t_i)\rangle\right)$$
$$= \hbar^2k^2\left(\langle n\rangle + \sum\nolimits_{i\neq j=1}^{n}0\right) = \hbar^2k^2N = \hbar^2k^2\Gamma_p\cdot(t-t_0)$$

In the 1D case the rates of increase of the variance, which correspond to heating, due to absorption and scattering are therefore:

$$\begin{cases}\left.\dfrac{d\sigma_{\vec{p}}^2}{dt}\right|_{\text{abs}} = \hbar^2k^2\Gamma_p \\ \left.\dfrac{d\sigma_{\vec{p}}^2}{dt}\right|_{\text{sca}} = \hbar^2k^2\Gamma_p\end{cases}$$

Fluctuations with an optical molasses

We assume that the intensity of all **six beams** is very low, therefore the probability of having photons from the two beams hitting the atom at the same time is negligible.

If <u>we are considering the linear momentum of the atom along one of the axes</u>, $\vec{p}(t) = \vec{p}_i(t)$ with $i = x, y, z$, than only the two laser beams in that direction can modify it by absorption. Scattering instead is isotropic, therefore each of the six laser beams gives a $1/3$ contribution in our direction.

$$\begin{cases}\left.\dfrac{d\sigma_{p_i}^2}{dt}\right|_{\text{abs}} = 2\hbar^2k^2\Gamma_p \\ \left.\dfrac{d\sigma_{p_i}^2}{dt}\right|_{\text{sca}} = 6\dfrac{1}{3}\hbar^2k^2\Gamma_p = 2\hbar^2k^2\Gamma_p\end{cases}$$

Indeed

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$$\sigma_{p_z}^2\Big|_{\text{sca}} = \left\langle \sum_{j=1}^{n} \hbar k \vec{u}_{\text{sca}}(t_j)\Big|_z \cdot \sum_{i=1}^{n} \hbar k \vec{u}_{\text{sca}}(t_j)\Big|_z \right\rangle = \hbar^2 k^2 \langle n \cos^2\theta \rangle = \hbar^2 k^2 \langle n \rangle \langle \cos^2\theta \rangle$$
$$= \hbar^2 k^2 N \frac{1}{3} = \frac{1}{3}\hbar^2 k^2 \Gamma_p (t - t_0)$$

where we were able to separate the two expectation values since n and θ are independent stochastic variables.

Heating contributions

Heating contributions are all those contributions that cause an increasing in the chaotic motion of the atoms. Both absorption and scattering are heating contributions.

$$\frac{d\sigma_{p_i}^2}{dt}\bigg|_{\text{heat}} = \frac{d\sigma_{p_i}^2}{dt}\bigg|_{\text{abs}} + \frac{d\sigma_{p_i}^2}{dt}\bigg|_{\text{sca}} = 2\hbar^2 k^2 \Gamma_p + 2\hbar^2 k^2 \Gamma_p = 4\hbar^2 k^2 \Gamma_p$$

Cooling contributions

We now want to calculate the cooling contributions.

$$\frac{d\vec{p}}{dt} = \vec{F}_{OM} = -\alpha \vec{v} = -\frac{\alpha}{M}\vec{p}$$
$$\frac{dp^2}{dt} = \frac{d(\vec{p}\cdot\vec{p})}{dt} = \frac{d\vec{p}}{dt}\cdot\vec{p} + \vec{p}\cdot\frac{d\vec{p}}{dt} = 2\vec{p}\cdot\frac{d\vec{p}}{dt} = 2\vec{p}\cdot\left(-\alpha\frac{\vec{p}}{M}\right) = -2\frac{\alpha}{M}p^2$$

where under the condition $s_0 \ll 1$ it holds

$$\alpha(\Delta\omega) = -\frac{8\hbar k^2 s_0}{\Gamma}\frac{\Delta\omega}{\left[1 + \left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]^2}$$

Notice that the time evolution of the square of the linear momentum is similar to the time evolution of the momentum except for a factor 2.

We are considering the cooling effect on the <u>stochastic linear momentum</u>

$$\vec{p}(t) = \langle \vec{p}(t) \rangle + \Delta\vec{p}(t)$$

In the stationary case $\langle \vec{p}(t) \rangle = 0 = const.$ we have

$$\begin{cases} \dfrac{d\vec{p}}{dt} = \dfrac{d\langle \vec{p}(t)\rangle}{dt} + \dfrac{d\Delta\vec{p}(t)}{dt} = \dfrac{d\Delta\vec{p}(t)}{dt} \\ \dfrac{d\vec{p}}{dt} = \vec{F}_{OM} = -\dfrac{\alpha}{M}\vec{p}(t) = -\dfrac{\alpha}{M}\left(\langle \vec{p}(t)\rangle + \Delta\vec{p}(t)\right) = -\dfrac{\alpha}{M}\Delta\vec{p}(t) \end{cases}$$
$$\frac{d\Delta\vec{p}(t)}{dt}\bigg|_{\text{cool}} = -\frac{\alpha}{M}\Delta\vec{p}(t)$$

where we attribute this change to cooling since we have obtained this from the formula for the force $\vec{F}_{OM}$ of the optical molasses, which is the device responsible for cooling.

Along one component

$$\frac{d\Delta p_i(t)}{dt}\bigg|_{\text{cool}} = -\frac{\alpha}{M}\Delta p_i$$

In analogy to what we have obtained before, we can also show that

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$$\frac{d(\Delta p_i)^2}{dt} = -2\frac{\alpha}{M}(\Delta p_i)^2$$
$$\frac{d\langle(\Delta p_i)^2\rangle}{dt} = -2\frac{\alpha}{M}\langle(\Delta p_i)^2\rangle$$

Since $\sigma_{p_i}^2 = \langle(\Delta p_i)^2\rangle$, we get

$$\left.\frac{d\sigma_{p_i}^2}{dt}\right|_{\text{cool}} = -2\frac{\alpha}{M}\sigma_{p_i}^2$$

Doppler cooling limit

Including both the heating and cooling contributions in the rate equation we get

$$\frac{d\sigma_{p_i}^2}{dt} = \left.\frac{d\sigma_{p_i}^2}{dt}\right|_{\text{heat}} + \left.\frac{d\sigma_{p_i}^2}{dt}\right|_{\text{cool}} = 4\hbar^2k^2\Gamma_p - 2\frac{\alpha}{M}\sigma_{p_i}^2$$

Under the condition $s_0 \ll 1$, the stationary solution is given by

$$0 = \frac{d\sigma_{p_i}^2}{dt} = 4\hbar^2k^2\Gamma_p - 2\frac{\alpha}{M}\sigma_{p_i}^2$$

$$\sigma_{p_i}^2 = 2\hbar^2k^2\Gamma_p\frac{M}{\alpha} = \frac{2\hbar^2k^2M}{-\frac{8\hbar k^2 s_0}{\Gamma}\frac{\Delta\omega}{\left[1+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]^2}}\frac{\Gamma}{2}s_0\frac{1}{\left[1+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]} = -\frac{\hbar M\Gamma^2}{8\Delta\omega}\left[1+\left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]$$

Under the maximising condition for α, that is $\Delta\omega = -\Gamma/2$, the variance of p_i at equilibrium between heating and cooling processes is:

$$\sigma_{p_i}^2 = -\frac{\hbar M\Gamma^2}{8(-\Gamma/2)}2 = \frac{\hbar M\Gamma}{2}$$

The important parameter to characterise the temperature of the atoms is the average kinetic energy along each direction. Assuming that the linear momentum is $\vec{p}(t) = \langle\vec{p}(t)\rangle + \Delta\vec{p}(t)$ and at equilibrium $\langle\vec{p}(t)\rangle = 0$, it holds

$$\langle E_{k,i}\rangle = \frac{\langle p^2(t)\rangle}{2M} = \frac{\langle(\Delta\vec{p})^2\rangle}{2M} = \frac{\sigma_{p_i}^2}{2M} = \frac{\hbar M\Gamma}{4M} = \frac{\hbar\Gamma}{4}$$

At the same time, since we are considering the kinetic energy along one direction, that is, one degree of freedom, it holds

$$\langle E_{k,i}\rangle = \frac{1}{2}kT$$

The minimum temperature that we can achieve with an optical molasses, called **Doppler cooling limit**, is thus

$$\boldsymbol{T_{\text{min}} = \frac{\hbar\Gamma}{2k}}$$

In the case of sodium atoms $\Gamma = 2\pi\cdot 10^7\ rad/s$ and we get $\boldsymbol{T_{\text{min}} \simeq 240\ \mu K}$. This is close to the temperature required to obtain Bose-Einstein condensation: this can be achieved adding a further cooling stage, evaporative cooling. The optical molasses device is very important for the implementation of many other experiments, for example atomic clocks. Since it can push down the average kinetic energy of the

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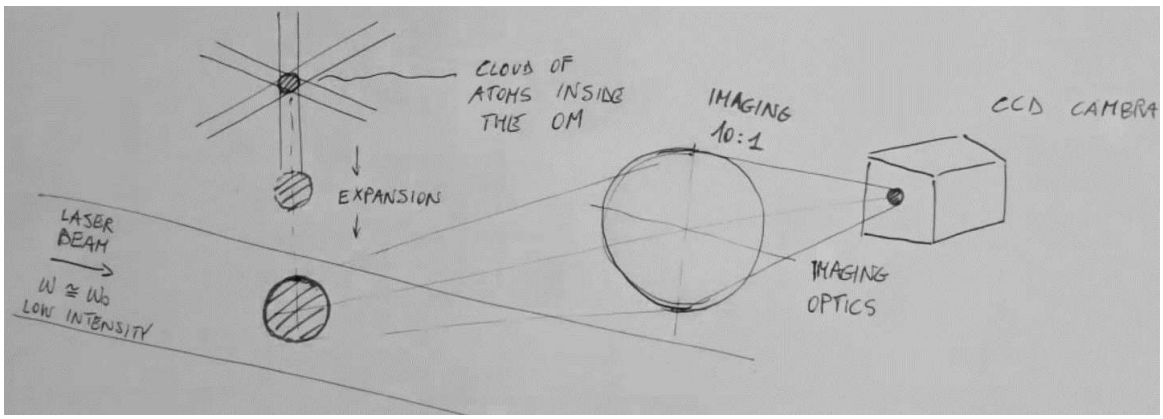
atoms well below $1\ mK$, it damps the motion of the atoms, and thus it allows also to make ultra-high resolution spectroscopy experiments.

Actually, researchers found that the temperature of sodium atoms inside optical molasses is one order of magnitude lower than the one estimated by the Doppler cooling limit. Initially, researchers were convinced that the Doppler cooling theory was right and there was some problem in the measurements. Therefore, they focused on improving the measurement of the temperature of the cloud of atoms, developing a more articulate technique, the time-of-flight measurement or TOF technique.

Time-of-flight measurement

An optical molasses holds a cloud of atoms of a certain dimension. At a certain point the laser beams are turned off and the cloud of atoms starts to fall because of gravity. Given the Maxwell-Boltzmann distribution of velocities, while falling the cloud also expands. After a certain distance there is another laser beam of low intensity and big cross-section. Imaging optics, consisting of a focusing lens and a CCD camera sensor, create an image of the cloud with a fixed magnification (e.g. 0.1).



Measuring the size of the image we can find very accurately the size of the cloud of atoms. A typical dimension for an atomic cloud in an optical molasses is of $4 \div 5\ mm$, which becomes more than $1\ cm$ after the fall, while the CCD can distinguish up to $10\ \mu m$ on the image. Once this measurement was invented it was possible to measure the profile of the cloud of atoms very accurately.

For the cloud of atoms inside the optical molasses we can assume that the velocity distribution is the Maxwell-Boltzmann one

$$n(v) \propto v^2 e^{-\frac{v^2}{2\tilde{v}^2}}$$

After the cloud is released it expands due to the Brownian motion of the atoms. The most important parameter to characterise the temperature of the cloud of atoms is the kinetic energy. Since we are imaging only a bidimensional profile of the cloud of atoms, we are interested in the kinetic energy along one direction.

$$\frac{1}{2}kT = \langle E_{k,i} \rangle = \frac{1}{2}M\langle v_i^2 \rangle = \frac{1}{2}M v_{rms,i}^2$$

$$v_{rms,i} = \sqrt{\frac{kT}{M}}$$

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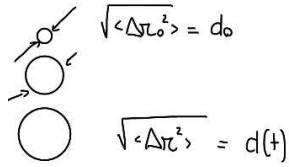
The dimension of the cloud increases in time following the formula:



$$\langle \Delta r^2 \rangle = \langle \Delta r_0^2 \rangle + \langle \Delta v^2 \rangle \cdot t^2 = \langle \Delta r_0^2 \rangle + \langle (\vec{v} - \langle \vec{v} \rangle)(\vec{v} - \langle \vec{v} \rangle) \rangle \cdot t^2$$

We can approximate $\langle \vec{v} \rangle \simeq 0$ since in the optical molasses the velocity is close to zero. This assumption is still quite good even after the velocity increases during the fall.

$$\langle \Delta r^2 \rangle = \langle \Delta r_0^2 \rangle + \langle v^2 \rangle \cdot t^2 = \langle \Delta r_0^2 \rangle + \frac{kT}{M} \cdot t^2$$

Once we fix the propagation distance from the optical molasses to the low intensity laser beam, the expansion of the cloud of atoms is therefore proportional to its temperature. This technique allowed to measure the temperature very accurately and prove that the temperature in an optical molasses is one order of magnitude lower than the Doppler limit. Researchers understood then that the problem was in the Doppler cooling theory and developed a new one, the polarization gradient cooling theory.

Polarisation gradient cooling (or Sisyphus cooling)

Polarization gradient cooling, or Sisyphus cooling, is a type of laser cooling of atoms used to reach temperatures below the Doppler cooling limit.

1D optical molasses

In real experiments, the two counter-propagating laser beams always have orthogonal polarisation. This is for practical reasons:

- they are usually produced from the same laser and separated using a polarising beam splitter;
- the orthogonal polarisation avoids interference between the counter-propagating beams, which would cause an optical feedback that would ruin the coherence of the beam.

There are two possible cases:

- linear $\pi - \pi$ polarisation, where the two beams are linearly polarised on orthogonal directions;
- circular $\sigma^+ - \sigma^-$ polarisation, where the two beams are circularly polarised with different rotation directions.

Linear $\pi - \pi$ polarisation

Consider a progressive linearly polarised wave

$$\vec{E}_x(z,t) = E_0 \cos(\omega t - kz)\, \vec{u}_x$$

and a regressive linearly polarised wave

$$\vec{E}_y(z,t) = E_0 \cos(\omega t + kz)\, \vec{u}_y$$

The dynamic of the interference between these two waves is characterised by a spatial scale of the order of the wavelength.

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$$\begin{aligned}\vec{E}(z,t) &= E_0\cos(\omega t - kz)\,\vec{u}_x + E_0\cos(\omega t + kz)\,\vec{u}_y \\ &= E_0[\cos(\omega t)\cos(kz) + \sin(\omega t)\sin(kz)]\vec{u}_x \\ &+ E_0[\cos(\omega t)\cos(kz) - \sin(\omega t)\sin(kz)]\vec{u}_y \\ &= E_0\cos(\omega t)\cos(kz)\left[\vec{u}_x + \vec{u}_y\right] + E_0\sin(\omega t)\sin(kz)\left[\vec{u}_x - \vec{u}_y\right]\end{aligned}$$

In $z = 0$ it holds

$$\vec{E}(0,t) = E_0\cos(\omega t)\left[\vec{u}_x + \vec{u}_y\right]$$

In $z = \lambda/4$ it holds $\cos\left(\frac{2\pi}{\lambda}\frac{\lambda}{4}\right) = \cos\left(\frac{\pi}{2}\right) = 0$ therefore

$$\vec{E}(\lambda/4,t) = E_0\sin(\omega t)\left[\vec{u}_x - \vec{u}_y\right]$$

In $z = \lambda/2$ it holds

$$\vec{E}(\lambda/2,t) = -E_0\cos(\omega t)\left[\vec{u}_x + \vec{u}_y\right] = -\vec{E}(0,t)$$

In $z = 3\lambda/4$ it holds

$$\vec{E}(3\lambda/4,t) = -E_0\sin(\omega t)\left[\vec{u}_x - \vec{u}_y\right] = -\vec{E}(\lambda/4,t)$$

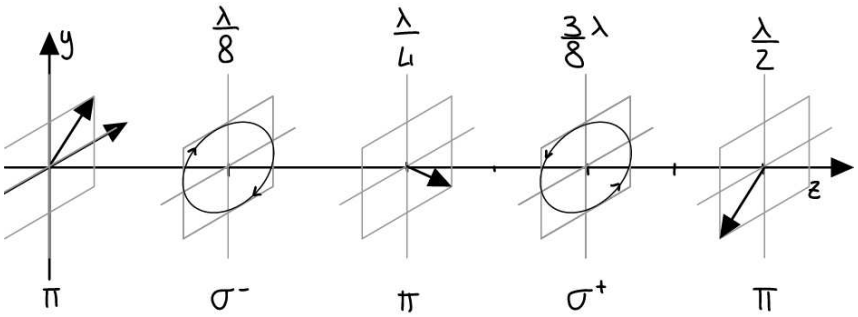
All of these results correspond to linear polarisations.

In $z = \lambda/8$ it holds

$$\begin{aligned}\vec{E}(\lambda/8,t) = \vec{E}_x(\lambda/8,t) + \vec{E}_y(\lambda/8,t) &= E_0\cos\left(\omega t - k\frac{\lambda}{8}\right)\vec{u}_x + E_0\cos\left(\omega t + k\frac{\lambda}{8}\right)\vec{u}_y \\ &= E_0\cos\left(\omega t - \frac{\pi}{4}\right)\vec{u}_x + E_0\cos\left(\omega t + \frac{\pi}{4}\right)\vec{u}_y \\ &= E_0\cos\left(\omega t - \frac{\pi}{4}\right)\vec{u}_x + E_0\sin\left[\frac{\pi}{2} - \left(\omega t + \frac{\pi}{4}\right)\right]\vec{u}_y \\ &= E_0\cos\left(\omega t - \frac{\pi}{4}\right)\vec{u}_x - E_0\sin\left(\omega t - \frac{\pi}{4}\right)\vec{u}_y\end{aligned}$$

This is a σ^- (left-handed) circular polarisation.

In $z = 3\lambda/8$ we have instead a σ^+ (right-handed) circular polarisation.



As a consequence, the kind of transitions that can be excited by the radiation depends on the position of the atom.

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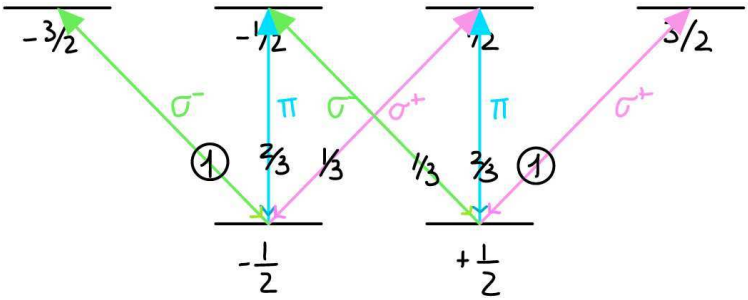
Let us consider the transitions with a ground state characterised by $J' = 1/2$ ($m_j' = \pm 1/2$) and an excited state characterised by $J'' = 3/2$ ($m_j'' = \pm 1/2, \pm 3/2$). The possible transitions of π and $\sigma^{\pm}$ kind and their strengths, $S/S_{\max}$, are shown in the figure.

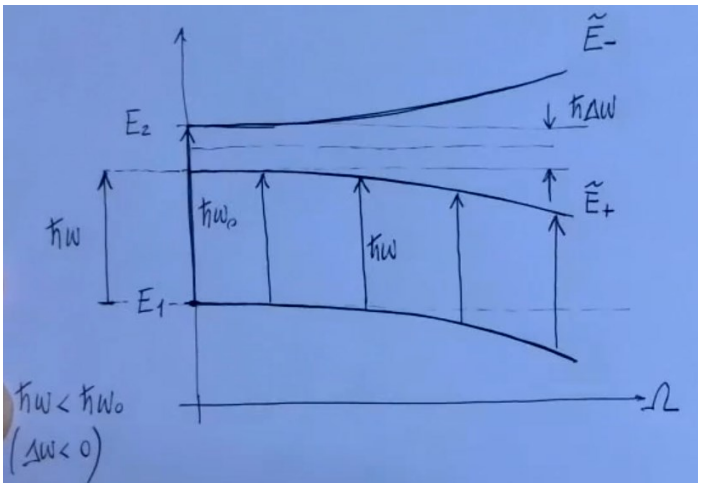


Light shift

Let us consider again the light shift, under the assumption of red detuning ($\Delta\omega < 0$).

In the rotating time frame

$$\tilde{E}_{\pm} = \pm\left(\frac{\hbar\Delta\omega}{2} + \frac{\hbar\Omega^2}{4\Delta\omega}\right)$$



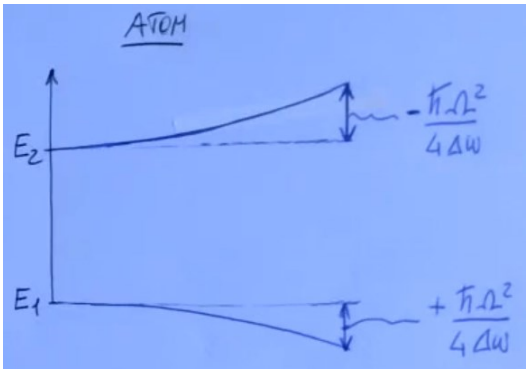
Since the quantum of energy of the electromagnetic field $\hbar\omega$ doesn't change with Ω, E_1 must have the same shape as the first excited energy level. The bending of both levels is given by

$$\Delta E_{e,g} = \pm\frac{\hbar\Omega^2}{4\Delta\omega}$$

Since $S = |\mu_{\varepsilon,12}|^2 \propto |\Omega_{\varepsilon,12}|^2$ it holds

$$\Delta E_{e,g} \propto \pm S$$

The deflection of the energy levels is therefore higher for the transitions characterised by an higher strength.



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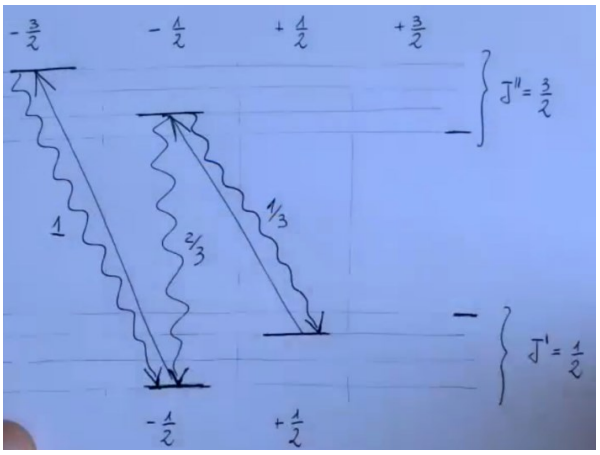
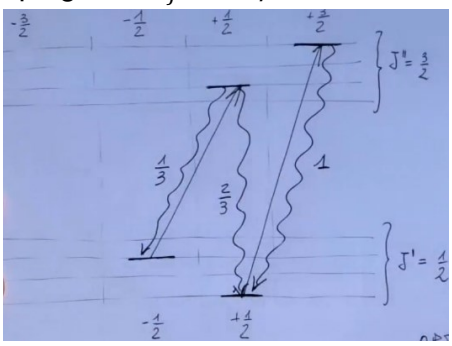
Let us consider first σ^- transitions, for which $\Delta m_j = -1$. Since the light shift is proportional to S, the one of the transition with $S = 2/3$ is double that of the transition with $S = 1/3$, and the one of $S = 1$ is three times that of $S = 1/3$.

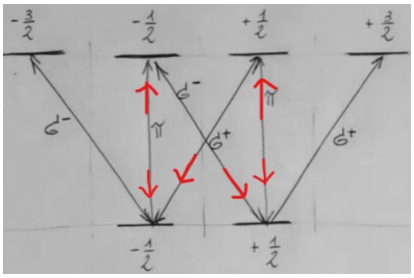


After absorption, there is fluorescence through one of the transitions represented in the figure. Assume that initially the atoms are equally distributed in the two sublevels of the $j' = 1/2$ state. While those in the state $m_j = -1/2$ always go back to that state, two thirds of those excited from $m_j = +1/2$ will go to the other state. The state with $m_j = -1/2$ is therefore favoured after a few fluorescence cycles: there is <u>optical pumping</u> into $m_j = -1/2$.

Let us consider now σ^+ transitions, for which $\Delta m_j = 1$. For the same reason as before, there is optical pumping into $m_j = +1/2$.



Finally, π transitions don't have a favoured state since the two ground sub-states behave symmetrically: an excited atom either goes back to the original state or goes to the other state.



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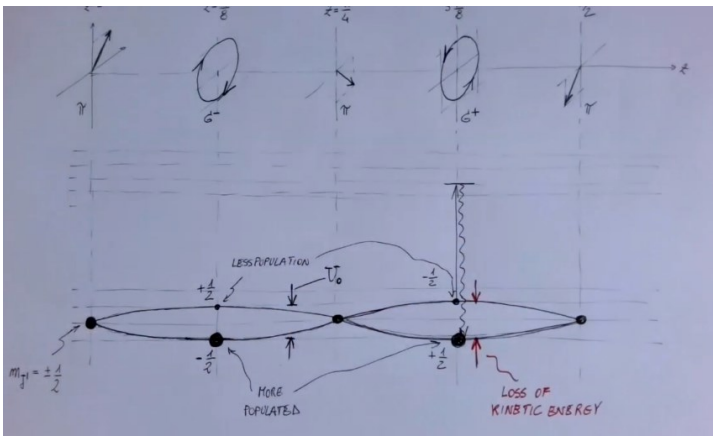
Principle of the Sisyphus cooling

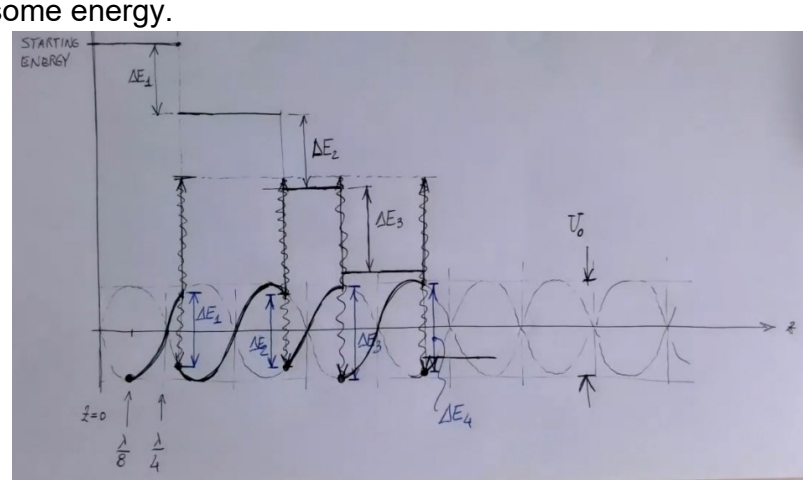
As the atoms move, the sub-levels of the ground state split in energy and their population changes according to the previous considerations. The circles in the figure represent the population.

- For $z = 0$ there is π polarisation of the electric field, thus only π transitions, which are highly symmetrical, can be excited. The energy levels are therefore degenerate and there is a symmetric distribution of the population between $1/2$ and $-1/2$.
- As the atoms propagate, σ^- is reached and there is light shift, which is stronger for $-1/2$ than for $1/2$. There is a splitting and optical pumping into $-1/2$.
- In $3\lambda/8$, there is again a different light shift for the two different m_j, but in this case the optical pumping is on $1/2$.



The atom at $-1/2$ feels a hill of potential: any time the atom gets on a hill, there is loss of kinetic energy, which leads to the cooling of the atom. Indeed, since the lower energy level is favoured, the <u>cycling transition causes a loss of kinetic energy</u>.

In the following figure the upper graph represents the energy of the atom, while the lower graph represents the light shift and the bold line the state of an atom. The energy of the state of an atom follows the light shift until a cycling transition brings the atom in a lower (in that position) energy state. In each of these transitions the atom loses some energy.



At some point the total energy of the atom is so low that it can't climb the hill of potential energy and no more transitions can occur. The typical residual energy of the atom is of the order of U_0 (shown in figure), therefore

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$$\frac{1}{2}kT_{\min} = \langle E_{k,min} \rangle \simeq U_0$$

$$\boldsymbol{T}_{\min} \simeq \frac{2U_0}{k} \simeq \mathbf{20}\ \boldsymbol{\mu K} \text{ (for Na atoms)}$$

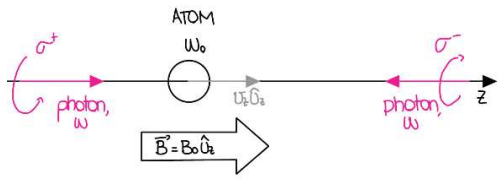
Researchers measured the minimum temperature in optical molasses using the time-of-flight technique for different alkali atoms and found very good agreement with this result. The explanation provided by the polarisation gradient cooling theory is therefore valid.

Magneto-optical trap (MOT)

The problem with optical molasses is that they are not a trap. The average velocity of the atoms is equal to zero, but the single atoms have a certain velocity and move in all directions with a Brownian motion: their free path is repeatedly deviated by collisions or scattering cycles (photon absorption and subsequent fluorescence). After each scattering event the linear momentum of the atom will have an aleatory direction. After a certain number of collisions an atom will exit from the trap region of the optical molasses and leave the cloud of atoms. The solution to this problem is represented by the adoption of magneto-optical traps.

Basic configuration of an MOT

An atom of resonance frequency ω_0 moves with a velocity $v_z\vec{u}_z$. Two counter-propagating laser beams have <u>opposite circular polarisations</u>. In addition, there is a magnetic field $B_0\vec{u}_z$.



We assume that:

1) we are working with red detuning, $\boldsymbol{\Delta\omega < 0}$;
2) the atoms have LS coupling with
   a. initial state $j' = 0,\ m_j' = 0$
   b. final state $j'' = 1,\ m_j'' = 0, \pm 1$

The Zeeman splitting is then

$$\Delta E_Z = \Delta E_Z'' - \Delta E_Z' = \mu_B\left(g_{j''}m_j'' - g_{j'}m_j'\right)B_0$$

where $\mu_B = \frac{e\hbar}{2m}$.

Obviously, when also $m_j'' = 0$, there is no sensitivity to Zeeman splitting to the first order. When $m_j'' = \pm 1$, instead, it holds $g_{j''}m_j'' - g_{j'}m_j' \sim \pm 1$, therefore

$$\Delta E_Z \sim \pm\mu_B B_0$$

The general result is

$$\Delta\omega_Z = \frac{\Delta E_Z}{\hbar} \sim \begin{cases} \pm\dfrac{\mu_B B_0}{\hbar} & m_j'' = \pm 1 \\ 0 & m_j'' = 0 \end{cases}$$

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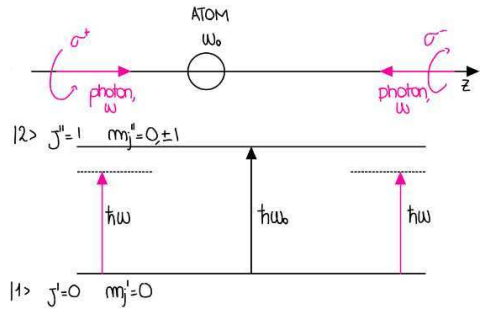
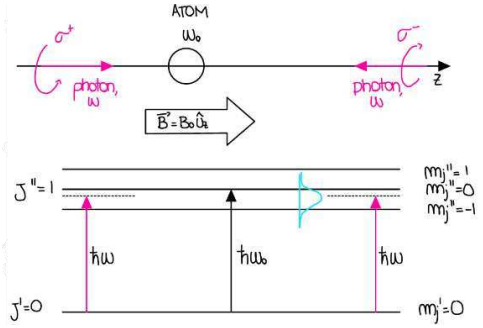
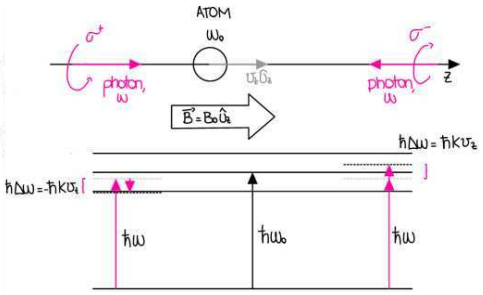
Let us consider first an <u>atom at rest</u>, with $\underline{B_0 = 0}$ and two counterpropagating circularly polarised lasers with red detuning ($\Delta\omega < 0$). Since the magnetic field is off, the excited states are degenerate. There is no resonance and therefore no absorption.



When the <u>atom is still at rest</u> but the <u>magnetic field is turned on</u>, the excited energy levels split. The lasers in use require $\Delta m_j = \pm 1$, but the levels that respect this condition are far from resonance. The state $m_j'' = 0$ is not too far from resonance, but doesn't respect the selection rule for circularly polarised light. Therefore, there is still no absorption.



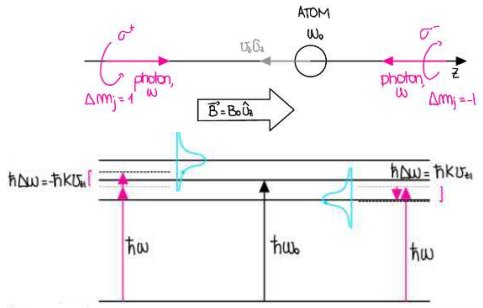
Let us consider now the <u>atom moving</u> at a <u>positive speed</u> $\vec{v} = v_{z1}\vec{u}_z, v_{z1} > 0$, with the magnetic field still turned on, $B_0 \neq 0$. The energy of the two photons changes due to the Doppler shift. However, due to the selection rules there is still no interaction even when there is a resonance.



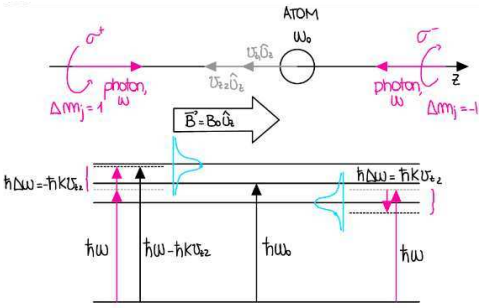
Finally, let us consider the <u>atom moving</u> at a <u>negative speed</u> $\vec{v} = v_{z1}\vec{u}_z, v_{z1} < 0$, with the magnetic field still turned on, $B_0 \neq 0$. The energy of the two photons changes due to the Doppler shift. This time, the Doppler shift is such that the achievable resonances respect the selection rules.

When one transition is on resonance and has a strong interaction, the other is close enough to resonance for the tail of the Lorentzian line broadening to cause a weak interaction.

When the speed is such that the force due to the σ^- photon is on resonance, this transition is much stronger than the one due to the σ^+ photon, so the atom velocity $|v_z|$ is increased.



A more negative velocity, $|v_{z2}| > |v_{z1}|$, can be used so that the σ^+ transition is on resonance and the σ^- transition is further from resonance. This way, the σ^+ transition is stronger than σ^- and the velocity of the atom $|v_z|$ is decreased.



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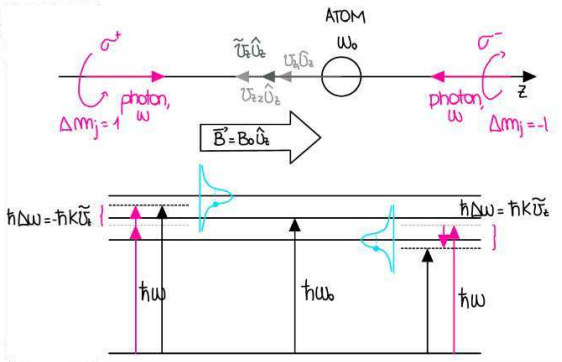
In conclusion, there exists a <u>stationary solution velocity</u> $\tilde{v}_z$ with $v_{z2} < \tilde{v}_z < v_{z1}$, which is achieved when the two forces have the same strength.



Evaluation of the stationary velocity

We now want to find the value of $\tilde{v}_z$. Keeping in mind that $v_z < 0$ and that σ^+ and σ^- interact with $m_j'' = +1$ and $m_j'' = -1$ respectively, we have

$$\sigma^+, \qquad \Delta\omega'_{(+)} = \omega' - \omega'_0 = \omega - kv_z - (\omega_0 + \Delta\omega_Z) = \omega - kv_z - \left(\omega_0 + \frac{\mu_B B_0}{\hbar}\right), \qquad m_j'' = +1$$

$$\sigma^-, \qquad \Delta\omega'_{(-)} = \omega' - \omega'_0 = \omega + kv_z - (\omega_0 - \Delta\omega_Z) = \omega + kv_z - \left(\omega_0 - \frac{\mu_B B_0}{\hbar}\right), \qquad m_j'' = -1$$

Defining

$$kv'_z = kv_z + \frac{\mu_B B_0}{\hbar}$$

$$v'_z = v_z + \frac{\mu_B B_0}{k\hbar}$$

we can write

$$\Delta\omega'_{(+)} = \Delta\omega - kv'_z$$
$$\Delta\omega'_{(-)} = \Delta\omega + kv'_z$$

When there is no Zeeman nor Doppler effect the resonant radiation pressure force is

$$\vec{F}_{RRP} = \hbar\vec{k}\frac{\Gamma}{2}\frac{s_0}{1 + s_0 + \left(\frac{\Delta\omega}{\Gamma/2}\right)^2}$$

In our case it becomes

$$\vec{F}_{MOT} = \vec{F}_{\sigma^+} + \vec{F}_{\sigma^-} = \hbar\vec{k}\frac{\Gamma}{2}\frac{s_0}{1 + s_0 + \left(\frac{\Delta\omega - kv'_z}{\Gamma/2}\right)^2} - \hbar\vec{k}\frac{\Gamma}{2}\frac{s_0}{1 + s_0 + \left(\frac{\Delta\omega + kv'_z}{\Gamma/2}\right)^2}$$
$$= [F_{RRP}(\Delta\omega - kv'_z) - F_{RRP}(\Delta\omega + kv'_z)]\vec{u}_z \simeq -2(kv'_z)\frac{\partial F_{RRP}}{\partial\Delta\omega}\vec{u}_z$$

This equation is formally identical to that of the optical molasses, but with v'_z instead of v_z.

We have already shown for an optical molasses that

$$\vec{F}_{OM} = -\alpha v_z\vec{u}_z, \qquad \alpha(\Delta\omega) = -\frac{8\hbar k^2 s_0}{\Gamma}\frac{\Delta\omega}{\left[1 + s_0 + \left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]^2}$$

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therefore for a magneto-optical trap the result is

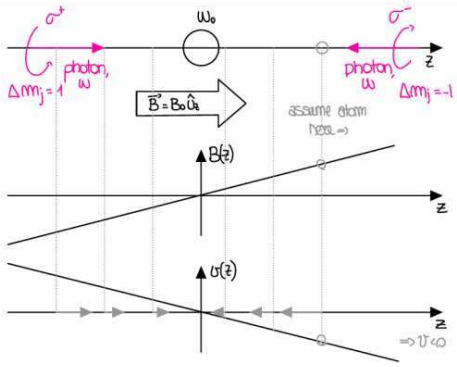
$$\vec{F}_{OM} = -\alpha v_z' \vec{u}_z, \qquad \alpha(\Delta\omega) = -\frac{8\hbar k^2 s_0}{\Gamma} \frac{\Delta\omega}{\left[1 + s_0 + \left(\frac{\Delta\omega}{\Gamma/2}\right)^2\right]^2}$$

We can solve this equation looking for a stationary solution

$$0 = M\frac{d\vec{v}_z}{dt} = \vec{F}_{OM} = -\alpha v_z' \vec{u}_z = -\alpha\left(v_z + \frac{\mu_B B_0}{k\hbar}\right)\vec{u}_z$$

$$\boldsymbol{\tilde{v}_z = -\frac{\mu_B B_0}{k\hbar}}$$

Adding a **non uniform magnetic field** $\vec{B} = B(z)\vec{u}_z$ we can change the speed of the atom.

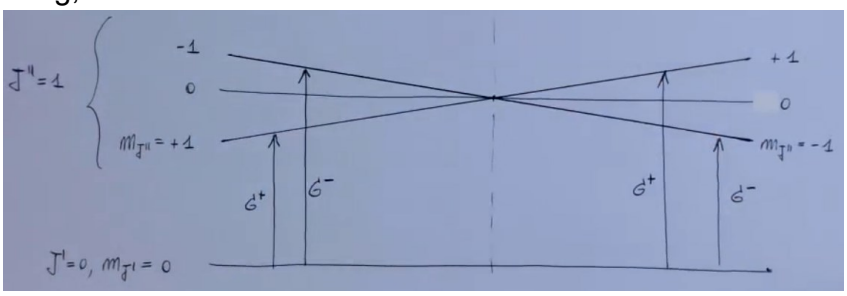


$$\tilde{v}_z = -\frac{\mu_B B(z)}{k\hbar}$$

The atom moves towards the centre of the trap, but goes slower the closer it gets to it: we have an adiabatic change of the velocity as the atom approaches the centre of the trap.

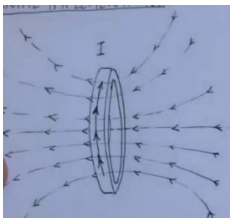
If for some reason the atom goes out of the central region of the trap, a <u>restoring force</u> brings it back towards the centre (unlike in the optical molasses).

The following figure shows the energy levels as a function of z, which change due to Zeeman splitting, and the transitions.



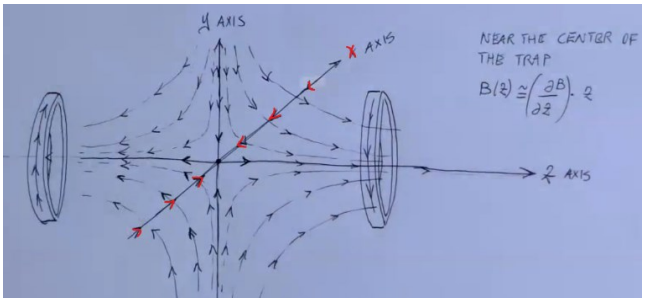
Experimental implementation

The current in a coil generates a magnetic field



$$\vec{B}(z) = \frac{\mu_0}{2}\frac{R^2 I}{[z^2 + R^2]^{3/2}}\vec{u}_z$$

Two coils with current flowing in different directions are used to produce the field shown in figure.



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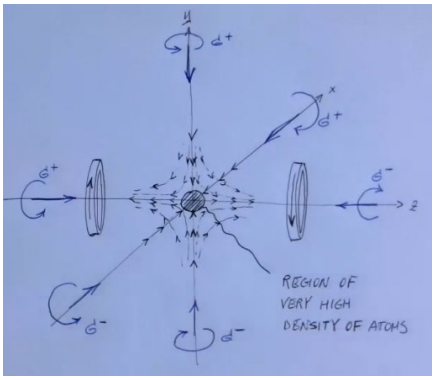
Near the centre of the trap

$$B(j) \simeq \left(\frac{\partial B}{\partial j}\right) j, \qquad j = x, y, z$$

Since

$$\left(\frac{\partial B}{\partial z}\right) > 0, \qquad \left(\frac{\partial B}{\partial x}\right) < 0, \qquad \left(\frac{\partial B}{\partial y}\right) < 0$$

the trapping effect is achieved on the z axis using the configuration of laser beams previously described



$$\tilde{v}_z = -\frac{\mu_B B(z)}{k\hbar} \simeq -\frac{\mu_B}{k\hbar}\left(\frac{\partial B}{\partial z}\right) z$$

while on the other axes it is achieved swapping the polarisation of the two laser beams

$$\tilde{v}_x = +\frac{\mu_B B(x)}{k\hbar} \simeq \frac{\mu_B}{k\hbar}\left(\frac{\partial B}{\partial x}\right) x$$

$$\tilde{v}_y = +\frac{\mu_B B(y)}{k\hbar} \simeq \frac{\mu_B}{k\hbar}\left(\frac{\partial B}{\partial y}\right) y$$

The combination of the electrical field of the optical molasses and the magnetic field of the coils creates therefore a region of very high density of atoms.

Considering the mono-dimensional case, the force is

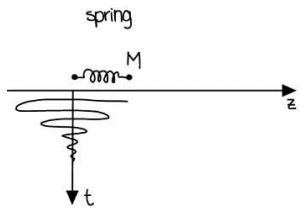
$$\vec{F}_{MOT} = -\alpha v_z' \vec{u}_z = -\alpha\left(v_z + \frac{\mu_B}{k\hbar}\frac{\partial B}{\partial z} z\right)\vec{u}_z = -\alpha v_z \vec{u}_z - \left(\alpha \frac{\mu_B}{k\hbar}\frac{\partial B}{\partial z}\right) z\vec{u}_z = -\alpha v_z \vec{u}_z - \beta z \vec{u}_z$$

where the first term is a damping force while the second one is a restoring force.

Alone, the restoring force would give

$$-\beta z \vec{u}_z = \vec{F}_{\mathrm{MOT,restoring}} = M a_z \vec{u}_z = M \ddot{z} \vec{u}_z$$

$$\ddot{z}(t) = -\frac{\beta}{M} z(t)$$



Therefore we have damped oscillations around the centre of the magneto-optical trap.

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Atomic clocks

Ramsey fringes and perturbation theory

We are going to see calculations specialised for the derivation of Ramsey fringes, which are an interferometric signal originating from the wavefunctions of the atom. This phenomenon is at the core of the operation of optical clocks.

Consider the temporal dynamic of the generic state function

$$|\boldsymbol{\psi}(\boldsymbol{t})\rangle = \sum_{\boldsymbol{k}} \boldsymbol{a_k}(\boldsymbol{t})\boldsymbol{e}^{-i\frac{\boldsymbol{E_k}}{\hbar}\boldsymbol{t}}|\boldsymbol{k}\rangle$$

$$i\hbar\frac{\partial}{\partial t}|\psi\rangle = \widehat{H}|\psi\rangle = \left(\widehat{H}_0 + \widehat{H}_I\right)|\psi\rangle$$

Multiplying by one of the N eigenstates, $|n\rangle$,

$$i\hbar\frac{\partial}{\partial t}\langle n|\psi\rangle = \langle n|\widehat{H}_0|\psi\rangle + \langle n|\widehat{H}_I|\psi\rangle$$

$$i\hbar\frac{\partial}{\partial t}\left(a_n(t)e^{-i\frac{E_n}{\hbar}t}\right) = E_n a_n(t)e^{-i\frac{E_n}{\hbar}t} + \sum_k a_k(t)e^{-i\frac{E_k}{\hbar}t}\langle n|\widehat{H}_I|k\rangle$$

$$i\hbar\dot{a}_n e^{-i\frac{E_n}{\hbar}t} + a_n E_n e^{-i\frac{E_n}{\hbar}t} = E_n a_n e^{-i\frac{E_n}{\hbar}t} + \sum_k a_k e^{-i\frac{E_k}{\hbar}t}H_{I,nk}$$

$$i\hbar\dot{a}_n e^{-i\frac{E_n}{\hbar}t} = \sum_k a_k e^{-i\frac{E_k}{\hbar}t}H_{I,nk}$$

$$\dot{a}_n(t) = -\frac{i}{\hbar}\sum_k a_k(t)e^{-i\frac{E_k-E_n}{\hbar}t}H_{I,nk}\,, \qquad n = 1,2,\dots,N$$

We introduce now the approximation of **perturbation theory**, based on the hypothesis

$$\widehat{H}_I = \lambda\widehat{H}_I', \qquad \lambda \ll 1$$

We can thus approximate the a_n coefficients as

$$a_n = a_n^{(0)} + \lambda a_n^{(1)} + \lambda^2 a_n^{(2)} + \lambda^3 a_n^{(3)} + \cdots$$

Therefore

$$\dot{a}_n^{(0)} + \lambda\dot{a}_n^{(1)} + \lambda^2\dot{a}_n^{(2)} + \cdots = -\frac{i}{\hbar}\sum_k\left(a_k^{(0)} + \lambda a_k^{(1)} + \lambda^2 a_k^{(2)} + \cdots\right)e^{-i\omega_{kn}t}\lambda H_{I,nk}'$$

We can separate the terms of different orders:

$$\dot{a}_n^{(0)}(t) = 0$$

$$\dot{a}_n^{(1)}(t) = -\frac{i}{\hbar}\sum_k a_k^{(0)}(t)e^{-i\omega_{kn}t}H_{I,nk}'$$

and so on.

Let us assume that at time t_0 the system is in the state $|i\rangle$: since $\dot{a}_n^{(0)}(t) = 0$, the zero order solution is

$$a_n^{(0)}(t) = \delta_{ni}, \qquad n = 1,2,\dots$$

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The first order solution is

$$\dot{a}_n^{(1)} = -\frac{i}{\hbar}\sum_k \delta_{ki} e^{-i\omega_{kn}t} H'_{I,nk} = -\frac{i}{\hbar} e^{-i\omega_{in}t} H'_{I,ni}, \qquad n = 1,2,\dots$$

$$\int_{a_n^{(1)}(t_0)}^{a_n^{(1)}(t)} da_n^{(1)} = -\frac{i}{\hbar}\int_{t_0}^{t} e^{-i\omega_{in}t'} H'_{I,ni} dt'$$

$$a_n^{(1)}(t) - a_n^{(1)}(t_0) = -\frac{i}{\hbar}\int_{t_0}^{t} e^{-i\omega_{in}t'} H'_{I,ni} dt'$$

For $n \neq i$

$$a_n^{(1)}(t_0) = 0$$

$$a_n^{(1)}(t) = -\frac{i}{\hbar}\int_{t_0}^{t} e^{-i\omega_{in}t'} H'_{I,ni} dt'$$

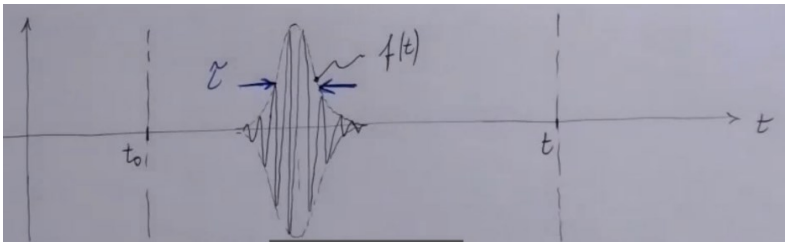
The probability of finding the system in the state $|n\rangle$ starting from the state $|i\rangle \neq |n\rangle$ is given, to the first order, by

$$P_{i\to n}(t) = |a_n(t)|^2 = \left|a_n^{(0)}(t) + \lambda a_n^{(1)}(t) + \lambda^2 a_n^{(2)}(t) + \cdots\right|^2 \simeq \left|\lambda a_n^{(1)}(t)\right|^2$$
$$= \frac{1}{\hbar^2}\left|\int_{t_0}^{t} e^{-i\omega_{in}t'} \lambda H'_{I,ni} dt'\right|^2 = \frac{1}{\hbar^2}\left|\int_{t_0}^{t} e^{-i\omega_{in}t'} H_{I,ni} dt'\right|^2$$

Let us focus on the **Ramsey fringes** problem. Let us consider a <u>two-level system</u> interacting with an <u>electromagnetic RF pulse</u>:

$$\boldsymbol{\hat{H}_I = -\hat{\vec{\mu}}_m \cdot \vec{B} = -\hat{\vec{\mu}}_m \cdot \vec{\varepsilon} B_0 f(t) \cos(\tilde{\omega} t)}$$

where the duration of the envelope $f(t)$ of the pulse is of order τ.



$$\hat{H}_I = -\hat{\mu}_{m\varepsilon} B_0 f(t)\cos(\tilde{\omega} t) = \hat{W} f(t)\cos(\tilde{\omega} t)$$

As usual, the matrix $[H_I]$ is antidiagonal:

$$[H_I] = \begin{bmatrix} 0 & W_{12} \\ W_{21} & 0 \end{bmatrix} f(t)\cos(\tilde{\omega} t)$$

$$P_{i\to n}(t) = \frac{1}{\hbar^2}\left|\int_{t_0}^{t} e^{-i\omega_{in}t'} H_{I,ni} dt'\right|^2 = \frac{1}{\hbar^2}\left|\int_{t_0}^{t} e^{-i\omega_{in}t'} W_{ni} f(t')\cos(\tilde{\omega} t')\, dt'\right|^2$$
$$= \frac{|W_{ni}|^2}{\hbar^2}\left|\int_{t_0}^{t} e^{-i\omega_{in}t'} f(t')\cos(\tilde{\omega} t')\, dt'\right|^2$$

Since the pulse goes to zero outside the time interval $[t_0, t]$, we can extend the integral to the whole time axis. This way we obtain an integral that corresponds to the Fourier transform of $f(t)$, translated by $\tilde{\omega}$ because of the cosine.

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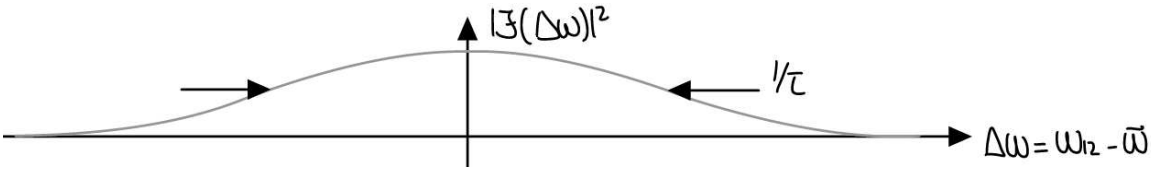
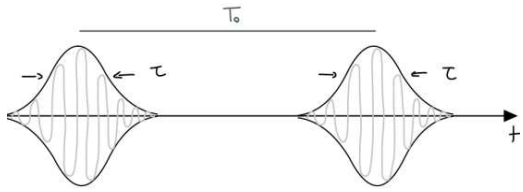
$$P_{1\to 2}(t) = \frac{|W_{21}|^2}{\hbar^2}\left|\int_{-\infty}^{+\infty} e^{-i\omega_{12}t'} f(t')\cos(\tilde{\omega}t')\,dt'\right|^2 = \frac{|W_{21}|^2}{\hbar^2}|F(\omega_{12}-\tilde{\omega})|^2$$

Since τ is small, the bandwidth is quite large.



From now on, we define $\omega = \omega_{12}$.

When there is an additional delayed version of the pulse, with the same carrier, we have two <u>coherent pulses</u>:



$$g(t) = [f(t) + f(t+T_0)]\cos(\tilde{\omega}t)$$

$$G(\omega) = \left[F(\omega) + F(\omega)e^{-i\omega T_0}\right] * \delta(\omega-\tilde{\omega}) = F(\omega-\tilde{\omega}) + F(\omega-\tilde{\omega})e^{-i(\omega-\tilde{\omega})T_0}$$

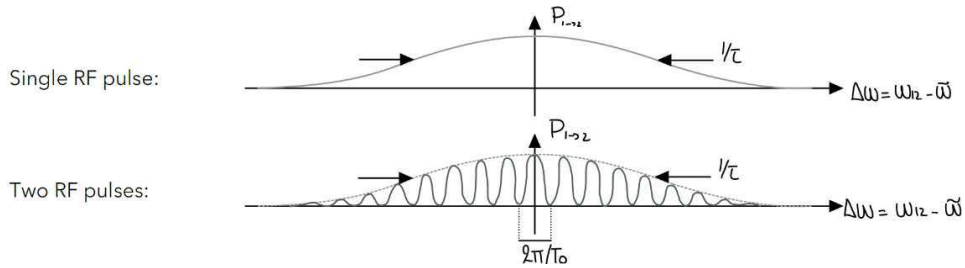
$$|G(\omega)|^2 = |F(\omega-\tilde{\omega})|^2\left|1+e^{-i(\omega-\tilde{\omega})T_0}\right|^2 = |F(\omega-\tilde{\omega})|^2\, 4\cos^2\left[\frac{(\omega-\tilde{\omega})T_0}{2}\right]$$

$$= |F(\omega-\tilde{\omega})|^2\, 2\left[1+\cos\left((\omega-\tilde{\omega})T_0\right)\right]$$

Therefore, the transition probability becomes

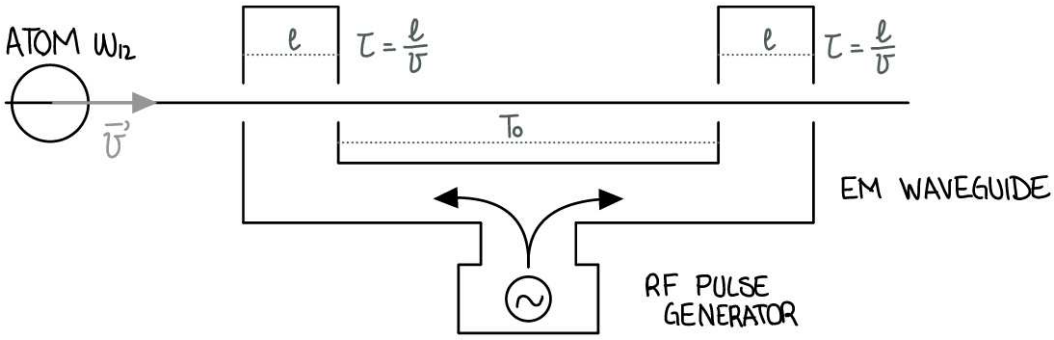
$$P_{1\to 2}(t) = \frac{|W_{12}|^2}{\hbar^2}|F(\omega-\tilde{\omega})|^2 2\left[1+\cos\left((\omega-\tilde{\omega})T_0\right)\right]$$

In the frequency domain, the transition probability oscillates with periodicity $2\pi/T_0$: these oscillations are the Ramsey fringes. These fringes allow for an improvement of the resolution in the determination of $\omega_{12} = \frac{E_1-E_2}{\hbar}$ by a factor $T_0/\tau \gg 1$.



Practical implementation: the Ramsey cavity

In the figure is represented the operational scheme of a Ramsey cavity.



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The electromagnetic wave generated by the pulse generator propagates in both metallic branches of an RF waveguide. The field in the two branches is therefore coherent. Let τ be the time it takes the atom to go through each cavity.

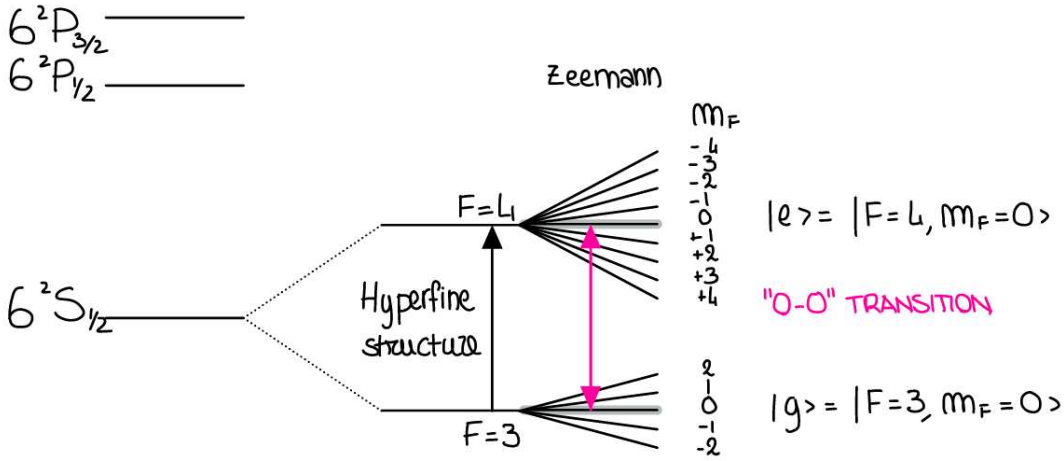
From a physical point of view, the first pulse, experienced by the atom while going through the first cavity, brings the atom into a superposition of states. While the atom travels between the two cavities it is not coupled to the RF field: the atom and RF field evolve according to their characteristic frequencies, ω_{12} and $\tilde{\omega}$ respectively. When the atom enters the second cavity, there is an accumulated phase difference of $(\omega_{12} - \tilde{\omega})T_0$. There are two significant cases:

- when $(\omega_{12} - \tilde{\omega})T_0 = 2\pi$ the interaction continues from where it left off and the overall pulse duration becomes 2τ, while $P_{1\to2}$ becomes four times higher than in the case of a single pulse because of the term $2[1 + \cos(2\pi)] = 4$;
- when $(\omega_{12} - \tilde{\omega})T_0 = \pi$ the interaction it time-reversed, meaning that the two pulses work in the opposite direction and the atom returns to its initial state, as can also be seen mathematically since $P_{1\to2} \propto 2[1 + \cos(\pi)] = 0$;

Caesium atomic clocks

Caesium atomic clocks are very important since they are used for the definition of the unit of time: the second is the duration of 9192631770 periods of the radiation corresponding to the transition between the <u>two hyperfine levels of the ground state of Cs^{133}</u>.

In figure are shown the energy levels of $\boldsymbol{Cs^{133}}$ $(\boldsymbol{I = 7/2})$ with the hyperfine structure and the Zeeman splitting for increasing magnetic field. Since it is an alkali atom, its properties are determined by the outer electron and the state can be described using the Russell-Saunders notation.



The relevant transition for the definition of the unit of time is the one with the smallest sensitivity to magnetic fields, that is $m'_F = m''_F = 0$, thus called "$\mathbf{0-0}$ **transition**": $|g\rangle = |F = 3, m_F = 0\rangle$, $|e\rangle = |F = 4, m_F = 0\rangle$.

Preparation of the atom: the Stern-Gerlach magnet

Both <u>hyperfine levels of the ground state of Cs^{133} are equally populated</u>, because the energy corresponding to the transition is very low (the transition is in the RF domain).

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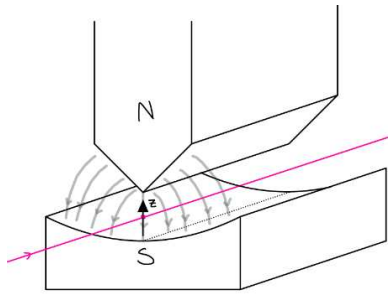
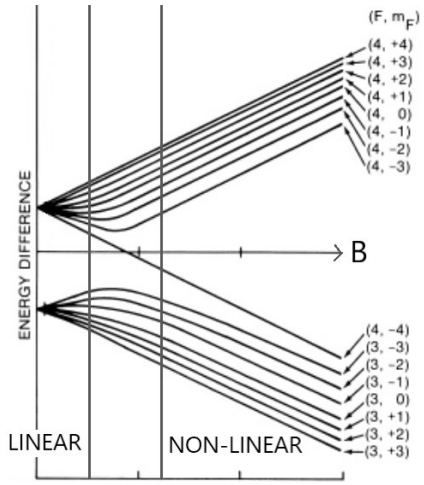
We need therefore a device to separate the $|g\rangle$ state from the $|e\rangle$ state, like the **Stern-Gerlach magnet** shown in figure.



A caesium atomic beam enters the device and is subjected to a magnetic field that can be approximated as

$$\vec{B}(z) \simeq \left[B_0 + \left(\frac{\partial B}{\partial z}\right) z\right] \vec{u}_z$$

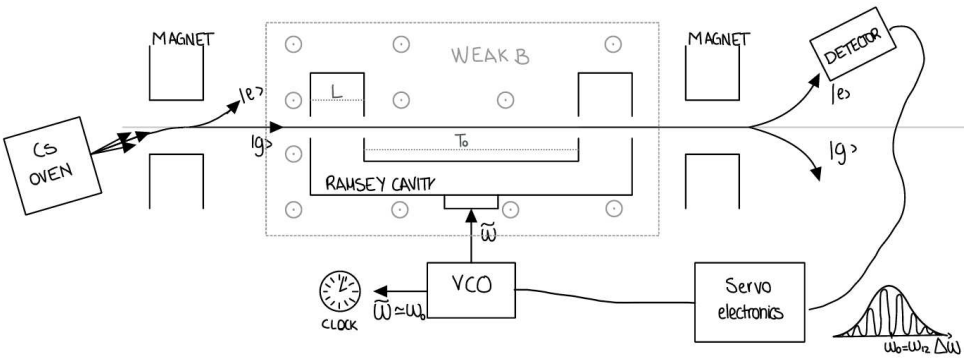
where B_0 is very large. The field is so strong that it gives rise to <u>nonlinear Zeeman splitting</u>, as shown in figure.



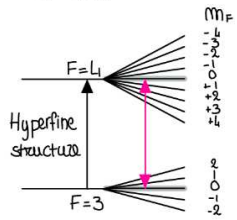
The force acting on a caesium atom passing through the device is then

$$\vec{F} = -\vec{\nabla} E_p \simeq -\frac{\partial E_p}{\partial z}\vec{u}_z = -\frac{\partial E_p}{\partial B}\frac{\partial B}{\partial z}\vec{u}_z$$

In nonlinear Zeeman splitting regime, $\frac{\partial E_p}{\partial B}$ can only have two values, one positive and one negative. Since $\frac{\partial B}{\partial z} > 0$ in both cases, there are two possible deflection directions: the states with $F = 4$ and $m_F \neq -4$ (including $|e\rangle$) are deflected in one direction, while all other states (including $|g\rangle$) are deflected in the other.



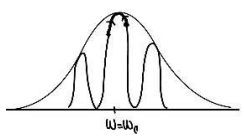
After the deflection, the ground state beam enters a Ramsey cavity driven by a voltage-controlled oscillator, that is an oscillator whose frequency can be selected by changing the voltage.



A <u>weak magnetic beam is applied in the Ramsey cavity to keep the degeneracy lifted</u>, so as to work only with the pure $0 - 0$ transition.

A second magnet at the output splits the atoms based on their state. A detector measures the atoms in the excited state.

Suppose that $\tilde{\omega}$ of the cavity is not perfectly tuned to the transition frequency ω_0, then the number of atoms in the excited state decreases. Based on the Ramsey fringe, the servo electronics provides a feedback to the voltage-controlled oscillator to adjust the frequency.



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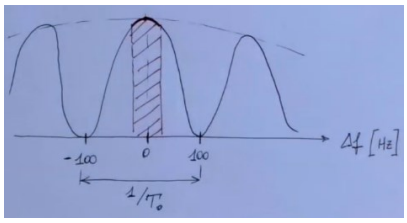
In the end, the frequency of the oscillator is stabilised against the frequency of the transition $|g\rangle \rightarrow |e\rangle$ of the Cs^{133} isotope: the oscillator provides a clock signal.

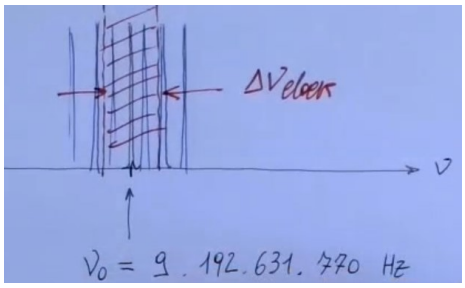
Let us look closely at the Ramsey fringe signal: the frequency of the oscillator stabilises around the transition frequency. The final precision and accuracy of the clock depends therefore on $1/T_0$:



$$\Delta\nu_{\text{clock}} = a\frac{1}{T_0}$$

where $a \sim 10^{-2} \div 10^{-3}$.

The fundamental limit to the frequency stability of the clock is therefore determined by the transit time T_0. The velocity of caesium atoms was $v_{Cs,\text{oven}} \simeq 100\ m/s$ and the distance between cavities is $d = 0.5\ m$, thus $T_0 = d/v_{Cs,\text{oven}} \simeq 5\ ms$, $1/T_0 = 200\ Hz$ and $\Delta\nu_{\text{clock}} \simeq 10^{-3} \cdot 200\ Hz = 0.2\ Hz$. This gives a good relative stability

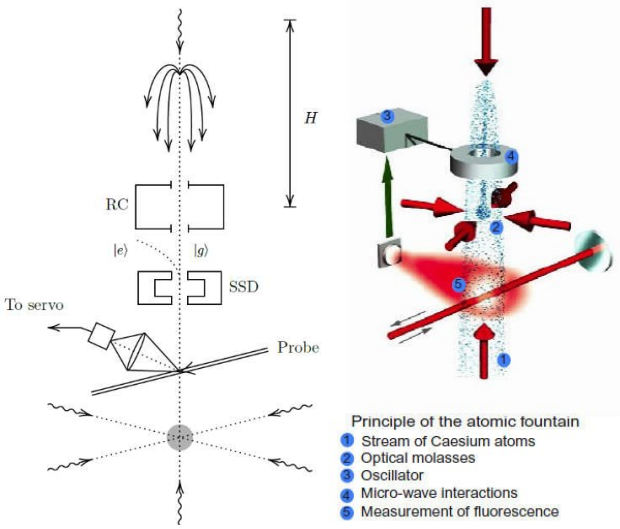


$$\frac{\Delta\nu}{\nu_0} \sim \frac{0.1\ Hz}{10 \cdot 10^9\ Hz} = 10^{-11}$$

To further improve the stability, the distance between the Ramsey cavities can be increased, but not enough to change the order of magnitude of T_0. An alternative solution consists in reducing the transit time by using so called cold atoms, as is done in atomic fountain clocks.

Atomic fountain clocks

Initially, six laser beams form an optical molasses. At some point, the laser beams are turned off and the cold atoms are released. Then there is a region with a probe laser which can be turned on to determine the amount of excitation with a detector. Then a state selecting device (similar to the Stern-Gerlach magnet) separates the atoms based on their state. Then the atoms go through a Ramsey cavity, travel a certain height and fall back inside the Ramsey cavity because of gravity.



Assuming that T_0 is the time of a complete transit within the fountain:

$$H = z(T_0/2) = \frac{1}{2}g\left(\frac{T_0}{2}\right)^2 \quad \Rightarrow \quad T_0 = 2\sqrt{\frac{2H}{g}}$$

Considering a fountain with $H = 0.3\ m$, we get $T_0 \simeq 0.5\ s$, which is two orders of magnitude higher than before.

Caesium fountain clocks can reach accuracies of 10^{-16}. This corresponds to an error of the clock of $1\ s$ every 300 million years.

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